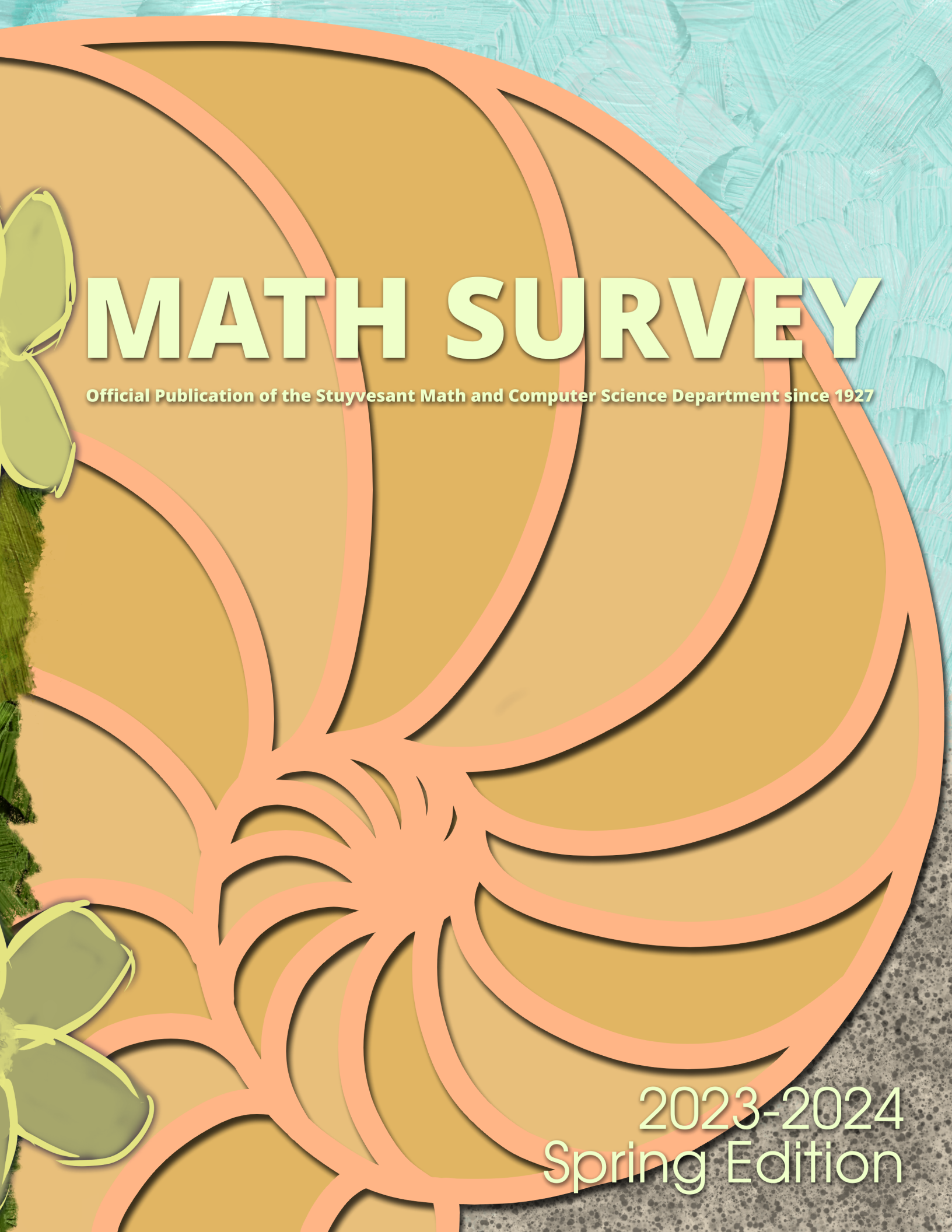


A large, stylized orange flower graphic with multiple layers of petals, centered on the cover. The petals are outlined in a darker orange and filled with a lighter orange color. The background is a mix of light blue and light green textures.

MATH SURVEY

Official Publication of the Stuyvesant Math and Computer Science Department since 1927

2023-2024
Spring Edition

Contents

Page No.

Contributor's List	i
Editor's Note	ii
1 Puzzles	1
Dissection Puzzle <i>by Ethan Sharma</i>	1
Solutions	5
Set <i>by Mikayla Lin</i>	9
Solutions	10
Mino Puzzle <i>by Patrick Was</i>	12
Solution	13
Spring Kakuro <i>by Vincent Chen</i>	14
Solutions	15
2 Mathematics Articles	16
Quadratic Number Theory <i>by Tanvir Ahmed</i>	16
Catalan Numbers <i>by Ivy Zhu</i>	32
Primitive Roots <i>by Patrick Was</i>	44
Bayesian Inference and Tossing Coins <i>by Narnia Poddar</i>	53
The Black–Scholes Equation: Transforming Quantitative Finance <i>by Anastasia Lee</i>	61
Introduction to Mathematical Modeling <i>by Ethan Sharma</i>	65
Mario Kart Score Distributions <i>by Elijah White</i>	69
The Math Behind Western Standoffs: Bayesian Games <i>by Shaon Anwar</i>	75
Lucas Numbers <i>by Elizabeth Burnell</i>	78
Cake Cutting <i>by Elizabeth Burnell</i>	83
3 Computer Science Articles	86
Wasserstein Generative Adversarial Networks (WGANs) <i>by Han Xiao</i>	86
4 Features	92
Stuyvesant Math and Computer Science Achievements	92

Stuyvesant Math Survey

2023-2024 Spring Edition

The Official Mathematics and Computer Science Publication
of the Stuyvesant High School Math Department since 1927

Principal

Mr. Seung Yu

Department Chairman

Mr. Eric Smith

Faculty Advisor

Mr. Gary Rubinstein

Editor-in-Chiefs

Amanda Louie
Daniel Potievsky

Associate Editors

Calvin Zhang
David Jiang

Math Editor

Shaon Anwar

Puzzles Editor

Aditya Pahuja

CS Editor

Han Xiao

Math Staff

Ethan Sharma
Patrick Was
Ivy Zhu
Anatasia Lee
Tanvir Ahmed
Narnia Poddar
Elizabeth Burnell
Elijah White

Puzzles Staff

Mikayla Lin
Ethan Sharma
Patrick Was
Vincent Chen

Art Editor

Patrick Cao

Layout Staff

Eric Ye
Vincent Chen

Business Manager

Anastasia Lee

Sponsor

Stuyvesant Parents' Association

Special thanks to Mr. Gary Rubinstein for his support for this edition.



Visit <https://stuymathsurvey.com> for the online version.
345 Chambers Street, New York, NY 10282 | stuymathsurvey@gmail.com

Editor's Note

This spring, we are delighted to share with you the 2023-2024 Spring Edition of *Math Survey*! This edition is filled with articles exploring new topics (and revisiting some!) in math and computer science as well as engaging puzzles. Thank you so much to our writers for their hard work and dedication to research in math and computer science as well as to our publication.

We would like to say a big thank you to our faculty advisor, Mr. Rubinstein, for his guidance and support. We would also like to extend our appreciation to the Parents' Association, for without their generous support we would not be able to print hard copies.

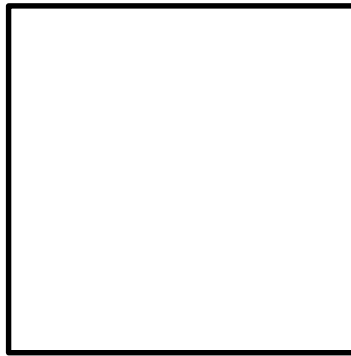
Looking back on our roles as Co-Editor-in-Chiefs, we want to thank everyone we've had the privilege of working with for their enthusiasm and passion for math and computer science. We admire the students, teachers, and alumni that have contributed through this publication's success, for the hours they have sacrificed to showcase the beauty of problem solving and we are grateful that you have allowed us to lead you this past year. To our fellow members of the Class of 2024, congratulations, and to next year's team, we wish you the best of luck in continuing to expand Math Survey's legacy!

Amanda Louie, Daniel Potievsky
June 2024

Dissection Puzzle

Ethan Sharma

Welcome to the Dissection Puzzle! For each problem, you will be given a starting figure and a desired figure. Your task is to make a finite number of cuts and rearrange the pieces from the starting figure to the desired figure. For 2-dimensional figures, a cut consists of a line that intersects the figure at two points (you will only be given convex 2-dimensional figures). You are allowed to add/remove segments from the figure as long as the figure remains closed. For a general n -dimensional figure, a cut is a $(n - 1)$ -dimensional hyperplane that cuts the figure into more than one n -dimensional figures. Then, you must put the pieces back together to obtain the desired figure. With this, rearrangement like rotations are allowed. As an example:



(a) Starting Figure (Square)



(b) Desired Figure

Figure 1: Example Dissection

The solution cuts are shown on the next page.

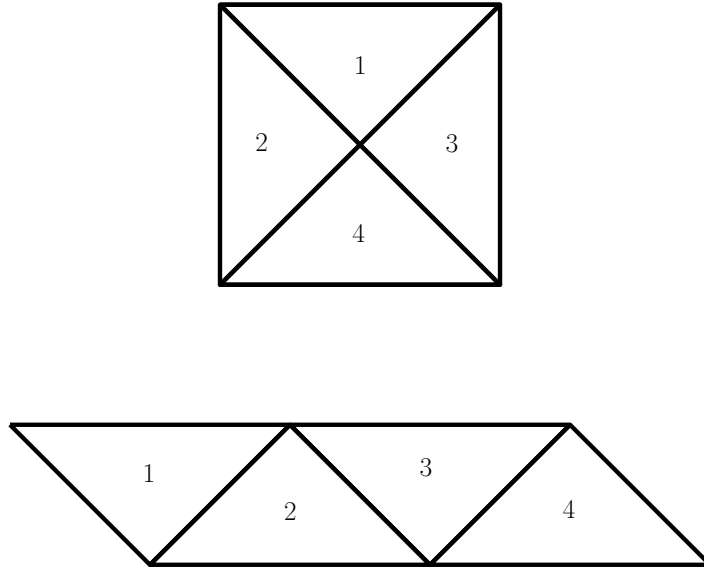


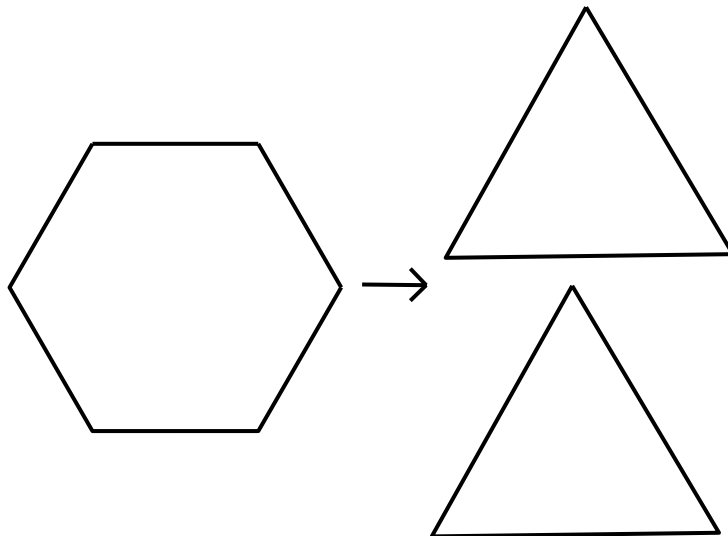
Figure 2: Example Dissection Cut Solution

Note that the order we number the segments here does not matter since they are all congruent.

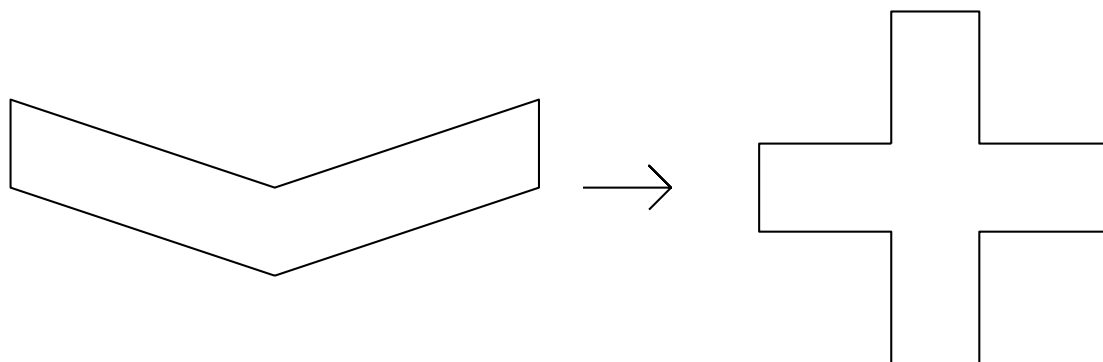
The solution is on page 5.

Problem 1

The starting figure is a regular hexagon, and the desired figure is two congruent equilateral triangles.

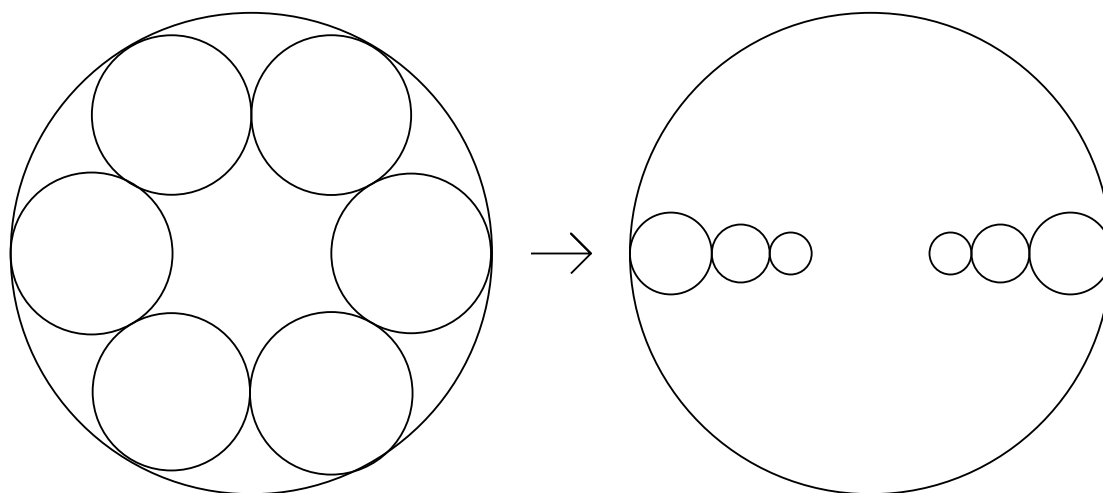


Problem 2



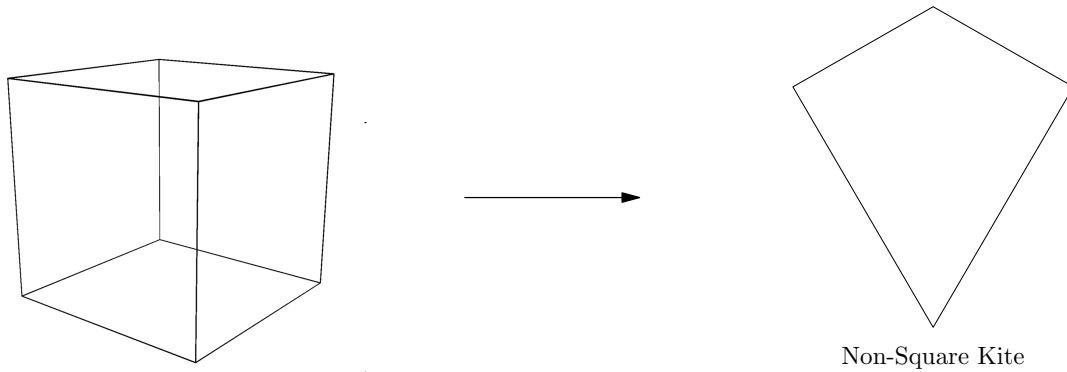
Problem 3

For this problem, you are allowed to perform one geometric inversion, but you must undo it before getting to the final figure (that is, invert again about the same reference circle). The figure isn't directly to scale, but try to get the same configuration.



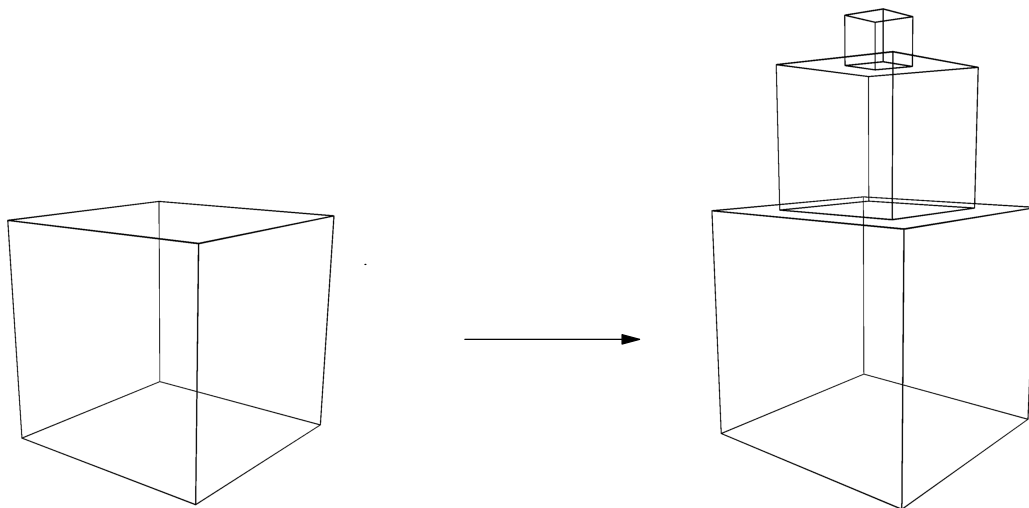
Problem 4

For this problem, you are allowed to perform an orthogonal projection onto a plane along with cuts and rearrangements. The initial figure is a cube, and the desired figure is a non-square kite.



Problem 5

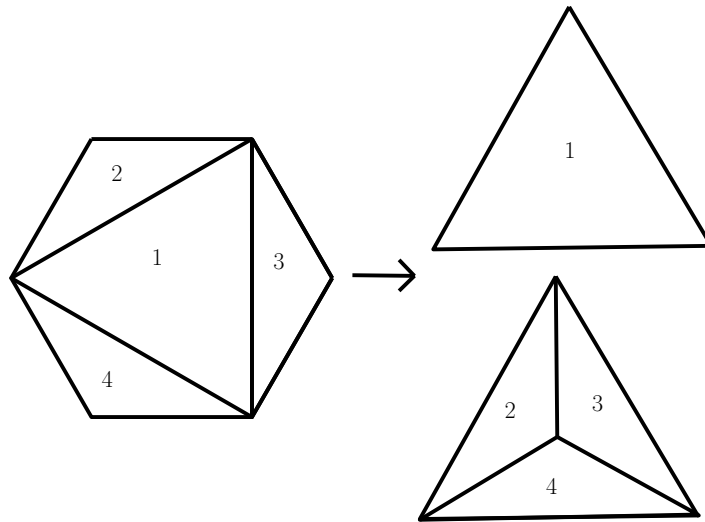
The initial figure is a cube, and the desired figure shows three cubes of descending side length stacked on each other.



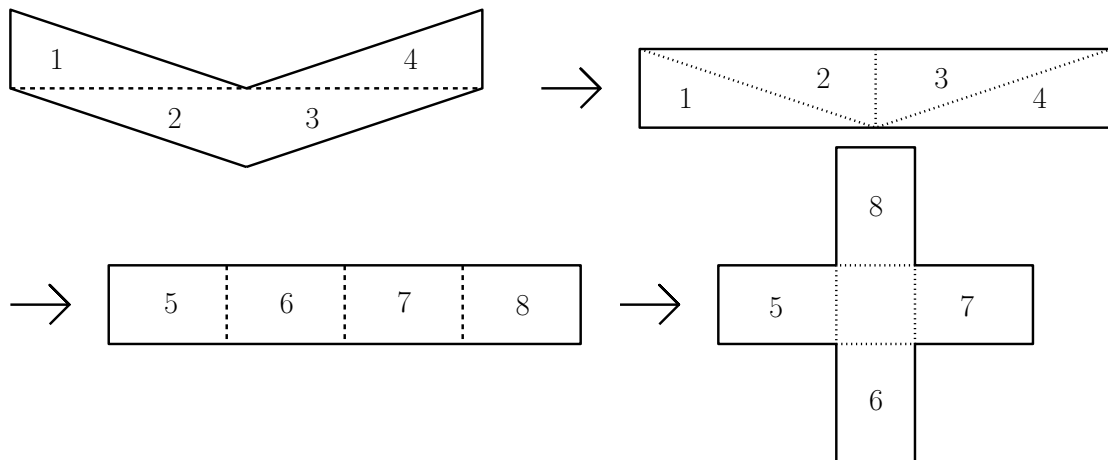
Dissection Puzzle Solutions

Ethan Sharma

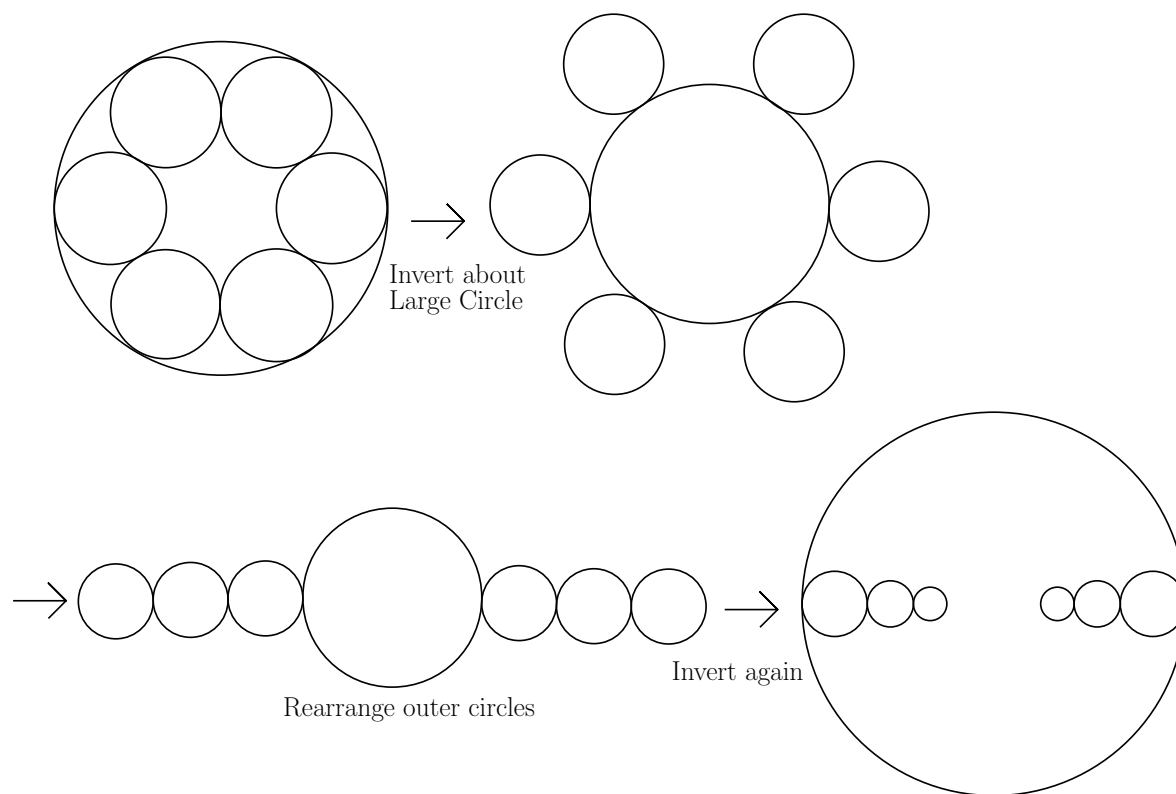
Problem 1



Problem 2

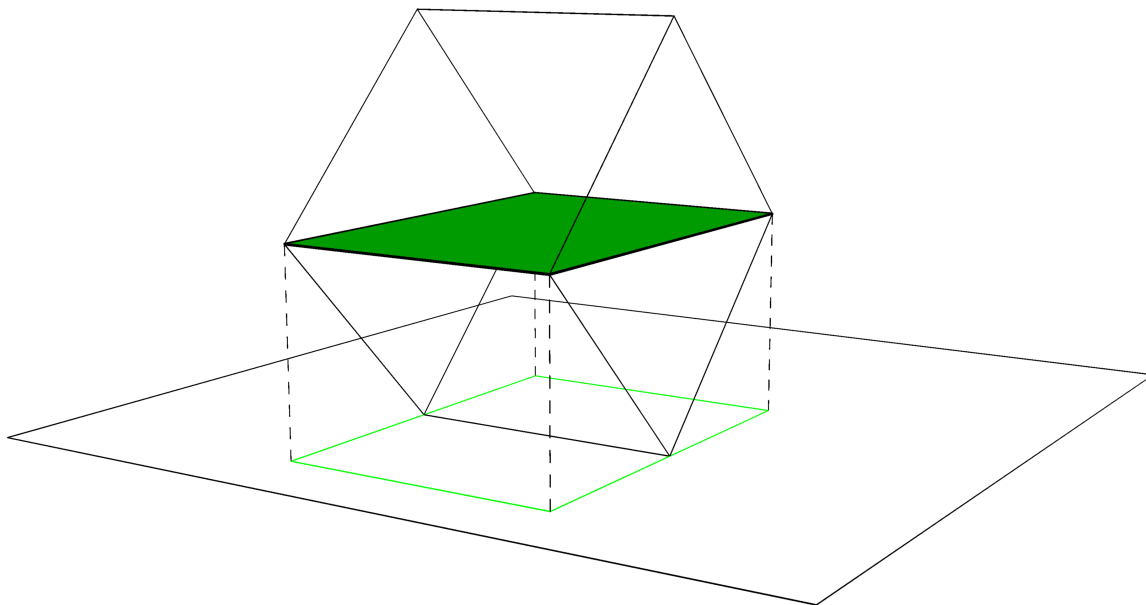


Problem 3



Problem 4

First, rotate the cube so that a rectangular cross section is perpendicular to the plane we plan on projecting onto. Alternatively, we can just choose the plane that'll give us a rectangular projection.



After projection, we are left with a rectangle with side lengths 1 and $\sqrt{2}$. We then cut along the main diagonal and rotate one piece, as shown, and we are left with a non-square kite.

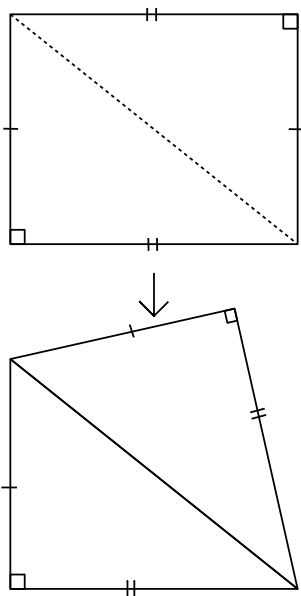


Figure 1: Rectangle to Non-Square Kite

Problem 5

We can perform cuts with two dimensional planes parallel to a face of the cube. This is a face view perpendicular to those planes to see the cuts. We can slightly bypass the requirement of cuts being full planes by just putting back together the dotted/dashed borders to reform the full cube.

Assume the initial cube is $5 \times 5 \times 5$ (for instructive purposes, it could be anything). The first four cuts is along the larger dashed lines, then we take out the $3 \times 3 \times 5$ rectangular prism out the middle and perform the four cuts along the dotted lines. Again, we put the edges back together and take out a $1 \times 1 \times 5$ rectangular prism. Then, we only show part of each prism to give an illusion of three cubes stacked on top of each other.

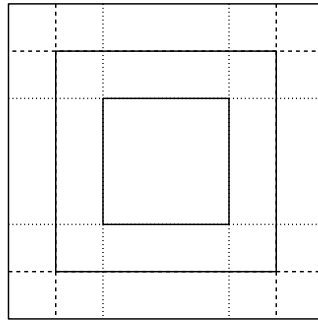
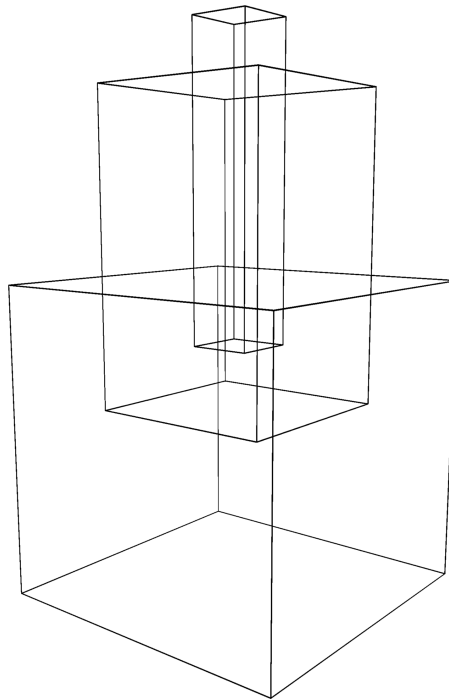


Figure 2: Face view. Dotted/Dashed lines are cut lines

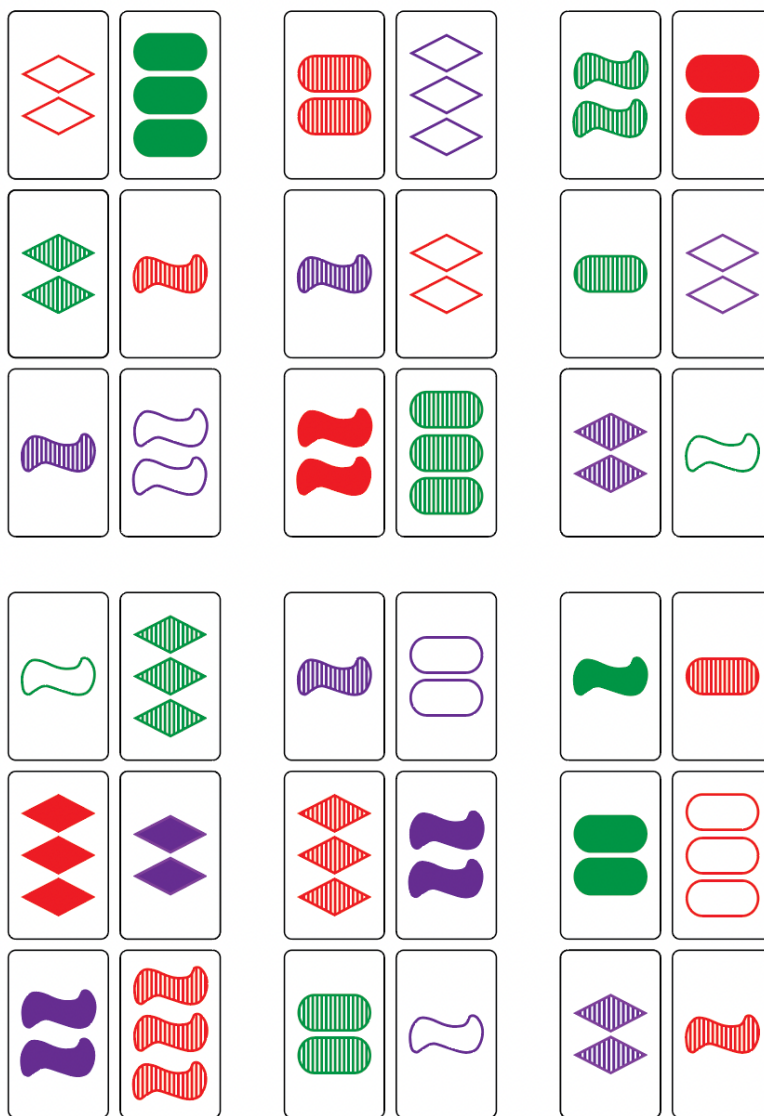
Here is what the figure looks like with the insides shown:



Set

Mikayla Lin

Daniel has gotten distracted from his math homework with a game of **set**. However, without reading glasses he's practically **blind** and unable to see the entire screen. Perhaps you can figure out what topic he should return to.

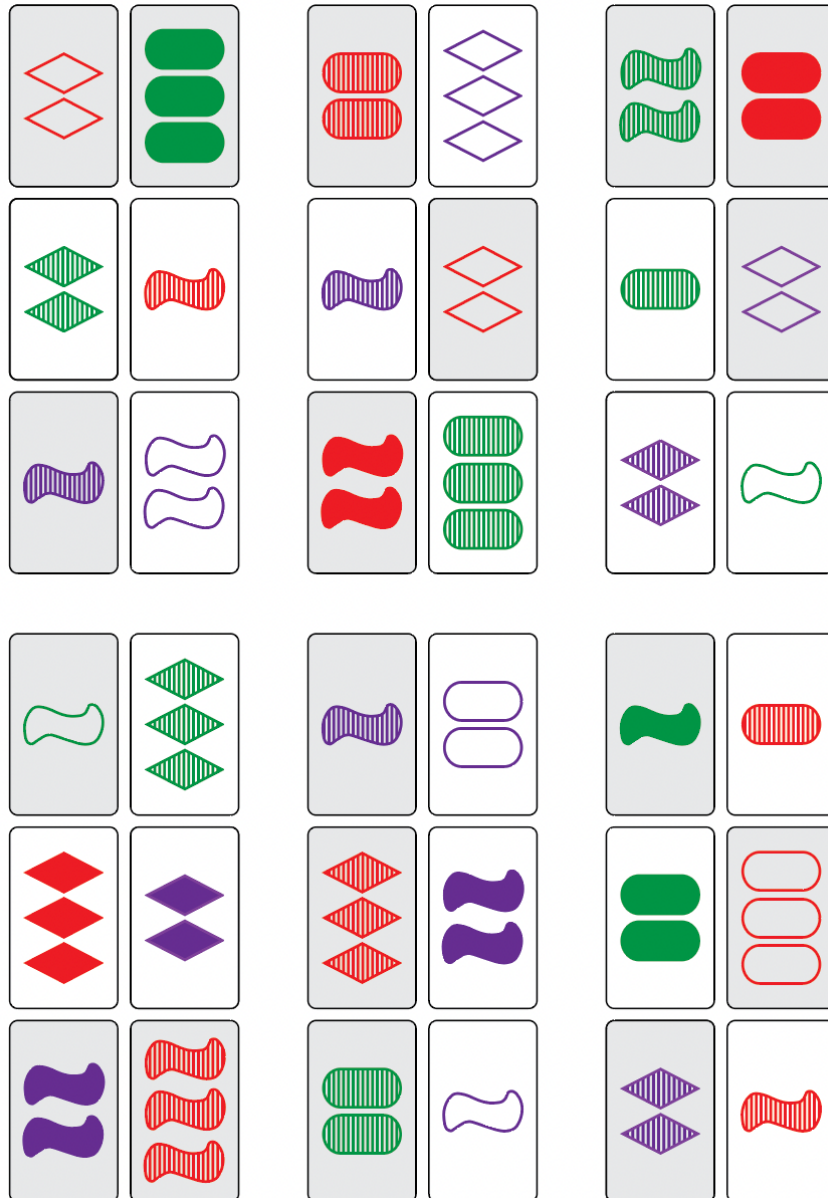


The solution is on page 10.

Solution to Set

Mikayla Lin

First, find the sets within each 2 by 3 grid.



The “blind” in the flavor text and the 2 by 3 grids clue the usage of braille.

a	b	c	d	e	f	g	h	i	j
●○ ○ ○	●○ ●○ ○	●● ○ ○	●● ○ ○	●○ ○ ○	●● ●○ ○	●● ●● ○	●○ ●● ○	○● ●○ ○	○● ●● ○
k	l	m	n	o	p	q	r	s	t
●○ ○ ●	●○ ●○ ●	●● ○ ●	●● ○ ●	●○ ○ ●	●● ●○ ●	●● ●● ●	●○ ●● ●	○● ●○ ●	○● ●● ●
u	v	w	x	y	z				
●○ ○ ●●	●○ ●○ ●●	○● ●● ○	●● ○ ●●	●● ○ ●●	●○ ○ ●●				

Using this key, you should get **MODULO**.

Mino Puzzle

Patrick Was

The Game of Tetris

Tetris is one of those games that transcends the wearing down of time—it seems it has never been more popular than now, played by nerdy Stuyvesant students in the halls and computer labs, despite its first version being developed in 1984.

Its history isn't terribly significant in any mathematical way, but its original creator, Alexey Pajitnov, originally had an alternative idea to the premise of the game. Though the fundamentals would have been the same—stacking, clearing lines, getting points—he had originally planned on **pentominoes** instead of the **tetrominoes** that we know of.

But... why would he *change* that facet of the game? Why not keep it as pentominoes? Sure, there's the obvious notion of decreasing piece count to not make the game overbearing, but surely there must have been something else. Maybe exploring **minos** could help one gain some insight on this.

It helps to define the following:

- A **mino** is a square cell.
- A **polyomino** is a connection of minos such that each is connected to at least one other mino by an edge.
- An n -**mino** is a polyomino composed of n minos (though **pentominoes**, made of 5 minos, and **tetrominoes**, made of 4 minos, will be referred to by these names).

The solution is on page 13.

Part I

1. Consider a 7×4 rectangle. Is it possible to fill this square using one of each tetromino (namely, the I, O, L, J, S, Z, and T piece)?
2. Consider placing a T-tetromino on the bottom of a limitless board. What placement of a follow T-tetromino would best accommodate the remaining pieces?
3. Is it possible to tile a 3×20 board using each pentomino exactly once?
4. Is it possible to tile two 6×5 rectangles using each pentomino exactly once?

Part II

1. Let $M(n)$ be the number of pieces of an n -mino. For example, $M(4) = 7$ because there are 7 distinct tetrominoes. Try to come up for an expression of $M(n)$ in terms of n .

Solution to Mino Puzzle

Patrick Was

Part I

1. **Answer:** Such an arrangement is not possible.

Solution: This question introduces the idea of **parity**, which is a topic of interest in both stacking mechanics and identifying **perfect clears**.

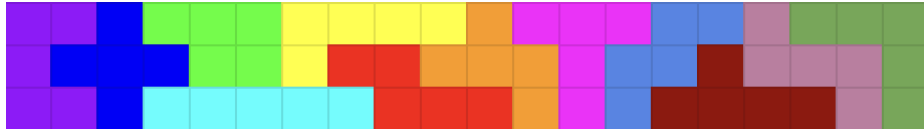
Consider the rectangle described in the problem. Shade each of the 28 cells alternating black or white. Because the number of cells within the rectangle is even, there will be 14 black and 14 white squares no matter the coloration.

However, coloring each tetromino with the same pattern, it appears that there are always 2 black and 2 white cells except for the T piece. Depending on the coloration, it will have 3 black cells or 3 white cells, with only 1 of the alternative cell color. Since the parities of the 7 pieces together and the rectangle are not equal, such an arrangement is not possible.

2. **Answer:** An even number of spaces away.

Solution: Though the cleanest solution would be to place one T-piece on top of another, as this will most easily result in an even number of black and white squares, placing the second T-piece anywhere an even number of pieces away will make them result in the same parity, hence resulting in an even parity.

3. **Answer:**



4. **Answer:**



Part II

1. **Answer/Solution:** Despite many efforts, there is no general formula for an n -mino, though there are estimates to an upper and lower bound for a relation between n -minos with different rotational properties.

Spring Kakuro

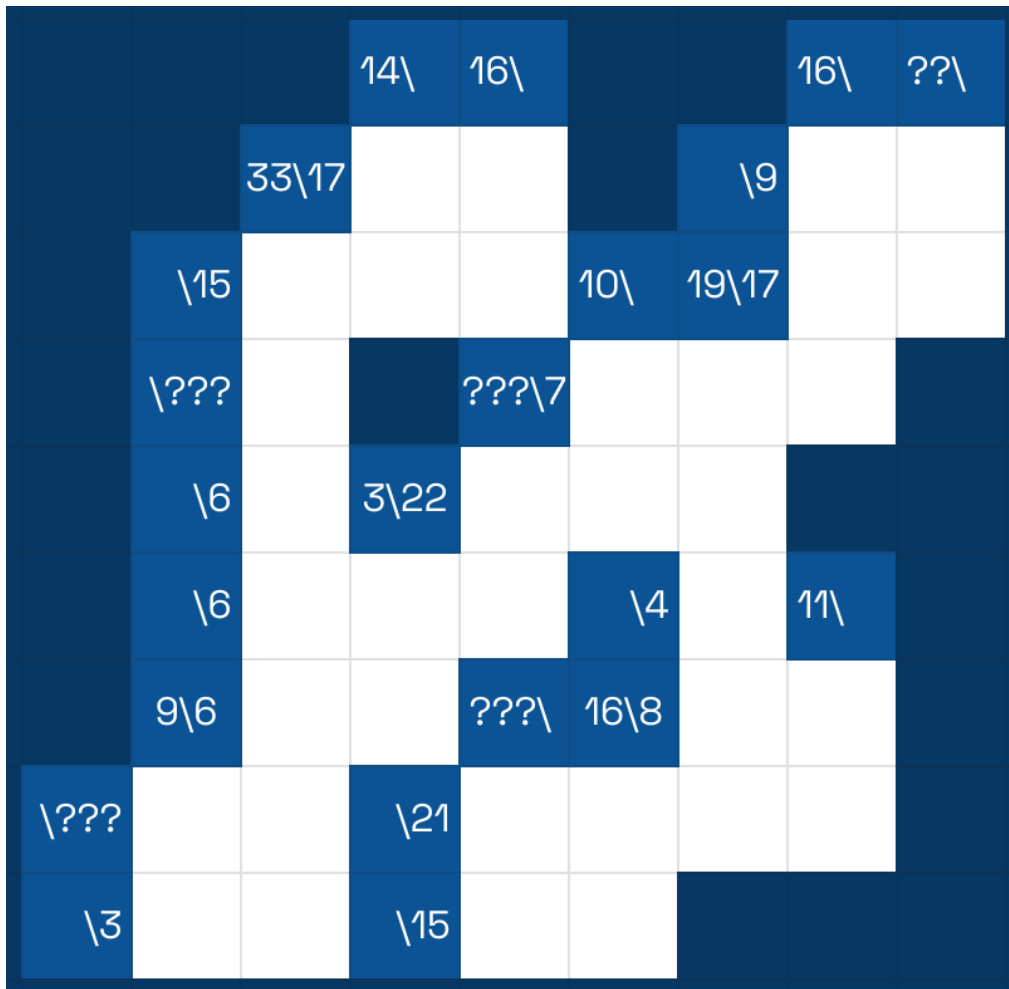
Vincent Chen

This game essentially mixes the rules of Kakuro and Sudoku:

- You must put in digits from 1-9 in each cell (you may NOT put in 0)
- No number may appear more than once in each row or column (see below for an example).
- The diagonals indicate the sum of the digits in its row / column.

The only difference in this version is that you have to pay attention to the entire row and column, not just the hint. For example, normally the sequence “3 4 ——— 3 4 7” would be legal in normal Kakuro, where “3 4” and “3 4 7” correspond to different hints on the same row, but not in this version.

The solution is on page 15.



Solution to Spring Kakuro

Vincent Chen

			14\	16\			16\	14\
		33\17	8	9		\9	3	6
	\15	2	6	7	10\	19\17	9	8
	\9	9		7\7	2	1	4	
	\6	6	3\22	5	8	9		
	\6	3	1	2	\4	4	11\	
	9\6	4	2	12\	16\8	3	5	
\15	7	8	\21	4	9	2	6	
\3	2	1	\15	8	7			

Quadratic Number Theory

Tanvir Ahmed

1 Introduction

While “quadratic” number theory isn’t a conventional name for a particular subject in number theory, it’s helpful to think of number theory that is related to quadratics/squares as related to each other. In this article, we will look at the notion of quadratic residues and devise different ways to check if something is a quadratic residue, and then apply this knowledge to the square-sum problem, which in general concerns representing positive integers as a sum of perfect squares. In the first section, we will introduce the notion of a field as it is a helpful way to think about congruences and modular arithmetic.

2 The Finite Fields of Order p

2.1 Definition of a Field

It is first important to establish that the set of numbers from 0 to $p-1$ for a prime number p forms a field under addition and multiplication mod p . We first have to define a field, which is as follows:

Definition 2.1. A field F is a set of elements, along with two binary operations $+$ and $\cdot$ (addition and multiplication) such that the following are true for all $a, b, c \in F$:

- $a + b = b + a$
- $a \cdot b = b \cdot a$
- $a + (b + c) = (a + b) + c$
- $a \cdot (b \cdot c) = (a \cdot b) \cdot c$
- $a \cdot (b + c) = a \cdot b + a \cdot c$
- $a + 0 = a$ for some fixed element $0 \in F$ (called the additive identity)
- $a \cdot 1 = a$ for some fixed element $1 \in F$ (called the multiplicative identity)
- There is an element $-a \in F$ such that $a + (-a) = 0$
- If $a \neq 0$, there is an element $a^{-1} \in F$ such that $a \cdot a^{-1} = 1$

Some common examples of fields are $\mathbb{R}$, $\mathbb{Q}$, and $\mathbb{C}$, under the usual operations of addition and multiplication. Now, we let $\mathbb{Z}_n$ be the set of numbers from 0 to $n-1$ inclusive, where the operations of addition and multiplication are defined mod n . For example, in $\mathbb{Z}_6$, $3 + 4 = 1$, because once you reduce $3 + 4 \bmod 6$, you get 1, and multiplication is defined similarly. For the rest of this paper, we will often call the elements of $\mathbb{Z}_n$ residues. It is easy to check that all the properties of a field except for the last one hold for all positive integers n for $\mathbb{Z}_n$. Considering when the last property is true leads to the following theorem.

Theorem 2.1. $\mathbb{Z}_n$ is a field if and only if n is prime.

Proof. The reverse direction is easy to prove. Suppose n is composite, and let a be a divisor of n not equal to 1 or n . Then, for all integers b , ab will be a multiple of a , a property that is preserved mod n because n is also a multiple of a . In other words, ab cannot be equal to 1 for any b , so a can't have an inverse mod n .

Suppose n is prime and let a be an integer such that $a \not\equiv 0 \pmod{n}$. We wish to show that a has an inverse mod n . The condition that $a \not\equiv 0 \pmod{n}$ is equivalent to saying that a is relatively prime to n since n is prime. Thus, by Bezout's Lemma, there are integers b and k such that $ab + nk = 1$. Reducing this mod n , we get $ab \equiv 1 \pmod{n}$, which shows that b is the inverse of a mod n . $\square$

This implies that $\mathbb{Z}_p$ is a field for prime p . These fields are known as finite fields, as opposed to $\mathbb{R}$, $\mathbb{Q}$, $\mathbb{C}$, etc. which are infinite. The second part of this proof also implies that in general, if $\gcd(a, n) = 1$, a has a multiplicative inverse mod n , which is also important.

2.2 Polynomials

Definition 2.2. A polynomial $p(x)$ in the field F is of the form $a_n x^n + a_{n-1} x^{n-1} + \cdots + a_1 x + a_0$ where $a_i \in F$, $a_n \neq 0$, and x is a variable. Furthermore, the degree of $p(x)$ is n . If $a \in F$, then $p(a)$ is the result of substituting a for x in the expression $p(x)$ and evaluating it. The set of polynomials over F is denoted by $F[x]$, and is called the ring of polynomials over F .

Most of the time, when we deal with polynomials, we are most concerned with solving for the values of x which make the polynomial evaluate to 0, also called the roots of the polynomial. For real numbers, it is well known that if a polynomial has degree n , it has at most n roots, and this is true for fields of the form $\mathbb{Z}_p$ as well, which we will now prove.

Theorem 2.2. If $p(x) \in F[x]$ for an arbitrary field F has degree n , $p(x)$ has at most n roots.

Proof. We prove this by induction on n . If $n = 0$, $p(x)$ is always equal to a nonzero constant, so it has no solutions. If $n = 1$, $p(x) = ax + b$ for $a, b \in F$. In this case, $ax + b = 0$ implies $x = \frac{-b}{a}$, so $p(x)$ only has one solution.

Now suppose that all polynomials of degree n have at most n roots. Let $p(x)$ be a polynomial of degree $n + 1$. We can express it as $p(x) = a_{n+1} x^{n+1} + a_n x^n + \cdots + a_1 x + a_0$. If $p(x)$ has no roots in F , we are done. otherwise, suppose x_1 is a root. Then, we have the following two equations for all $x \in F$:

$$\begin{aligned} p(x) &= a_{n+1} x^{n+1} + a_n x^n + \cdots + a_1 x + a_0 \\ 0 &= a_{n+1} x_1^{n+1} + a_n x_1^n + \cdots + a_1 x_1 + a_0 \end{aligned}$$

Subtracting these equations gives us:

$$p(x) = a_{n+1}(x^{n+1} - x_1^{n+1}) + a_n(x^n - x_1^n) + \cdots + a_1(x - x_1)$$

Importantly, for each i , we have $x^i - x_1^i = (x - x_1)(x^{i-1} + x^{i-2}x_1 + \cdots + x_1^{i-1})$. We will call that second factor $q_i(x)$ because it's not important what it is. Therefore, $x^i - x_1^i = (x - x_1)q_i(x)$. Plugging this back into the last equation, we get

$$\begin{aligned} p(x) &= a_{n+1}(x - x_1)q_{n+1}(x) + a_n(x - x_1)q_n(x) + \cdots + a_1(x - x_1)q_1(x) \\ &= (x - x_1)(a_{n+1}q_{n+1}(x) + \cdots + a_1q_1(x)) \end{aligned}$$

If we call the second factor $q(x)$, we get $p(x) = (x - x_1)q(x)$, where $q(x)$ has degree n . Thus, if x is a root of $p(x)$, x is a root of $x - x_1$ or of $q(x)$. $x - x_1$ has only one root, and $q(x)$ has at most n roots by the induction hypothesis, so altogether, $p(x)$ has at most $n + 1$ roots. □

3 Quadratic Residues

3.1 QR's and QNR's

The properties of a field are such that one can easily solve linear equations (in the context of $\mathbb{Z}_p$, they would be linear congruences), and we do this by using the standard four operations (addition, subtraction, multiplication, division). Quadratic equations, however, are not as easy, and it is not guaranteed that they always have a solution. In the next few sections, we are interested in when quadratic equations have solutions in the field $\mathbb{Z}_p$. Since equations over $\mathbb{Z}_p$ are actually congruences, we will refer to these as quadratic congruences.

The simplest quadratic congruence is $x^2 \equiv a \pmod{p}$ for some $a \in \mathbb{Z}_p$. We say that a is a quadratic residue mod p if $a \not\equiv 0 \pmod{p}$ and this congruence has a solution. Since $\mathbb{Z}_p$ is a field and this congruence is a polynomial of degree 2, the congruence has at most 2 solutions in x . In fact, if $a \not\equiv 0 \pmod{p}$, and the congruence has one solution, then it has 2 solutions, because if $x^2 \equiv a \pmod{p}$, $(-x)^2 \equiv a \pmod{p}$ as well. We will use the abbreviations QR and QNR to stand for quadratic residue and quadratic non-residue respectively.

The first important thing we can establish about quadratic residues is that there is always an equal numbers of QR's and QNR's for any odd prime p .

Proposition 3.1. If p is an odd prime, there are exactly $\frac{p-1}{2}$ QR's and $\frac{p-1}{2}$ QNR's.

Proof. Every quadratic residue corresponds to two "square roots", and these square roots must be distinct. To see this, suppose that the congruence $x^2 \equiv a \pmod{p}$ has two identical roots for some QR a and say that root is r . We saw how these roots are negatives of each other, which implies that $r \equiv -r \pmod{p} \implies 2r \equiv 0 \pmod{p}$. Since a is a QR, $a \not\equiv 0 \pmod{p}$, so $r \not\equiv 0 \pmod{p}$. By reducing $r \pmod{p}$, we may then assume $1 \leq r \leq p-1 \implies 2 \leq 2r \leq 2p-2$. By the congruence $2r \equiv 0 \pmod{p}$, we have that $2r$ is a multiple of p , but the only multiple of p in the range 2 to $p-2$ is p , so $2r = p$, but p is odd, leading to a contradiction.

We also know that every residue from 1 to $p-1$ is a square root of some QR: it is a square root of its square. Therefore, if we consider all the square roots of all QR's, not only do we get all residues from 1 to $p-1$, but we also get twice the number of QR's, implying that the number of QR's is $\frac{p-1}{2}$, and the number of QNR's is $p-1 - \frac{p-1}{2} = \frac{p-1}{2}$. □

3.2 The Legendre Symbol

The Legendre symbol, denoted by $\left(\frac{a}{p}\right)$ is defined for odd primes p as follows:

$$\left(\frac{a}{p}\right) = \begin{cases} 1 & a \text{ is a QR mod } p \\ -1 & a \text{ is a QNR mod } p \\ 0 & a \equiv 0 \pmod{p} \end{cases}$$

One important property of the Legendre Symbol is that it is multiplicative, which we will now prove.

Proposition 3.2. For an odd prime p and for integers a and b , $\left(\frac{ab}{p}\right) = \left(\frac{a}{p}\right) \left(\frac{b}{p}\right)$.

Proof. If $a \equiv 0 \pmod{p}$ or $b \equiv 0 \pmod{p}$, then $\left(\frac{ab}{p}\right) = \left(\frac{0}{p}\right) = 0$ and $\left(\frac{a}{p}\right) \left(\frac{b}{p}\right) = 0$, so the two expressions are equal.

If $\left(\frac{a}{p}\right) = \left(\frac{b}{p}\right) = 1$, then $a \equiv x^2 \pmod{p}$, $b \equiv y^2 \pmod{p}$, so $ab \equiv (xy)^2 \pmod{p}$, so $\left(\frac{ab}{p}\right) = 1$, making the two expressions equal.

If one of a and b is a QR and the other is a QNR, then without loss of generality, say $a \equiv x^2 \pmod{p}$ and b is QNR. If ab is a QR, so that $ab \equiv y^2 \pmod{p}$, then $a \equiv \frac{ab}{b} \equiv \left(\frac{x}{y}\right)^2 \pmod{p}$, which would imply that b is a QR, which we assumed that it was not. Thus, $\left(\frac{a}{p}\right) \left(\frac{b}{p}\right) = 1(-1) = -1 = \left(\frac{ab}{p}\right)$.

In the case that a and b are both QNR's, then we have to show that ab is a QR so that $\left(\frac{ab}{p}\right) = 1$. If we multiply a by all nonzero residues mod p , then we must get back all nonzero residues mod p , and we know that multiplying a by all the QR's give us all the QNR's, which constitute half the residues mod p . The other half of the residues mod p are QR's by Theorem 3.1. Thus, multiplying a by QNR's will give us QR's. In other words, if both a and b are QNR's, then $\left(\frac{a}{p}\right) \left(\frac{b}{p}\right) = (-1)(-1) = 1 = \left(\frac{ab}{p}\right)$. $\square$

3.3 Primitive Roots

$\mathbb{Z}_p$ for primes p is special because it is a field, but it is also special because it always has a primitive root. To understand what a primitive root is, we first have to introduce the idea of an order of an number mod n .

Definition 3.1. If a and n are relatively prime integers, then the order of a mod n , denoted by $\text{ord}_n(a)$ is the smallest positive integer r such that $a^r \equiv 1 \pmod{n}$.

If $\text{ord}_n(a) = r$, then the result of taking consecutive powers of a mod n will only leave you with r numbers, because after the r th power, it will cycle back. Whenever there is an integer a such that a raised to consecutive powers will produce all residues relatively prime to n mod n , we call a a primitive root mod n . The reason it has to be relatively prime residues is that a is relatively prime to n itself, so any powers of it also must be relatively prime to n . We can phrase this condition more concisely if we use the Euler totient function.

Definition 3.2. The Euler totient function evaluated at a positive integer n , denoted by $\phi(n)$ is the number of positive integers less than or equal to n relatively prime to n .

Then, a is a primitive root mod n if $\text{ord}_n(a) = \phi(n)$. For a prime number p , this would imply that a is a primitive root of order $\text{ord}_p(a) = \phi(p) = p - 1$. It turns out that all prime numbers have primitive roots. To prove this, we first need two small lemmas.

Lemma 3.3.1. For relatively prime integers a and b , $\phi(ab) = \phi(a)\phi(b)$.

Proof. If a positive integer n is relatively prime to ab , then n is relatively prime to a and to b , and being relatively prime to a and to b are independent conditions since a and b are themselves relatively prime. Thus, $\phi(ab) = \phi(a)\phi(b)$. □

Lemma 3.3.2. For all positive integers n , we have:

$$\sum_{d|n} \phi(d) = n$$

Proof. We first consider the case where n has one prime factor, so it is of the form p^k and all of the divisors of p^k are of the form p^j . Therefore, we have:

$$\sum_{d|p^k} \phi(d) = \sum_{j=0}^k \phi(p^j)$$

It is easy to see that $\phi(p^j) = p^j - p^{j-1}$ for $j > 0$ because the only numbers relatively prime to p^j less than or equal to p^j are those that are not multiples of p and are in that range. Therefore, the sum becomes:

$$\sum_{j=0}^k \phi(p^j) = \sum_{j=0}^k p^j - p^{j-1} = 1 + p - 1 + p^2 - p + \cdots + p^k - p^{k-1} = p^k = n$$

Now suppose $n = p_1^{r_1} p_2^{r_2} \cdots p_k^{r_k}$. Then, all we have to do is multiply the sums $\sum_{d|p_i^{r_i}} \phi(d)$ for $i = 1, 2, \dots, k$, and we will obtain $\sum_{d|n} \phi(d)$. This is because the Euler totient function is multiplicative, product of these k sums is a unique product of k terms, each one from one of the sums $\sum_{d|p_i^{r_i}} \phi(d)$. Therefore,

$$\begin{aligned} \sum_{d|n} \phi(d) &= \left(\sum_{d|p_1^{r_1}} \phi(d) \right) \left(\sum_{d|p_2^{r_2}} \phi(d) \right) \cdots \left(\sum_{d|p_k^{r_k}} \phi(d) \right) \\ &= p_1^{r_1} p_2^{r_2} \cdots p_k^{r_k} \\ &= n \end{aligned}$$

□

Theorem 3.1. For every prime p , there exists a primitive root mod p .

Proof. For every divisor d of $p - 1$, let $\psi(d)$ be the number of residues mod p which have an order of d . On the one hand, every nonzero residue mod p has an order which is a divisor of $p - 1$ because by Fermat's Little Theorem, every nonzero residue raised to the $p - 1$ power is congruent to 1 mod p . Therefore, we can conclude that $\sum_{d|p-1} \psi(d) = p - 1$.

Suppose for some $d \mid p - 1$ that a has order d . Then, consider the polynomial $x^d - 1 \pmod{p}$, which by 2.2, has at most d solutions. However, $1, a, \dots, a^{d-1}$ are distinct solutions of that polynomial (we know they're distinct because d is the *order* of a). Obviously, any elements of order d must then be one of $1, a, \dots, a^{d-1}$, and those elements are the ones where a is raised to a power relatively prime

to d because otherwise, the order of that element would be less than d . Therefore, $\psi(d) = \phi(d)$ if there exists an element of order d at all.

What we have just shown is that if there exists an element of order d , there are $\phi(d)$ elements of order d . In other words, if $\psi(d) \neq 0$, $\psi(d) = \phi(d)$, so $\psi(d) = 0$ or $\psi(d) = \phi(d)$. This can be summarized as follows:

$$p - 1 = \sum_{d|n} \psi(d) \leq \sum_{d|n} \phi(d) = p - 1$$

with equality holding only when $\psi(d) = \phi(d)$ for all divisors of $p-1$. Since we can't have $p-1 < p-1$, we must have $\psi(d) = \phi(d)$ for all divisors of $p-1$. Thus, the number of primitive roots, which is $\psi(p-1)$ is equal to $\phi(p-1) > 0$. □

3.4 Euler's Criterion

Usually, we denote a primitive root mod p with the letter g . Thus, all the residues mod p are just $1, g, g^2, \dots, g^{p-1}$. If p is an odd prime, $p-1$ is even, which implies that the QR's are the even powers of g , and the QNR's are the odd powers of g . This provides another way to prove that the product of two QNR's is a QR as the product of two odd powers of g will obviously be an even power of g .

Euler's criterion uses primitive roots in a clever way to determine if a given number is QR mod p . It says the following:

Theorem 3.2. If a and p are relatively prime and p is prime, then

$$\left(\frac{a}{p}\right) \equiv a^{\frac{p-1}{2}} \pmod{p}$$

Proof. Let g be a primitive root and let $a \equiv g^r \pmod{p}$. If a is a QR, $\left(\frac{a}{p}\right) = 1$ and r is even, so $a^{\frac{p-1}{2}} \equiv (g^{p-1})^{\frac{r}{2}} \equiv 1 \pmod{p}$ since $\frac{r}{2}$ is an integer. On the other hand, if a is a QNR and r is odd, then $a^{\frac{p-1}{2}} \equiv (g^{r-1} \cdot g)^{\frac{p-1}{2}} \equiv 1 \cdot g^{\frac{p-1}{2}} \pmod{p}$. Since g is a primitive root, this is not congruent to 1. However, if we square $g^{\frac{p-1}{2}}$, we get 1, and the only other thing that squares to 1 is -1 , so $a^{\frac{p-1}{2}} \equiv g^{\frac{p-1}{2}} \equiv -1 \equiv \left(\frac{a}{p}\right) \pmod{p}$. □

If we wanted an algorithm to calculate the Legendre symbol with a computer, using Euler's criterion is a pretty good way to do that. Calculating exponents mod a given number are really fast to compute for a computer since we can repeatedly square the base and multiply those intermediate results together appropriately to obtain the particular power we want. This process closely resembles converting a number into binary. In particular, we are converting $\frac{p-1}{2}$ into binary. While this is a fast algorithm, we will see that using quadratic reciprocity will be even faster and more manageable to use to work out these problem by hand.

Finally, we can use primitive roots and Euler's criterion to figure out when there is a square root of -1 mod a prime p , or when $\left(\frac{-1}{p}\right) = 1$. By Euler's criterion, we know that $\left(\frac{-1}{p}\right) \equiv (-1)^{\frac{p-1}{2}} \pmod{p}$. When $p-1$ is a multiple of 4, then the exponent of -1 is even, so $\left(\frac{-1}{p}\right) = 1$. When $p-1$ is not a multiple of 4, then the exponent of -1 will be odd, and $\left(\frac{-1}{p}\right) = -1$. In other words, this is equivalent to saying that -1 is QR when $p \equiv 1 \pmod{4}$ and QNR when $p \equiv 3 \pmod{4}$.

3.5 When is 2 a QR?

From the end of the last section, we saw that -1 is a QR mod a prime p if and only if p is 1 mod 4. There is a similar rule for when 2 is a QR. In particular, suppose that $s = \frac{p-1}{2}$, and consider the following equations:

$$\begin{aligned} 1 &= (-1)(-1)^1 \\ 2 &= 2(-1)^2 \\ 3 &= (-3)(-1)^3 \\ &\vdots \\ s &= (\pm s)(-1)^s \end{aligned}$$

Now we will multiply these equations and reduce mod p . On the left, we obviously get $s!$. On the right hand side, notice that $-1 \equiv 2 \left(\frac{p-1}{2} \right) \equiv 2s \pmod{p}$. Similarly, $-3 \equiv 2(s-1) \pmod{p}$. Something similar occurs for odd numbers where they are congruent to 2 times s minus something. On the other hand, $2 \equiv 2 \cdot 1 \pmod{p}$, $4 \equiv 2 \cdot 2 \pmod{p}$, and so on. The numbers $-1, 2, -3, \dots, \pm s$ are actually congruent to $2 \cdot 1, 2 \cdot 2, \dots, 2 \cdot s$ in some order (all mod p). Overall, we have:

$$\begin{aligned} s! &\equiv (2 \cdot 1)(2 \cdot 2) \cdots (2 \cdot s)(-1)^{1+2+\cdots+s} \\ s! &\equiv 2^s s! (-1)^{\frac{s(s+1)}{2}} \\ 1 &\equiv 2^s (-1)^{\frac{s(s+1)}{2}} \end{aligned}$$

Recall that $s = \frac{p-1}{2}$, so $2^s \equiv \left(\frac{2}{p} \right)$ by Euler's criterion. Furthermore, $\frac{s(s+1)}{2} = \frac{(p-1)(p+1)}{8}$. After a bit of rearranging, we thus get the identity:

$$\left(\frac{2}{p} \right) = (-1)^{\frac{(p-1)(p+1)}{8}} = (-1)^{\frac{p^2-1}{8}}$$

We can notice that when p is 1 or 7 (mod 8), the whole expression $\frac{p^2-1}{8}$ is even, but when p is 3 or 5 (mod 8), that expression is not even. In other words, 2 is a QR only when p is 1 or 7 mod 8.

4 Quadratic Reciprocity Law

4.1 Proof

Now, we will finally prove the Law of Quadratic Reciprocity, which says the following:

Theorem 4.1. Given distinct odd primes p and q , then

$$\left(\frac{p}{q} \right) \left(\frac{q}{p} \right) = (-1)^{\frac{(p-1)(q-1)}{4}}$$

where $\left(\frac{p}{q} \right)$ and $\left(\frac{q}{p} \right)$ are Legendre symbols.

There are many different proofs of this theorem, but none of them are completely straightforward.

Proof. First, we will consider all the residues mod pq which are coprime to pq . There are exactly $\phi(pq) = (p-1)(q-1)$ of these. We can pair each residue with its negative. In particular, we can create $\frac{(p-1)(q-1)}{2}$ sets of the form $\{a, -a\}$ where a is a residue such that the union of these sets gives us all the residues mod pq coprime to pq .

We will now multiply one element from each set in 3 different ways. For each set, we have the choice of flipping a sign, so all 3 of these products will differ by at most a negative sign. Here are the three products:

- The first product will contain all coprime residues mod pq which are less than $\frac{p}{2}$ when reduced mod p . We can write this as:

$$A = \prod_{k=1}^{\frac{p-1}{2}} \left(\prod_{j=1}^{q-1} pj + k \right)$$

- The second product will contain all coprime residues mod pq which are less than $\frac{q}{2}$ when reduced mod q . We can write this as:

$$B = \prod_{k=1}^{\frac{q-1}{2}} \left(\prod_{j=1}^{p-1} qj + k \right)$$

- The third product will contain all coprime residues mod pq which are less than $\frac{pq}{2}$. We can write this as:

$$C = \prod_{\substack{a \bmod pq \\ \gcd(a, pq)=1 \\ a < \frac{pq}{2}}} a$$

We will calculate $A \bmod q$, $B \bmod p$, and $C \bmod$ both p and q , as well as the sign difference between A and $B \bmod pq$. Notice that in the innermost product for A , we have the product of $q-1$ nonzero residues which are separated by p each. Therefore, these numbers, in some order must just be equivalent to 1 to $q-1 \bmod q$, so the inner product is equivalent to $(q-1)! \bmod q$. Therefore, we can write:

$$\begin{aligned} A &\equiv \prod_{j=1}^{\frac{p-1}{2}} (q-1)! \\ &\equiv (q-1)!^{\frac{p-1}{2}} \pmod{q} \end{aligned}$$

Similarly, $B \equiv (p-1)!^{\frac{q-1}{2}} \pmod{p}$. Now, we can compare A and B . First, we will define a function $\lambda(r, s)$ where r is a residue mod p , s is a residue mod q , and $\lambda(r, s)$ gives us the residue mod pq which r and s determine (this is unique due to the Chinese Remainder Theorem). Then, A can be rewritten as:

$$\begin{aligned} A &= \prod_{r=1}^{\frac{p-1}{2}} \prod_{s=1}^{q-1} \lambda(r, s) \\ &= \left(\prod_{r=1}^{\frac{p-1}{2}} \prod_{s=1}^{\frac{q-1}{2}} \lambda(r, s) \right) \left(\prod_{r=1}^{\frac{p-1}{2}} \prod_{s=\frac{q+1}{2}}^{q-1} \lambda(r, s) \right) \end{aligned}$$

Similarly, we can write B as:

$$B = \left(\prod_{r=1}^{\frac{q-1}{2}} \prod_{s=1}^{\frac{p-1}{2}} \lambda(r, s) \right) \left(\prod_{r=1}^{\frac{q-1}{2}} \prod_{s=\frac{p+1}{2}}^{p-1} \lambda(r, s) \right)$$

Notice that the first factors for each of these expressions are equal; the order of the products is just flipped. For the second factor, we can use the fact that $\lambda(-r, -s) = -\lambda(r, s)$ and flip the indices, say for the second factor of A (let's call the second factor for A A_2 and the second factor B B_2). In A_2 , we are taking the product from $r = 1$ to $r = \frac{p-1}{2}$. If we instead take the product from $\frac{p+1}{2}$ to $p-1$, it has the effect of flipping the sign of $r \bmod p$, and we can do the same thing for s going from $\frac{q+1}{2}$ to $q-1$. In particular,

$$\begin{aligned} A_2 &\equiv \prod_{r=1}^{\frac{p-1}{2}} \prod_{s=\frac{q+1}{2}}^{q-1} \lambda(r, s) \\ &\equiv \prod_{r=\frac{p+1}{2}}^{p-1} \prod_{s=1}^{\frac{q-1}{2}} \lambda(-r, -s) \\ &\equiv \prod_{r=\frac{p+1}{2}}^{p-1} \prod_{s=1}^{\frac{q-1}{2}} -\lambda(r, s) \\ &\equiv (-1)^{\frac{(p-1)(q-1)}{4}} \cdot B_2 \end{aligned}$$

The exponent of -1 here comes from finding the total of number of factors multiplied out by the product. Finally, we have that $\frac{A}{B} = \frac{A_2}{B_2} = (-1)^{\frac{(p-1)(q-1)}{4}}$ or $A = (-1)^{\frac{(p-1)(q-1)}{4}} \cdot B$.

To calculate $C \bmod p$, notice that C is equivalent to the following:

$$C = \frac{(1 \cdot 2 \cdots (p-1)) \cdot ((p+1)(p+2) \cdots (2p-1)) \cdots \left((p(\frac{q-1}{2}) + 1)(p(\frac{q-1}{2}) + 2) \cdots (p(\frac{q-1}{2}) + \frac{p-1}{2}) \right)}{q \cdot 2q \cdot 3q \cdots \left(\frac{p-1}{2} \right) q}$$

In this fraction, the numerator contains all the residues mod pq less than $\frac{pq}{2}$ except for the multiples of p . However, we need to get rid of the multiples of q as well, and the denominator contains all the multiples of q in this range, canceling them out from the numerator. We can simplify this mod p , getting

$$\begin{aligned} C &\equiv \frac{(p-1)!^{\frac{q-1}{2}} \left(\frac{p-1}{2} \right)!}{q^{\frac{p-1}{2}} \left(\frac{p-1}{2} \right)!} \\ &\equiv \frac{(p-1)!^{\frac{q-1}{2}}}{q^{\frac{p-1}{2}}} \pmod{p} \end{aligned}$$

Similarly, we can reduce mod q to get:

$$C \equiv \frac{(q-1)!^{\frac{p-1}{2}}}{p^{\frac{q-1}{2}}} \pmod{q}$$

Now, comparing A and B with C , we get:

$$\begin{aligned}\frac{A}{C} &\equiv p^{\frac{q-1}{2}} \equiv \left(\frac{p}{q}\right) \pmod{q} \\ \frac{B}{C} &\equiv q^{\frac{p-1}{2}} \equiv \left(\frac{q}{p}\right) \pmod{p}\end{aligned}$$

Since A and C differ by at most a sign, $\frac{A}{C}$ is ± 1 when taken mod pq . Since $\left(\frac{p}{q}\right)$ is already ± 1 , this means that $\frac{A}{C} \equiv \left(\frac{p}{q}\right) \pmod{pq}$, and likewise, $\frac{B}{C} \equiv \left(\frac{q}{p}\right) \pmod{pq}$. Dividing these congruences, we get:

$$\frac{A}{B} \equiv \left(\frac{p}{q}\right) \left(\frac{q}{p}\right) \equiv (-1)^{\frac{(p-1)(q-1)}{4}} \pmod{pq}$$

Here, we used the fact that the Legendre symbols are ± 1 , so dividing them is the same as multiplying them. Now, we have Quadratic Reciprocity as a congruence mod pq , but we can turn it into an equation under real numbers because the congruence only involves powers of $-1 \pmod{pq}$, and powers of -1 behave exactly the same mod pq and in the real numbers, at least when multiplied. This proves the theorem. $\square$

4.2 Application

Quadratic Reciprocity can be written in a more useful way, which provides more insight on how we can use it to calculate Legendre symbols efficiently. In particular:

- If either p or q is congruent to 1 mod 4, then $\left(\frac{p}{q}\right) = \left(\frac{q}{p}\right)$. This is because this would imply that the power of -1 in Theorem 4.1 is even, and so $\left(\frac{p}{q}\right) \left(\frac{q}{p}\right)$, from which it is clear that the two Legendre symbols are equal.
- If both p and q are congruent to 3 mod 4, then $\left(\frac{p}{q}\right) = -\left(\frac{q}{p}\right)$. The proof of this case is basically the same as the last case, except the exponent of -1 ends up being odd.

With this in mind, we can calculate Legendre symbols with large numbers very fast. For example, consider $\left(\frac{1001}{99991}\right)$ (99991 is a prime). To use quadratic reciprocity, we need the numerator to be prime, so we can factor $1001 = 7 \cdot 11 \cdot 13$. Thus, we have:

$$\begin{aligned}\left(\frac{1001}{99991}\right) &= \left(\frac{7}{99991}\right) \left(\frac{11}{99991}\right) \left(\frac{13}{99991}\right) \\ &= \left(-\left(\frac{99991}{7}\right)\right) \left(-\left(\frac{99991}{11}\right)\right) \left(\left(\frac{99991}{13}\right)\right) \\ &= \left(\frac{3}{7}\right) \left(\frac{1}{11}\right) \left(\frac{8}{13}\right) \\ &= \left(-\left(\frac{7}{3}\right)\right) \left(\left(\frac{2}{13}\right)\right) \\ &= -\left(\frac{1}{3}\right) \left(\frac{2}{13}\right) \\ &= (-1) \cdot (-1) \\ &= 1\end{aligned}$$

This tells us that 1001 is in fact a QR mod 99991, which would be more tedious to calculate using Euler's criterion, at least by hand. All we do in each step of this process is for every Legendre symbol, we check if either the top or bottom is 1 mod 4, in which case we can flip it without any sign, and otherwise we need to add a negative sign. After that, we can reduce the top mod the bottom to get a smaller number. Of course, along the way, we also need to take care of factors of 2 and -1 , which aren't really applicable to Quadratic Reciprocity. We also need to make sure that the top of any Legendre symbol we use is a prime, so at the very least we need to factor the top of the Legendre symbol we are evaluating into primes.

We can also use Quadratic Reciprocity to obtain certain rules for when a fixed number (for now we will just consider a prime) is a QR mod an arbitrary prime, similar to how we obtained a rule for -1 and 2. For example, we might be interested in when 5 is a QR. Since $5 \equiv 1 \pmod{4}$, we know that $\left(\frac{5}{p}\right) = \left(\frac{p}{5}\right)$, so 5 is a QR mod p when p is a QR mod 5. In other words, p must be 1 or 4 mod 5.

As another example, we might want to know when 3 is a QR mod p . Because $3 \equiv 3 \pmod{4}$, we have two cases: when p is 1 mod 4 or 3 mod 4.

If $p \equiv 1 \pmod{4}$, we can flip $\left(\frac{3}{p}\right)$ to obtain $\left(\frac{p}{3}\right)$, which is 1 when p is 1 mod 3. Thus, if $p \equiv 1 \pmod{4}$ and $p \equiv 1 \pmod{3}$, which is equivalent to $p \equiv 1 \pmod{12}$, 3 is a QR mod p .

If $p \equiv 3 \pmod{4}$, $\left(\frac{3}{p}\right) = -\left(\frac{p}{3}\right)$, so we need $p \equiv 2 \pmod{3}$. Thus, if $p \equiv -1 \pmod{4}$ and $p \equiv -1 \pmod{3}$, which is equivalent to $p \equiv -1 \equiv 11 \pmod{12}$, we have that 3 is a QR mod p . Overall, 3 is a QR mod p if and only if p is 1 or 11 mod 12.

4.3 The Jacobi Symbol

To alleviate the problem of having to prime factorize the top of Legendre symbol before applying Quadratic Reciprocity, we can introduce the Jacobi Symbol.

Definition 4.1. For an integer a and a positive odd integer b , we define the Jacobi symbol of a over b , written $\left(\frac{a}{b}\right)$, where if $b = p_1 p_2 \cdots p_k$ where the p_i 's are primes (not necessarily distinct):

$$\left(\frac{a}{b}\right) = \left(\frac{a}{p_1 p_2 \cdots p_k}\right) = \left(\frac{a}{p_1}\right) \left(\frac{a}{p_2}\right) \cdots \left(\frac{a}{p_k}\right)$$

where the symbols on the right are Legendre symbols.

It is important to note that the Jacobi symbol does not tell us if a is a QR mod b . However, by definition, we know that if p is an odd prime, $\left(\frac{a}{p}\right)$ as a Jacobi symbol coincides in value with $\left(\frac{a}{p}\right)$ as a Legendre symbol. Therefore, $\left(\frac{a}{p}\right)$ as a Jacobi symbol is $+1$, then so is $\left(\frac{a}{p}\right)$ as a Legendre symbol, and this would tell us that a is a QR mod p . However, Jacobi symbols are easier to compute than Legendre symbols because we don't need to factorize anything, and Quadratic Reciprocity still holds.

Theorem 4.2. If a, b are coprime positive odd integers, then:

$$\left(\frac{a}{b}\right) \left(\frac{b}{a}\right) = (-1)^{\frac{(a-1)(b-1)}{4}}$$

Proof. Suppose $a = p_1 p_2 \cdots p_r$ for primes p_i (not necessarily distinct), and suppose $b = q_1 q_2 \cdots q_s$ for primes q_j (not necessarily distinct). Then, we have:

$$\begin{aligned}
\left(\frac{a}{b}\right) \left(\frac{b}{a}\right) &= \left(\frac{p_1 p_2 \cdots p_r}{q_1 q_2 \cdots q_s}\right) \left(\frac{q_1 q_2 \cdots q_s}{p_1 p_2 \cdots p_r}\right) \\
&= \prod_{i=1}^r \left(\frac{p_i}{q_1 q_2 \cdots q_s}\right) \left(\frac{q_1 q_2 \cdots q_s}{p_i}\right) \\
&= \prod_{i=1}^r \prod_{j=1}^s \left(\frac{p_i}{q_j}\right) \left(\frac{q_j}{p_i}\right) \\
&= \prod_{i=1}^r \prod_{j=1}^s (-1)^{\frac{(p_i-1)(q_j-1)}{4}} \\
&= (-1)^{\sum_{i=1}^r \sum_{j=1}^s \frac{(p_i-1)(q_j-1)}{4}}
\end{aligned}$$

Notice that

$$\begin{aligned}
\sum_{i=1}^r \sum_{j=1}^s \frac{(p_i-1)(q_j-1)}{4} &= \frac{1}{4} \left(\sum_{i=1}^r p_i - 1 \right) \left(\sum_{j=1}^s q_j - 1 \right) \\
&= \frac{1}{4} (p_1 + p_2 + \cdots + p_r - r)(q_1 + q_2 + \cdots + q_s - s)
\end{aligned}$$

We now claim the following:

$$\frac{p_1 + p_2 + \cdots + p_r - r}{2} \equiv \frac{a-1}{2} \pmod{2}$$

Since a is odd, $a \equiv 1, 3 \pmod{4}$. Suppose $a \equiv 1 \pmod{4} \implies \frac{a-1}{2} \equiv 0 \pmod{2}$. Furthermore, a must have an even number of prime factors which are 3 mod 4, so suppose that $2x$ of the p_i 's are 3 mod 4 while y of them are 1 mod 4. Then, we have

$$\begin{aligned}
p_1 + p_2 + \cdots + p_r - r &\equiv 3 \cdot 2x + 1 \cdot y - 2x - y \\
&\equiv 4x \\
&\equiv 0 \pmod{4}
\end{aligned}$$

This congruence implies that $\frac{p_1 + p_2 + \cdots + p_r - r}{2} \equiv 0 \equiv \frac{a-1}{2} \pmod{2}$.

The other case is when $a \equiv 3 \pmod{4} \implies \frac{a-1}{2} \equiv 1 \pmod{2}$. Furthermore, an odd number of the p_i 's must be 3 mod 4, say $2x+1$. Then, suppose there are y p_i 's which are 1 mod 4. Then, we have

$$\begin{aligned}
p_1 + p_2 + \cdots + p_r - r &\equiv 3(2x+1) + 1 \cdot y - 2x - 1 - y \\
&\equiv 4x + 2 \\
&\equiv 2 \pmod{4}
\end{aligned}$$

Dividing the congruence by 2 gives us $\frac{p_1 + p_2 + \cdots + p_r - r}{2} \equiv 1 \equiv \frac{a-1}{2} \pmod{2}$.

In a similar fashion, we have that $\frac{q_1+q_2+\dots+q_s}{2} \equiv \frac{b-1}{2} \pmod{2}$. Combining these two facts together, we get

$$\left(\frac{p_1+p_2+\dots+p_r-r}{2}\right)\left(\frac{q_1+q_2+\dots+q_s-s}{2}\right) \equiv \left(\frac{a-1}{2}\right)\left(\frac{b-1}{2}\right) \pmod{2}$$

Finally, this implies that

$$\begin{aligned} \left(\frac{a}{b}\right)\left(\frac{b}{a}\right) &= (-1)^{\sum_{i=1}^r \sum_{j=1}^s \frac{(p_i-1)(q_j-1)}{4}} \\ &= (-1)^{\left(\frac{p_1+p_2+\dots+p_r-r}{2}\right)\left(\frac{q_1+q_2+\dots+q_s-s}{2}\right)} \\ &= (-1)^{\left(\frac{a-1}{2}\right)\left(\frac{b-1}{2}\right)} \\ &= (-1)^{\frac{(a-1)(b-1)}{4}} \end{aligned}$$

This proves the identity. □

Now we can try evaluating that Legendre symbol from the last section without having to factorize anything. In particular, we have

$$\begin{aligned} \left(\frac{1001}{99991}\right) &= \left(\frac{99991}{1001}\right) = \left(\frac{892}{1001}\right) = \left(\frac{4}{1001}\right) \left(\frac{223}{1001}\right) = \left(\frac{1001}{223}\right) = \left(\frac{109}{223}\right) = \left(\frac{223}{109}\right) = \left(\frac{114}{109}\right) \\ &= \left(\frac{2}{109}\right) \left(\frac{57}{109}\right) = (-1) \left(\frac{109}{57}\right) = -\left(\frac{52}{57}\right) = -\left(\frac{13}{57}\right) = -\left(\frac{57}{13}\right) = -\left(\frac{5}{13}\right) = -\left(\frac{13}{5}\right) \\ &= -\left(\frac{3}{5}\right) = -\left(\frac{5}{3}\right) = -\left(\frac{2}{3}\right) = 1 \end{aligned}$$

In this case, we have more steps than when we just used the standard Legendre symbol method, but since we don't have to prime factor anything, this actually works better for larger numbers.

5 Sums of Squares

The main question we will be addressing in this section is when a number can be written as the sum of multiple of perfect squares, so naturally, we would first like to know which numbers can be written as the sum of exactly 2 perfect squares. To gain some intuition for this, it is useful to consider the Gaussian integers.

5.1 Gaussian Integers

The Gaussian Integers, denoted $\mathbb{Z}[i]$, are the set of complex numbers $a + bi$, where $a, b \in \mathbb{Z}$, which can also be written as:

$$\mathbb{Z}[i] = \{a + bi \mid a, b \in \mathbb{Z}\}$$

With Gaussian integers, we can interpret the sum of two squares easily. In particular, for integers x and y , we have that $x^2 + y^2 = (x + yi)(x - yi)$, which is the product of two complex conjugates. This leads to the following proposition:

Proposition 5.1. If a and b are positive integers which can both be written as the sum of 2 squares, then so can ab .

Proof. Suppose that

$$\begin{aligned}a &= x^2 + y^2 = (x + yi)(x - yi) \\ b &= z^2 + w^2 = (z + wi)(z - wi)\end{aligned}$$

Multiplying a and b gives us the following:

$$\begin{aligned}ab &= (x + yi)(z + wi)(x - yi)(z - wi) \\ &= (xz - yw + i(xw + yz))(xz - yw - i(xw + yz)) \\ &= (xz - yw)^2 + (xw + yz)^2\end{aligned}$$

which is a sum of two integer squares. □

Although this theorem can be stated purely algebraically without Gaussian integers, it provides some more insight into their structure. In particular, we used the fact that the product of the conjugates of two complex numbers is conjugate of their products, which is why $(x - yi)(z - wi)$ turns out to be the conjugate of $(x + yi)(z + wi)$.

5.2 Some Terminology

In many ways, $\mathbb{Z}[i]$ can be treated like the integers. In particular, it is known as a Euclidean Domain, which also implies that it is a Unique Factorization Domain. We won't worry about these terms too much, but we will discuss some of the properties they imply for $\mathbb{Z}[i]$, but we won't go deep into their proofs, because it is more algebraic than number theoretic.

First of all, the norm of a Gaussian integer x , denoted $\|x\|$, is the square of its absolute value, or $|x|^2$. Also, if a Gaussian integer has a norm of 1, we call it a unit. The only units in the Gaussian integers are ± 1 and $\pm i$.

Furthermore, while we won't prove this, the Gaussian integers have unique factorization. In particular, if $a = x_1 x_2 \dots x_n = y_1 y_2 \dots y_m$ where the x_i 's and y_i 's are Gaussian integers which are not units, then each $m = n$, and each x_i is equal to some y_j times some unit. In other words, factorization is unique up to multiplying by units. For example, $5 = (2 + i)(2 - i) = (i(2 + i))(-i(2 - i)) = (-1 + 2i)(-1 - 2i)$, but these are not different factorizations because each factor in the different factorizations are just off by units (i and $-i$).

5.3 Fermat's Two Square Theorem

Now we are ready to prove the famous Two Square Theorem, which is often attributed to Fermat. It is also often called Fermat's Christmas Theorem.

Theorem 5.1. If p is a prime which is congruent to 1 mod 4, then it can be written as the sum of two perfect squares.

Proof. Since $p \equiv 1 \pmod{4}$, $\left(\frac{-1}{p}\right) = 1$, so there exists an integer x such that $x^2 \equiv -1 \pmod{p} \implies x^2 + 1 = kp$ for some integer k . $x^2 + 1$ can be factored over the Gaussian integers into primes, say $q_1, q_1, \dots, q_r$, giving

$$q_1 q_2 \cdots q_r = kp$$

Remember that that the q_i 's are primes over $\mathbb{Z}[i]$, not over $\mathbb{Z}$. Furthermore, none of the q_i 's can be a pure real or imaginary number because this would scale the product of the q_i 's by some real number and make the imaginary part of their product greater than 1, which it isn't. Therefore, we can choose one of the q_i 's such that it divides p . Suppose that $q_i = z + wi$.

In other words, we have that $z + wi \mid p$, so $\|z + wi\| \mid \|p\| \implies z^2 + w^2 \mid p^2$. Since $z + wi$ is not a unit, we also have that $z^2 + w^2 > 1$. Furthermore, since $z + wi$ is a prime factor of p not equal to p , $z^2 + w^2 < p^2$. The only divisor of p between 1 and p^2 is p itself, so $z^2 + w^2 = p$. Thus, every prime which is 1 mod 4 can be written as the sum of 2 squares. □

We can also prove that every such prime can be written in exactly one way as the sum of two squares, up to signs and ordering.

Theorem 5.2. If p is a prime such that $p = a^2 + b^2 = c^2 + d^2$ such that and $0 < a \leq b$, $0 < c \leq d$, then $a = c$ and $b = d$.

Proof. p can be factored in $\mathbb{Z}[i]$ as $(a + bi)(a - bi)$. Since $\|a + bi\| = \|a - bi\| = p$, we must have that $a \pm bi$ are primes in $\mathbb{Z}[i]$ (since if it could factor further, one of the factors would have a norm of 1), so $(a + bi)(a - bi)$ is the prime factorization of p in $\mathbb{Z}[i]$. Likewise, $(c + di)(c - di)$ is also the prime factorization of p . Therefore, these prime factorizations must only differ by units, and because of the ordering we've imposed on a, b, c, d , we must have that $a + bi = c + di$ and $a - bi = c - di$. In other words, we don't have to worry about anything differing by units. Thus, we have that $a = c$ and $b = d$. □

5.4 Examples

Here are just a couple of examples of primes which are 1 mod 4 and their representations as a sum of two squares.

$$5 = 2^2 + 1^2$$

$$13 = 3^2 + 2^2$$

$$17 = 4^2 + 1^2$$

$$29 = 5^2 + 2^2$$

$$37 = 6^2 + 1^2$$

$$41 = 5^2 + 4^2$$

$$53 = 7^2 + 2^2$$

In the more general study of representing two numbers, it is useful to note that $2 = 1^2 + 1^2$ and also that no prime which is 3 mod 4 can be written as the sum of two squares. This is simply because any square is either congruent to 0 or 1 mod 4, so adding 2 squares will give you something 0, 1, or 2 mod 4, but never 3.

5.5 The General Case

We will now classify all positive integers which can be written as the sum of two squares.

Theorem 5.3. A positive integer n can be written as a sum of two squares if and only if all primes $3 \bmod 4$ in its factorization appear with an even exponent.

Proof. Suppose $n = 2^k \cdot p_1^{a_1} \cdots p_r^{a_r} \cdot q_1^{b_1} \cdots q_s^{b_s}$, where the p_i 's are distinct primes $1 \bmod 4$, and the q_j 's are distinct primes $3 \bmod 4$.

To prove the backward direction, suppose that all the b_j 's are even. All the primes $2, p_1, \dots, p_r$ can be written as the sum of 2 squares, so by Proposition 5.1, $2^k p_1^{a_1} \cdots p_r^{a_r}$ can as well. Let's say it is equal to $X^2 + Y^2$. Furthermore, $q_1^{b_1} \cdots q_s^{b_s}$ is a perfect square, say Z^2 . Thus, $(XZ)^2 + (YZ)^2 = n$.

To prove the forward direction, suppose that one of the b_j 's are odd. Without loss of generality, we can take b_1 to be odd. Suppose that $x^2 + y^2 = n$. Reducing mod q_i , we get

$$\begin{aligned} x^2 + y^2 &\equiv 0 \pmod{q_i} \\ x^2 &\equiv -y^2 \pmod{q_i} \end{aligned}$$

If it is not the case that $x \equiv y \equiv 0 \pmod{q_i}$, then we have that $-y^2$ is a QR mod q_i , so $\left(\frac{-y^2}{p}\right) = \left(\frac{-1}{p}\right) = 1$, but this is not true because $p \equiv 3 \pmod{4}$. Thus, $x = x_1 q$ and $y = y_1 q$ for integers x_1, y_1 . This gives that $x_1^2 + y_1^2 = \frac{n}{q_1^2}$. $\frac{n}{q_1^2}$ still has an odd exponent of q_1 , so we can continue this process until we reach a number n' which has an exponent of 1 on q_1 , at which point we will have integers x' and y' such that $x'^2 + y'^2 = n'$. If we try this process again, we will see that $q_1 \mid x'$ and $q_1 \mid y'$, so $q_1^2 \mid x'^2 + y'^2 = n'$, but this is not true because n' only has an exponent of 1 on q_1 . Thus, we have contradicted the existence of a representation of n as the sum of two squares, proving the theorem. □

5.6 Some Related Results

Here are some results related to the Two Square Theorem, but which we will not prove here. First of all, we might want to know what integers can be written as the sum of 3 or 4 squares. In fact, we have the following:

Theorem 5.4. An integer n can be written as the sum of 3 perfect squares if and only if it is not of the form $4^m(8k+7)$ for integers m and k .

Theorem 5.5. All positive integers can be written as the sum of 4 perfect squares.

A similar result to Theorem 5.6 exists for triangular numbers, where the n th triangular number is $\frac{n(n+1)}{2}$.

Theorem 5.6. All positive integers can be written as the sum of 3 perfect squares.

Catalan Numbers

Ivy Zhu

1 Introduction

The Catalan numbers are a sequence of natural numbers that appear frequently in counting problems. The first n Catalan numbers for $n = 0, 1, 2, 3, \dots$ are $1, 1, 2, 5, 14, 42, 132, 429, \dots$. Upon first glance, there doesn't seem to be a pattern relating consecutive numbers, but they can be expressed recursively and with an explicit formula! We will start by talking about the problems these numbers appear in, how they were solved, and a solution for the general formula to derive the n^{th} Catalan number.

2 Applications in Problems

The n^{th} Catalan number is often denoted with C_n which is what we will be using in this article. These problems all ultimately satisfy Segner's Recurrence Formula:

$$E_n = E_2E_{n-1} + E_3E_{n-2} + \dots + E_{n-1}E_2$$

In terms of the Catalan numbers, this formula is often rewritten as

$$C_{n-1} = C_0C_n + C_1 + C_{n-1} + \dots + C_nC_0 = \sum_{k=0}^n C_kC_{n-k}, C_0 = 1$$

2.1 Catalan's Problem

How many different ways can a product of n different factors be calculated by pairs if their order is prescribed? Let's understand this problem through an example where we have 4 factors x_1, x_2, x_3, x_4 . There are 5 different ordered multiplications, namely $((x_1 \cdot x_2) \cdot x_3) \cdot x_4, (x_1 \cdot (x_2 \cdot x_3)) \cdot x_4, (x_1 \cdot x_2) \cdot (x_3 \cdot x_4), x_1 \cdot ((x_2 \cdot x_3) \cdot x_4), x_1 \cdot (x_2 \cdot (x_3 \cdot x_4))$

Catalan came up with a solution and named it after himself as mathematicians liked to do.

$$C_{n-1} = \frac{(4n-6)!!!!}{n!}$$

The four exclamation marks mean that $n!!!! = n(n-4)(n-8) \dots$. This idea is equivalent to the parenthesis problem that we will go over later. To connect this to Segner's Recurrence Formula, we can think about having $n+1$ total numbers and n total operations. Let M_n be the number of multiplications of n numbers. The first i pairs can be grouped in M_i ways, and the remaining $n-i-1$ pairs in M_{n-i-1} so for each i , there are M_iM_{n-i-1} ways for the situation to happen. Thus, the total would be

$$M_n = M_1M_{n-1} + M_2M_{n-2} + \dots + M_{n-1}M_1$$

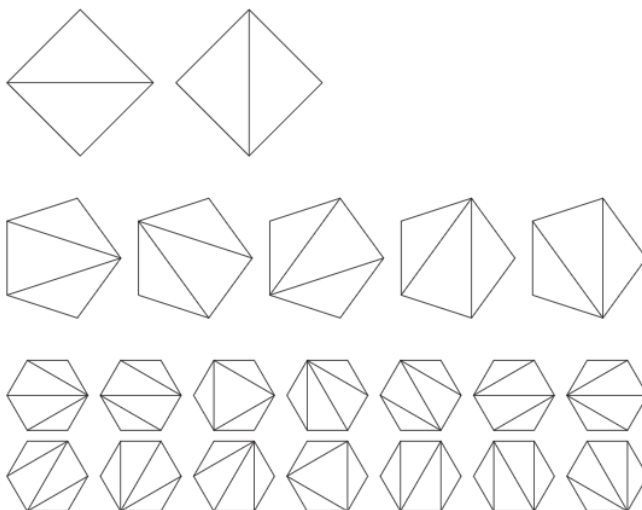
which is the same as the formula for C_{n-1}

2.2 Euler's Polygon Division Problem

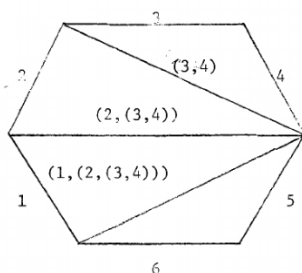
Segner's Recurrence Formula was used to solve this problem! In 1751, Euler asked how many ways a convex polygon with n sides can be divided into $n - 2$ triangles using their diagonals. Symmetry, rotations, and whether the triangles are congruent or not don't matter. The formula used was

$$E_n = \frac{2(n-2)!}{(n-1)!(n-2)!}$$

Here are the solutions to the first few polygons.



Notice that at least two of the triangles have a diagonal and two sides of the polygon as its sides. We can choose one of these triangles and remove the two sides of the polygon to break the case down into the same problem but with $n - 1$ sides. In the hexagon below, we first remove sides 3 and 4. Instead of thinking about the hexagon, we have reduced to thinking about a pentagon with the diagonal, denoted with $(3, 4)$ becoming a side. Then we remove side 2 and side $(3, 4)$. The final division is $((1, (2, (3, 4))), 5), 6$. This shares some similarities with Catalan's problem!



Section 2.6 will explain another way to think about this problem using binary trees.

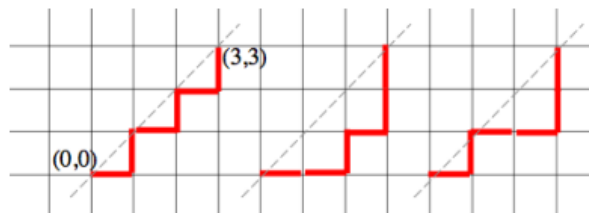
2.3 Dyck Paths

A Dyck Path is a series of up and down steps. The path goes from left to right. It begins and ends at the same level, and does not go below the starting point at any point. What is the total number of Dyck Paths of length $2n$? Once again, Catalan numbers begin to show up

$n = 0$:	*	1 way
$n = 1$:	$\wedge \backslash$	1 way
$n = 2$:	$\wedge \backslash \wedge \backslash, \wedge \backslash \wedge \backslash$	2 ways
$n = 3$:	$\wedge \backslash \wedge \backslash \wedge \backslash, \wedge \backslash \wedge \backslash \wedge \backslash, \wedge \backslash \wedge \backslash \wedge \backslash, \wedge \backslash \wedge \backslash \wedge \backslash, \wedge \backslash \wedge \backslash \wedge \backslash$	5 ways

2.3.1 Into A Formula

Let's think of Dyck Paths with the line $y = x$ as the level they start on. Now, they are a series of steps from $(0,0)$ to (n,n) that lie below the line.



We can approach this problem by thinking of every up step in the path as a 1 in the sequence, and every down step with -1 . There can never be more -1 s than 1 s since we can't pass the line $y = x$. Thus the number of unique Dyck Paths becomes the number of sequences with non-negative partial sums that can be formed from n copies of 1 s and n copies of -1 s. Look to section 2.4.1 for a proof to get a closed form for C_n

2.4 Parenthesis Problem

How many ways are there to count permutations of n pairs of parentheses such that each open parenthesis is followed by a closed one?

There is a recursive relationship here where we could add $()$ to the left, right, or around the whole thing. Going from $()$ to $()()$ could be seen as adding $()$ to the left or the right. Going from $()$ to $((()))$ would be adding parentheses around. Let's try small cases and list things out.

$n = 1$

- $()$

$n = 2$

- $()()$
- $((()))$

$n = 3$

- $()()()$
- $((()))()$
- $()((()))$
- $((())())$
- $((()))()$

With this, we can see that $C_1 = 1, C_2 = 2$, and $C_3 = 5$! This can be solved with coding! Here is a solution written in java

```

public List<String> generateParenthesis(int n) {
    List<String> list = new ArrayList<String>();
    backtrack(list, "", 0, 0, n);
    return list;
}

public void backtrack(List<String> list, String str, int open, int close, int max){

    if(str.length() == max*2){
        list.add(str);
        return;
    }

    if(open < max)
        backtrack(list, str+"(", open+1, close, max);
    if(close < open)
        backtrack(list, str+")", open, close+1, max);
}

```

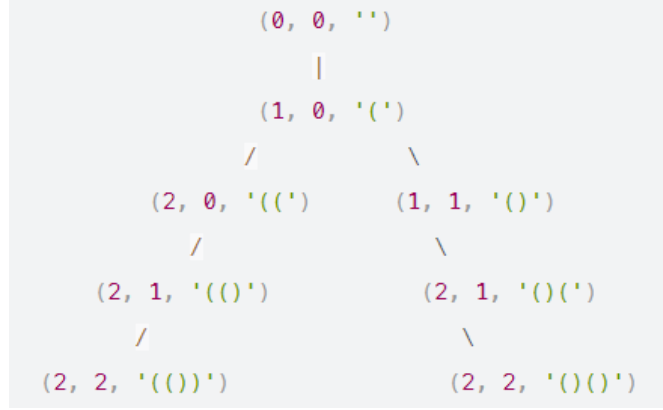
Let's analyze the lines under the line starting with **public void backtrack** in basic terms.

Let's look at **if (str.length() == max * 2)**. The word **str.length()** is actually a number, counting the amount of parenthesis we have added so far. **max** is a variable with the value of n where n is still the pairs of parentheses we want to find the permutations of. As we go through the rest of the code, keep in mind that $\text{max} = n$. With the **==**, we are checking if the number of total parentheses is equal to n . Since there are 2 parenthesis in each pair, we check if the amount of parentheses we have currently is equal to $n \cdot 2$, the final number of parentheses we want. If that is true, we stop adding more parenthesis.

Now let's observe **if (open < max)**. The word **open** is a variable counting the number of open parentheses we have, and we want n of them in total (n accounts for taking one from each of the n pairs we have). If we don't have enough of (, this means **open** is less than n , we want to proceed by adding another (, increasing the number of open parentheses we have, and branching off by going back to the first line under **public void backtrack**.

Taking a look at **if (close < open)**, the word **close** is a variable that counts the number of closed parentheses. Since we don't want to end up with situations like **)**(or **(())**(, the number of closed parentheses at any given moment should be less than or equal to the number of open ones. If there are less closed than open, we can add another closed parenthesis.

In the end, we will end up accounting for all the permutations, since at every point, we choose to add (or) until we reach our goal. If we take a look at the picture below, each line in the form (open, closed, parenthesis added so far), we can see how the branching off doesn't affect each result. Finally, in the last line, we end up with the permutations we got above.



2.4.1 Into a Formula

It's nice that a computer can run it, but what if we wanted to compute by hand? Let's approach finding a formula by looking at the parenthesis problem.

The total number of sequences would be $\binom{2n}{n} = \frac{2n!}{n!n!}$. There are $2n$ total parenthesis and we divide by $n!$ twice to account for the parenthesis being indistinguishable from one another. Let's represent the open parenthesis $($ with 1 and closed parenthesis $)$ with -1 and each number in the sequence as a_i with the entire sequence being $a_1, a_2, a_3, \dots, a_{2n}$. Every partial sum must be non-negative, that is

$$a_1 + a_2 + a_3 + \dots + a_k \geq 0$$

and $k = 1, 2, 3, \dots, 2n$. This means at any point in the sequence when looking at the numbers before, going from left to right, the number of 1s is greater than or equal to the number of -1 s. This makes sense when we think about the original context of the parenthesis problem: there are never more closing parenthesis than open ones.

Let A_n be the number of acceptable sequences and U_n be unacceptable sequences.

$$A_n + U_n = \binom{2n}{n} = \frac{2n!}{n!n!}$$

Let's try to count U_n by finding the first part the sequence breaks, that is to find the smallest k such that $a_1 + a_2 + a_3 + \dots + a_k < 0$. Since we are looking for the smallest k , there is an equal amount of 1s and -1 s before a_k . If that wasn't true, there would be a number before k where the sequence breaks, and k wouldn't be minimal. This means that k must be an odd number and $a_1 + a_2 + a_3 + \dots + a_{k-1} = 0$ and $a_k = -1$.

In the sequence up to a_k , which is $a_1, a_2, a_3, \dots, a_k$, there are $\frac{k-1}{2}$ copies of $+1$ s and $\frac{k+1}{2}$ copies of -1 s. In the remaining numbers, $a_{k+1}, a_{k+2}, \dots, a_{2n}$, there are $(n - \frac{k-1}{2})$ copies of $+1$ s and $(n - \frac{k+1}{2})$ copies of -1 s.

Let's consider replacing a_i where $i = 1, 2, 3, \dots, k$ with $-a_i$. This will make the sum of the entire sequence equal to 2: since the partial sum up to a_k was -1 and the rest of the sequence contains an extra copy of 1. Through this process, we made the sum up to a_k equal to 1. Now, in the whole sequence, there are $n+1$ copies of 1s and $n-1$ copies of -1 s. The number of possible sequences made by rearranging these 1s and -1 s would be $\frac{2n!}{(n+1)!(n-1)!}$. This number is equal to U_n since we can transform any sequence of this form into an unacceptable sequence by finding the point where there are more 1s than -1 s and negating all the numbers up to and including that point.

Now that we have $U_n = \frac{2n!}{(n+1)!(n-1)!}$, we can find A_n by substitution.

$$\begin{aligned}
 A_n &= \frac{2n!}{n!n!} - U_n &&= \frac{2n!}{n!n!} - \frac{2n!}{(n+1)!(n-1)!} \\
 &= \frac{2n!}{n!(n-1)!} \left(\frac{1}{n} - \frac{1}{n+1} \right) \\
 &= \frac{2n!}{n!(n-1)!} \left(\frac{1}{n(n+1)} \right) \\
 &= \frac{2n!}{n!n!} \frac{1}{n+1} \\
 &= \frac{1}{n+1} \binom{2n}{n}
 \end{aligned}$$

Thus, we get a closed form for C_n !

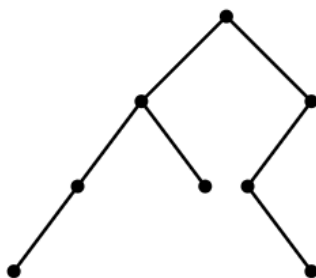
2.5 Ballot Problem

Suppose in an election, there are candidates A and B who each receive n votes. What is the number of sequences that A will never trail B when counting votes? For example, AAABBB does work, but AABBBB doesn't work because after counting 5 votes, A has 2 votes which is less than the 3 votes B has.

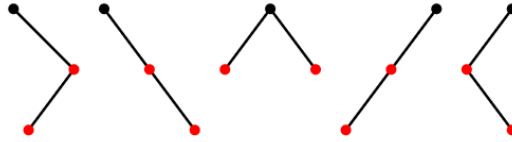
We can see a similarity between this and the parenthesis problem, especially when represented with 1s and -1 s. Since you can't have more closing parenthesis than open ones at any point in the sequence, the same idea persists in there never being more Bs than As. We can also see a parallel with the Dyck paths problem! Each 1 represents a step up and -1 represents a step down. Recall that we can't go lower than the starting height, so there can't be more steps down than steps up. Thus, the total number of sequences of all these situations would be C_n .

2.6 Rooted Binary Trees

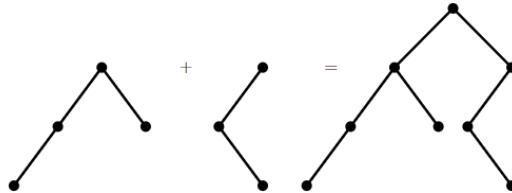
Rooted binary trees are a type of graph that appears in computer science. The tree is made up of vertices and edges where the vertices below a vertex connected by an edge are the children of that vertex. The root is the topmost vertex and a tree is binary when all vertices have either 0, 1, or 2 children. Below is an example of a rooted binary tree.



How many different rooted binary trees are there with n vertices (where the edges going in different directions count as different trees)? Try drawing out some small cases! There's 1 way for the tree to be empty, 1 way with one vertex, 2 ways with two, and 5 ways with three (the trees for 3 vertices are shown below).



These numbers look oddly familiar; they are Catalan numbers! In fact, we can think of making these trees by combining smaller ones.



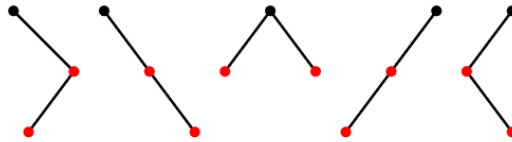
We start with n vertices, use 1 for a root vertex and its two children, and insert rooted binary trees on k and l vertices, with $k + l = n - 1$. We can write this as

$$C_n = \sum_{i=0}^{n-1} C_i C_{n-i-1}$$

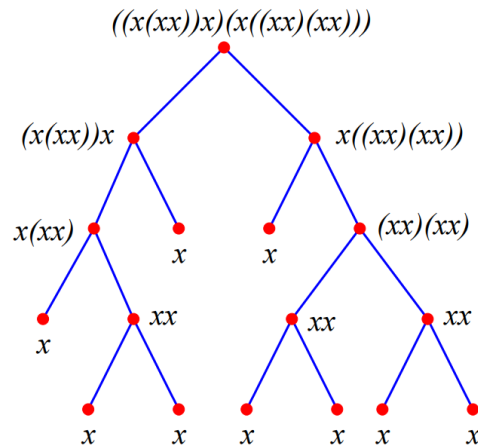
When we try to find C_3 with this formula, we get

$$C_3 = C_0 C_2 + C_1 C_1 + C_2 C_0 = 1 \cdot 2 + 1 \cdot 1 + 2 \cdot 1 = 5$$

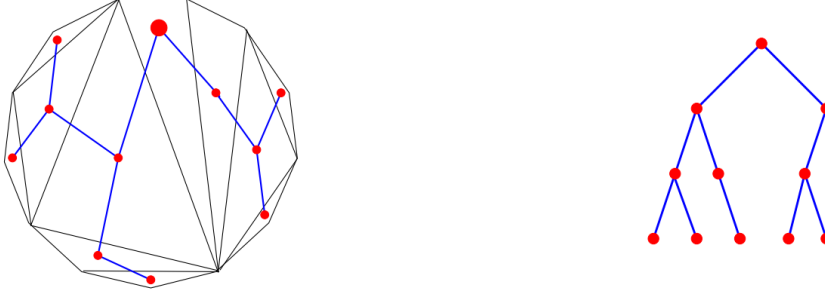
Now let's look at the trees with three vertices with this idea of combining trees. The two left trees have an empty left child from C_0 and a similar idea for the two right trees.



The parentheses problem can be mapped with a binary tree! Here's one that shows n pairs of parentheses.



Euler's polygon problem can similarly be mapped! Let each triangle represent a node and connect nodes based on what triangles share sides.



3 The n^{th} Catalan Number

As we have mentioned before, this is the explicit formula for C_n

$$C_n = \frac{1}{n+1} \binom{2n}{n} \quad (1)$$

Generating functions can be used to prove the explicit formula as well! Each term would be of the form $C_n x^n$ and let's define the function $C(x) = \sum_{n=0}^{\infty} C_n x^n$. We want the function to look like $1 + x + 2x + 5x^3 + \dots$. After squaring both sides, we get

$$(C(x))^2 = \sum_{n=0}^{\infty} C_n x^n \cdot \sum_{n=0}^{\infty} C_n x^n$$

Since these are two infinite series, we can use properties of the Cauchy product

$$\left(\sum_{i=0}^{\infty} a_i \right) \cdot \left(\sum_{j=0}^{\infty} b_j \right) = \sum_{k=0}^{\infty} c_k \text{ where } c_k = \sum_{l=0}^k a_l b_{k-l}$$

Then the expression is rewritten as

$$\begin{aligned} (C(x))^2 &= \sum_{n=0}^{\infty} \left(\sum_{k=0}^n C_k C_{n-k} \right) x^n \\ &= \sum_{n=0}^{\infty} C_{n+1} x^n = \frac{C(x) - 1}{x} \end{aligned}$$

Let $y = C(x)$, and we can rewrite the equation as

$$\begin{aligned} y^2 &= \frac{y-1}{x} \\ xy^2 - y + 1 &= 0 \end{aligned}$$

We end up having a quadratic in y , so let's apply the quadratic formula and see what we get!

$$y = C(x) = \frac{1 \pm \sqrt{1 - 4x}}{2x}$$

Is the plus or minus sign the right one in this situation? Let's try expanding using Taylor Series.

$$\begin{aligned} \frac{1 + (1 - 2x - 2x^2 + \dots)}{2x} &= \frac{1}{x} - 1 - x + \dots \\ \frac{1 - (1 - 2x - 2x^2 + \dots)}{2x} &= 1 + x + \dots \end{aligned}$$

Clearly, we want the minus sign here, but we can also look at it with another method. We know that $C_0 = 1$, but $C(0)$ gives us an indeterminate form. However, we can take the limit as x approaches 0 from the positive side and see that the limit approaches 1.

$$\begin{aligned} \lim_{x \rightarrow 0^+} C(x) &= \lim_{x \rightarrow 0^+} \frac{1 - \sqrt{1 - 4x}}{2x} \\ &= \lim_{x \rightarrow 0^+} \frac{2(1 - 4x)^{-1/2}}{2} \text{ by l'Hôpital's rule} \\ &= 1 \end{aligned}$$

We can use Newton's Binomial Theorem which says for any real number r that is not a non-negative integer when $-1 < x < 1$

$$(x + 1)^r = \sum_{i=0}^{\infty} \binom{r}{i} x^i$$

Now we can rewrite $\sqrt{1 - 4x}$ since $-1 < -4x < 1$ in order for the square root to be a real number.

$$\begin{aligned} \sqrt{1 - 4x} &= (1 + (-4x))^{1/2} = \sum_{n=0}^{\infty} \binom{1/2}{n} (-4x)^n \\ \frac{1 - \sqrt{1 - 4x}}{2x} &= \frac{1}{2x} \left(1 - \sum_{n=0}^{\infty} \binom{1/2}{n} (-4x)^n \right) \end{aligned}$$

Let's take a look at the coefficients of x^n in the sum

$$\begin{aligned} &\frac{\frac{1}{2}(\frac{1}{2} - 1) \cdots (\frac{1}{2} - n + 1)}{n!} (-4)^n \\ &= \frac{1(1 - 2) \cdots (1 - 2n + 2)}{n!} (-1)^n 2^n \\ &= \frac{(-1)(-3) \cdots (-(2n - 3))}{(n!)^2} (-1)^n 2^n (n)(n - 1) \cdots (1) \\ &= \frac{(-1)(-3) \cdots (-(2n - 3))}{(n!)^2} (-1)^n (2n)(2n - 2) \cdots (2) \\ &= -\frac{(2n)!}{(n!)^2 (2n - 1)} \\ &= -\binom{2n}{n} \frac{1}{2n - 1} \end{aligned}$$

Plugging this back into $C(x)$, we get

$$\begin{aligned}
C(x) &= \frac{1}{2x} \left(1 + \sum_{n=0}^{\infty} \binom{2n}{n} \frac{1}{2n-1} x^n \right) \\
&= \frac{1}{2x} \left(1 + (-1) + \sum_{n=1}^{\infty} \binom{2n}{n} \frac{1}{2n-1} x^n \right) \\
&= \frac{1}{2} \sum_{n=1}^{\infty} \binom{2n}{n} \frac{1}{2n-1} x^{n-1} \\
&= \frac{1}{2} \sum_{n=0}^{\infty} \binom{2(n+1)}{n+1} \frac{1}{2n-1} x^n \\
&= \sum_{n=0}^{\infty} \frac{2(n+1)(2n+1)}{2(n+1)^2} \binom{2n}{n} \frac{1}{2n+1} x^n \\
&= \sum_{n=0}^{\infty} \frac{1}{n+1} \binom{2n}{n} x^n
\end{aligned}$$

Finally, from this form, we can see that the coefficients of x^n , which are also C_n are

$$C_n = \frac{1}{n+1} \binom{2n}{n}$$

Alternatively, we can use the Generalized Binomial Theorem to see

$$\begin{aligned}
\sqrt{1-4x} &= \binom{1/2}{0} + \binom{1/2}{1}(-4x) + \binom{1/2}{2}(-4x)^2 + \dots + \binom{1/2}{k}(-4x)^k + \dots \\
\binom{1/2}{k} &= \frac{(1/2)(-1/2)(-3/2)\dots(1/2-k+1)}{k!} = \frac{(-1)^{k-1}1 \cdot 3 \cdot 5 \cdot (2k-3)}{2^k k!}
\end{aligned}$$

This means that the coefficient of x^k in $\sqrt{1-4x}$ is in the form

$$\frac{(-1)^{k-1}1 \cdot 3 \cdot 5 \cdot (2k-3)}{2^k k!} (-4)^k = \frac{(-1)1 \cdot 3 \cdot 5 \cdot (2k-3)2^k}{k!}$$

Let's use the substitution, $2^k k! = 2^k (1 \cdot 2 \cdot 3 \cdot \dots \cdot k) = 2 \cdot 4 \cdot 6 \cdot \dots \cdot k$, and multiply the number and denominator of the fraction on the left by $k!$

$$\begin{aligned}
\frac{(-1)1 \cdot 3 \cdot 5 \cdot (2k-3)2^k}{k!} &= \frac{(-1)1 \cdot 3 \cdot 5 \cdot (2k-3)2 \cdot 4 \cdot 6 \cdot \dots \cdot k}{k! k!} \\
&= \frac{(-1)(2k-2)!2k}{k! k!} \\
&= \frac{(-1)(2k)!}{(2k-1)k! k!} \\
&= \frac{-1}{2k-1} \binom{2k}{k}
\end{aligned}$$

Finally, we get that

$$\sqrt{1-4x} = -\sum_{k=0}^{\infty} \frac{1}{2k-1} \binom{2k}{k} x^k$$

$$C(x) = \frac{1 - \sqrt{1-4x}}{2x} = \frac{1 - \sum_{k=0}^{\infty} \frac{1}{2k-1} \binom{2k}{k} x^k}{2x}$$

From this form of $C(x)$, we can see that the coefficient of x^n in $C(x)$ is half of the coefficient of x^{n+1} in $\sqrt{1-4x}$ because of the $2x$ in the denominator.

$$\begin{aligned} \frac{1}{2} \binom{2(n+1)}{n+1} \frac{1}{2n+1} &= \frac{(2n+2)!}{2(n+1)!(n+1)!(2n+1)} \\ &= \frac{1}{2} \cdot \frac{2n+2}{n+1} \cdot \frac{(2n)!}{(n+1)!n!} \cdot \frac{2n+1}{2n+1} \\ &= \frac{1}{n+1} \frac{2n!}{n!n!} \end{aligned}$$

Now, we can use these coefficients to get the closed form!

$$C_n = \frac{1}{n+1} \frac{2n!}{n!n!} = \frac{1}{n+1} \binom{2n}{n}$$

4 Applications in the Real World

In addition to all the problems mentioned, Catalan numbers are seen in the fields of medicine, cryptography, geometric modeling, and more. Catalan numbers are used frequently to generate pseudo-random numbers. In cryptography, Catalan numbers are used to generate binary sequences which in turn produce keys. In steganography, concealing information, requires a set of constants which are generated with Catalan numbers. There are also applications in biometric identification, through triangulation and face recognition. There are still many problems that weren't mentioned, so feel free to explore more about this topic and do a deeper dive into its applications!

References

- Baracchini, Gabriella. "Dyck Paths and Up-down Walks," May 12, 2016.
https://math.mit.edu/~apost/courses/18.204-2016/18.204_Gabriella_Baracchini_final_paper.pdf.
- Corn, Patrick, and Alex Li. "Catalan Numbers — Brilliant Math & Science Wiki." brilliant.org, n.d. <https://brilliant.org/wiki/catalan-numbers/>.
- Guichard, David. "3.5 Catalan Numbers." www.whitman.edu, March 4, 2023.
https://www.whitman.edu/mathematics/cgt_online/book/section03.05.html.
- Morris, Joy. "Catalan Numbers." www.cs.uleth.ca, March 2023.
https://www.cs.uleth.ca/~morris/Combinatorics/html/sect_gen-fns-apps-Catalan.html.
- Selim, Aybeyan, and Muzafer Saračević. "Catalan Numbers and Applications," 2019.
https://www.researchgate.net/publication/335222209_Catalan_Numbers_and_Applications.

Spivey, Michael Z. “The Catalan Numbers from Their Generating Function.” A Narrow Margin, March 19, 2013. <https://mikespivey.wordpress.com/2013/03/19/the-catalan-numbers-from-their-generating-function/>.

Stanley, Richard. “Catalan Numbers,” 2015. <https://math.mit.edu/~rstan/transparencies/china.pdf>.

Trietsch, Dan. “Euler’s Problem of Polygon Division and Full Steiner Topologies—A Duality,” 1984. <https://www.kellogg.northwestern.edu/research/math/papers/625.pdf>.

Primitive Roots

Patrick Was

1 Primitive Roots

1.1 Definition

Definition 1.1.1 — An integer a is said to be a *primitive root* mod n if for every number coprime to n , there is an integer b such that $a^b \equiv k \pmod{n}$

Note: By standard of mod, $k < n$.

However, when talking about primitive roots, it is more useful in general to consider a group theory-related definition. This is helpful when proving certain theorems as well as helping recognize patterns within primitive roots.

Definition 1.1.2 — Define

$$\mathbb{Z}_n^\times = \{a \in \mathbb{N} \mid 1 \leq a < n, \gcd(a, n) = 1\}$$

Then, a *primitive root* mod n is an element $g \in \mathbb{Z}_n^\times$ such that every element $b \in \mathbb{Z}_n^\times$ can be written as $g^k \pmod{n}$, for some $k \in \mathbb{Z}$

With this definition, we could say that a primitive root is a group generator of $\mathbb{Z}_n^\times$.

1.2 Examples

On the left side is an example of a primitive root mod 5, and on the right is an example of an integer that is not a primitive root mod 5.

$$3^1 \equiv 3 \pmod{5}$$

$$3^2 \equiv 4 \pmod{5}$$

$$3^3 \equiv 2 \pmod{5}$$

$$3^4 \equiv 1 \pmod{5}$$

$$4^1 \equiv 4 \pmod{5}$$

$$4^2 \equiv 1 \pmod{5}$$

$$4^3 \equiv 4 \pmod{5}$$

$$4^4 \equiv 1 \pmod{5}$$

Because the set of integers less than 5 relatively prime to it is $\{1, 2, 3, 4\}$, each of these numbers must appear as a k in the definition. This is true for 3:

$$3^4 \equiv 1 \pmod{5}$$

$$3^3 \equiv 2 \pmod{5}$$

$$3^1 \equiv 3 \pmod{5}$$

$$3^2 \equiv 4 \pmod{5}$$

But we see that for 4, we only achieve 1 and 4. This can be explained using fundamental number theory: knowing that $4 \equiv -1 \pmod{5}$, we know that $4^k \equiv (-1)^k \pmod{5}$, so the value will be alternating between $1 \pmod{5}$ and $-1 \pmod{5}$, or $4 \pmod{5}$.

With this logic, we know that $n - 1$ will never be a primitive root mod n .

2 Chapter ϕ ve

2.1 Computing Euler's Totient Function

Definition 2.1.1 — Let *Euler's totient function* $\phi : \mathbb{N} \rightarrow \mathbb{N}$ (sometimes written as φ) be the number of positive integers less than or equal to n that are relatively prime to n .

The function is clearly defined for prime inputs: if p is a prime,

$$\phi(p) = p - 1$$

because for all $1 \leq q < p$, $\gcd(p, q) = 1$.

A general formula for $\phi(p^k)$ with prime p is also very approachable. It is significantly easier to do complementary counting here; the only $1 \leq q < p$ for which $\gcd(p, q) \neq 1$ are multiples of p . There are $\frac{p^k}{p} = p^{k-1}$ of these multiples, so

$$\phi(p^k) = p^k - p^{k-1}$$

Though it's nice to have formulas for prime numbers, the function is defined for all natural numbers, so it would be nice to have a general formula for any natural n .

A relatively simple observation to make occurs when we take the product of two primes. If we let this number be ab , with a and b being the two primes multiplied together, we can develop a pattern using complementary counting:

$$\begin{aligned}\phi(ab) &= ab - a - b + 1 \\ &= (a - 1)(b - 1) \\ &= \phi(a)\phi(b)\end{aligned}$$

We simply subtract the number of multiples of b (which is a), the number of multiples of a (which is b), and since we overcounted by 1 (we would have counted ab twice), we add 1. This factors to $(a - 1)(b - 1)$, which is equivalent to $\phi(a)\phi(b)$.

However, we can generalize this to any coprime a and b . We can generalize our previous result because the same logic will apply; we exclude the multiples of a , multiples of b , and fix any overlap (of which there will only be 1, since they are coprime), which just gives us $\phi(a)\phi(b)$.

Definition 2.1.2 — Let f be a function such that $f(ab) = f(a)f(b)$ for positive a, b , where $\gcd(a, b) = 1$. Then, f is called a *multiplicative function*.

Note: This is the standard definition in the branch of number theory. Generally, multiplicative functions do not have the restriction of two coprime factors, but here we will call multiplicative functions without this restriction *completely multiplicative functions*.

It's pretty clear that with this generalization, we can compute $\phi(n)$ for any $n \in \mathbb{N}$ —though the process might be daunting for some values, as we will get to later. However, we can further generalize our results from the previous conclusion to compute $\phi(n)$ without even using ϕ .

If we notice that we can write $\phi(n)$ as $\phi(p_1^{a_1} p_2^{a_2} p_3^{a_3} \dots)$, where $p_k^{a_k}$ is a prime factor of n , we can come up with an interesting computation.

Theorem 2.1.1

$$\phi(n) = n \left(1 - \frac{1}{p_1}\right) \left(1 - \frac{1}{p_2}\right) \left(1 - \frac{1}{p_3}\right) \dots \left(1 - \frac{1}{p_k}\right)$$

or equivalently

$$\phi(n) = n \prod_{p|n} \left(1 - \frac{1}{p}\right)$$

where p_k is a prime factor of n .

Proof.

$$\begin{aligned} \phi(n) &= \phi(p_1^{a_1} p_2^{a_2} p_3^{a_3} \dots p_k^{a_k}) \\ \phi(p_1^{a_1} p_2^{a_2} p_3^{a_3} \dots p_k^{a_k}) &= \phi(p_1^{a_1}) \phi(p_2^{a_2}) \phi(p_3^{a_3}) \dots \phi(p_k^{a_k}) \\ \phi(p_1^{a_1}) \phi(p_2^{a_2}) \dots \phi(p_k^{a_k}) &= (p_1^{a_1} - p_1^{a_1-1}) (p_2^{a_2} - p_2^{a_2-1}) \dots (p_k^{a_k} - p_k^{a_k-1}) \\ &= (p_1^{a_1-1} (p_1 - 1)) (p_2^{a_2-1} (p_2 - 1)) \dots (p_k^{a_k-1} (p_k - 1)) \\ &= p_1^{a_1-1} \left(1 - \frac{1}{p_1}\right) p_2^{a_2-1} \left(1 - \frac{1}{p_2}\right) \dots p_k^{a_k-1} \left(1 - \frac{1}{p_k}\right) \end{aligned}$$

Finally, because $p_1^{a_1} p_2^{a_2} \dots p_k^{a_k} = n$:

$$\phi(n) = n \left(1 - \frac{1}{p_1}\right) \left(1 - \frac{1}{p_2}\right) \left(1 - \frac{1}{p_3}\right) \dots \left(1 - \frac{1}{p_k}\right)$$

■

Corollary. $\phi(n)$ for any $n > 2$ is always even.

Proof.

If n is a power of 2 greater than 1 $\implies \phi(2^k) = 2^k - 2^{k-1}$, which is always even for $k > 1$.

If n is composed of an odd prime p , we can rewrite this factor in the formula as

$$1 - \frac{1}{p} = \frac{p-1}{p}$$

$p-1$ will always be even since p is prime, and since we are multiplying n by an even power, $\phi(n)$ must be even.

■

Though the computation has been greatly reduced, there is more we can do. Say that we know $\phi(8)$ and $\phi(10)$ and desire to compute $\phi(80)$. It wouldn't be too difficult to write $\phi(80)$ as a product of coprime factors; as a matter of fact, $\phi(16)$ and $\phi(5)$ would work just fine. However, as n increases, it will become more challenging to find coprime factors of n , especially if finding the prime factorization is too daunting.

Theorem 2.1.2

$$\phi(ab) = \phi(a)\phi(b) \frac{\gcd(a, b)}{\phi(\gcd(a, b))}$$

Proof.

$$\begin{aligned} \phi(ab) &= ab \prod_{p|ab} \left(1 - \frac{1}{p}\right) \\ &= ab \frac{\prod_{p|a} \left(1 - \frac{1}{p}\right) \prod_{p|b} \left(1 - \frac{1}{p}\right)}{\prod_{p|\gcd(a, b)} \left(1 - \frac{1}{p}\right)} \frac{\gcd(a, b)}{\gcd(a, b)} \\ &= \phi(a)\phi(b) \frac{\gcd(a, b)}{\phi(\gcd(a, b))} \end{aligned}$$

We use $\gcd(a, b)$ because there may be duplicate prime factors accounted for in the numerator, since a and b are not necessarily coprime. ■

2.2 Cardinality of the Set

The connection between $\mathbb{Z}_n^\times$ and $\phi(n)$ is pretty clear: $|\mathbb{Z}_n^\times| = \phi(n)$, since $\phi(n)$ simply counts the number of elements in the particular set.

Theorem 2.2.1

Considering our previous definition of $\mathbb{Z}_n^\times$:

$$|\mathbb{Z}_n^\times| = \phi(n)$$

3 More on Primitive Roots

3.1 Existence

Theorem 3.1

There are primitive roots mod n iff $n = 1, 2, 4, p^k$ or $2p^k$, where p is an odd prime and $k \in \mathbb{N}$.

In order to prove this theorem, it's most helpful to first address when there are no primitive roots and then prove for the remaining values of n that there do exist primitive roots.

Lemma 1. Let $m, n \in \mathbb{N}$. If $\gcd(m, n) = 1$ and $m, n \geq 3$, then there are no primitive roots mod mn .

Proof.

Consider some $a \in \mathbb{N}$ such that $\gcd(a, mn) = 1 \implies \gcd(a, m) = \gcd(a, n) = 1$.

Both $\phi(m)$ and $\phi(n)$ must be even. By Euler's Theorem:

$$\begin{aligned}
a^{\frac{\phi(m)\phi(n)}{2}} &= (a^{\phi(m)})^{\frac{\phi(n)}{2}} \equiv 1 \pmod{m} \\
&= (a^{\phi(n)})^{\frac{\phi(m)}{2}} \equiv 1 \pmod{n}
\end{aligned}$$

By the Chinese Remainder Theorem:

$$a^{\frac{\phi(n)\phi(m)}{2}} \equiv 1 \pmod{mn}$$

But,

$$\frac{\phi(m)\phi(n)}{2} = \frac{\phi(mn)}{2} < \phi(mn)$$

Because the order of a is less than $\phi(mn)$, a cannot be a primitive root mod mn . □

This helps us see that n does not have a primitive root if it is divisible by two odd primes or can be expressed as $2^k p^l$, where $k \geq 2$ and p is an odd prime.

Lemma 2. For any $k \in \mathbb{N}$, there is a primitive root mod $2p^k$ where p is an odd prime.

Proof.

$$\phi(2p^k) = \phi(p^k)$$

If g is odd:

$$\begin{aligned}
&\gcd(g, 2p^k) = 1 \\
&\text{ord}_{2p^k}(g) | \phi(p^k) \quad \text{and} \quad g^{\text{ord}_{2p^k}(g)} \equiv 1 \pmod{p^k} \\
&\implies \text{ord}_{2p^k}(g) = \phi(p^k) = \phi(2p^k)
\end{aligned}$$

So, g is a primitive root mod $2p^k$. If g is even, $g + p^k$ is odd:

$$g + p^k \equiv g \pmod{p^k}$$

$g + p^k$ is an odd primitive root and the previous steps follow. □

With these two lemmas, the theorem follows.

3.2 Order

Unfortunately, there is no direct way of computing primitive roots. It is generally reduced to trial and error, though there are methods that shorten this computation.

Definition 3.1.1 — Define $\text{ord}_n(g)$ to be the smallest $k > 0$ such that $g^k \equiv 1 \pmod{n}$.

An alternative definition to the one just constructed relates to *periods* of powers modulo n . Because the power k is strictly greater than 0, we can also define $\text{ord}_n(g)$ as the period of g , which simply means how many powers must be cycled through before finding an equivalent modulus.

By **Euler's Theorem** that $g^{\phi(n)} \equiv 1 \pmod n$, so by definition 3.1.1, $k \leq \phi(n)$, so $\phi(n)$ is the upper bound of k .

We know that $\text{ord}_n(g) | \phi(n)$. This is pretty clear to see, because $\text{ord}_n(g) \leq \phi(n)$ but it must be true that for some $y \in \mathbb{N}$, $y \text{ord}_n(g) = \phi(n)$, which ties back to the alternative definition.

Theorem 3.1

g is a primitive root mod n iff $\text{ord}_n(g) = \phi(n)$.

4 Other Useful Theorems

Theorem 4.1

$$\text{ord}_n(a^k) = \frac{\text{ord}_n(a)}{\gcd(k, \text{ord}_n(a))}$$

Proof.

Let $d = \gcd(k, \text{ord}_n(a))$

$$\begin{aligned} (a^k)^{\frac{\text{ord}_n(a)}{d}} &= \\ &= (a^{\text{ord}_n(a)})^{\frac{k}{d}} \\ &= 1^{\frac{k}{d}} \\ &= 1 \\ &\equiv 1 \pmod n \end{aligned}$$

$$\implies \text{ord}_n(a^k) \mid \frac{\text{ord}_n(a)}{d}$$

$$a^{k \text{ord}_n(a^k)} \equiv 1 \pmod n$$

$$\implies \text{ord}_n(a) \mid k \text{ord}_n(a^k)$$

$$\implies \frac{\text{ord}_n(a)}{d} \mid \frac{k}{d} \text{ord}_n(a^k)$$

Because $\frac{\text{ord}_n(a)}{d} \nmid \frac{k}{d}$, it must divide $\text{ord}_n(a^k)$.

Since we have divisibility both ways, $\frac{\text{ord}_n(a)}{d} = \text{ord}_n(a^k)$

■

Theorem 4.2

If there are primitive roots mod n , then there are exactly $\phi(\phi(n))$ primitive roots mod n .

This theorem is a corollary of the previous one, but influential enough to be its own theorem.

Proof.

Let g be a primitive root mod n .

Then, any other primitive root is in the set $\{1, r, r^2, \dots, r^{\phi(n)-1}\} \implies r^k, 0 < k < \phi(n)$

$$\text{ord}_n(r^k) = \frac{\text{ord}_n(r)}{\gcd(k, \text{ord}_n(r))} = \frac{\phi(n)}{\gcd(k, \phi(n))} = \phi(n)$$

$\implies \gcd(k, \phi(n)) = 1 \rightarrow$ There are $\phi(\phi(n))$ such numbers

■

4.1 Examples

As an example, we can find all primitive roots mod 18. Because $18 = 2(3^2)$, we know that there must exist primitive roots mod 18. If we look at

$$a^b \equiv k \pmod{n}$$

we know we are looking for a valid a , or a valid primitive root. Because our conditions are that $\text{ord}_n(a) | \phi(n) = 6$ and $\gcd(a, n) = 1$, we can test some possible values, denoted using hats:

$$\hat{k} \in \{1, 2, 3, 6\}$$

$$\hat{a} \in \{1, 5, 7, 11, 13, 17\} = \mathbb{Z}_{18}^\times$$

We can discount 1 because its order will always be 1. If we let each element of set S correspond to an element of $\mathbb{Z}_{18}^\times$, we get:

$$S = \{1, 6, 3, 6, 3, 2\}$$

Unsurprisingly, 17 doesn't work, as we've shown before. The primitive roots mod 18 are 5 and 11.

Using Theorem 4.2, we can see that there are exactly $\phi(\phi(8)) = 2$ primitive roots mod 8, which is exactly what we want. This is helpful because we know we could stop at 11 instead of computing all the way to 17.

As another example, we can attempt to find the primitive roots mod 8. We can immediately conclude that there are NO primitive roots using the existence theorem, but we can make an attempt to see what happens if we try.

$$\phi(8) = 4$$

$$\hat{k} \in \{1, 2, 4\}$$

$$\hat{a} \in \{1, 3, 5, 7\} = \mathbb{Z}_8^\times$$

We see that $3^2, 5^2$, and 7^2 are all 1 mod 8, which means that their orders are all 2. Because $2 \neq 4$, there are no primitive roots mod 8.

5 Discrete Logarithms

Definition 5.1 — If $a \equiv b^d \pmod{m}$, then d is the *discrete logarithm* of a modulo m to the base b ,

$$d = \log_b(a)$$

Properties of logarithms transcribe to discrete logarithms: if k is the order of $b \pmod{m}$,

$$\log_b(ac) \equiv \log_b(a) + \log_b(c) \pmod{k}$$

$$\log_b(a^r) \equiv r \log_b(a) \pmod k$$

This is generally very useful in **RSA encryption** (because, as we've seen before, there is no closed form for all primitive roots, especially as numbers get more complicated) and occasionally in solving some equivalencies.

$$\begin{aligned} x^4 &\equiv 9 \pmod{11} \\ \log_2(x^4) &\equiv \log_2(9) \pmod{10} \\ 2 \log_2(x) &\equiv 3 \pmod{5} \\ \log_2(x) &\equiv 4 \pmod{5} \implies 4, 9 \pmod{10} \\ x &\equiv 2^4, 2^9 \pmod{11} \rightarrow x \equiv 5, 6 \pmod{11} \end{aligned}$$

6 Primitive Roots of Unity

Definition 6.1 — ζ is a primitive n^{th} root of unity iff

$$\zeta^n = 1, \zeta^k \neq 1, 0 < k < n$$

From this definition, we know that 1 will only be a primitive root for $n = 1$, though generally, the first primitive root of unity is not particularly helpful.

The primitive 4th roots of unity are $\pm i$, because there is no $0 < k < n$ such that $i^k = 1$ except for $k = 4 = n$. 1 and -1 work for $k = 1$ and $k = 2$ respectively, and are thus not primitive roots.

Theorem 6.1

The primitive n^{th} roots of unity are of the form

$$\zeta_n = e^{\frac{2\pi i k}{n}}, k \in \mathbb{Z}_n^\times$$

and there are $\phi(n)$ primitive n^{th} roots of unity.

Proof.

ζ_n^k is only primitive if $k \mid n$ by the same order arguments as developed earlier.

$\gcd(k, n) = 1$ so that $(\zeta_n^k)^{\frac{n}{d}} = 1$

If

$$(\zeta_n^k)^r = 1, n \mid kr \implies n \mid r \implies \zeta_n^k$$

is a primitive root and there must be $\phi(n)$ of these. ■

We can also see pretty clearly from this that

$$\prod \zeta_n^k = 1, n \neq 2$$

using a method that could have easily proven why the totient function is always even: each root could be partitioned into $\{\zeta^k, \zeta^{n-k}\}$, which clearly multiplies to 1.

Theorem 6.2

The sum of the k^{th} powers of the primitive n^{th} roots of unity is

$$\mu(r) \frac{\phi(n)}{\phi(r)}$$

where $r = \frac{n}{\gcd(n,k)}$ and μ is the Möbius function

Proof.

$$\sum \zeta^k = \mu(k)$$

and there will be $\phi(g)$ of these terms. However, $\phi(n)$ of them are the primitive roots of unity, each being counted $\frac{\phi(g)}{\phi(n)}$ times.

$$\sum \zeta_n^k = \mu(k) \frac{\phi(g)}{\phi(n)}$$

■

7 Resources

Beirut, Wissam Raji. American University of, Wissam Raji, American University of Beirut, and Author Wissam Raji. “Elementary Number Theory (Raji).” Mathematics LibreTexts, July 7, 2021. [https://math.libretexts.org/Bookshelves/Combinatorics_and_Discrete_Mathematics/Elementary_Number_Theory_\(Raji\)](https://math.libretexts.org/Bookshelves/Combinatorics_and_Discrete_Mathematics/Elementary_Number_Theory_(Raji)).

Contents 2 discrete logarithms in cryptography. Accessed June 2024.
https://web.northeastern.edu/dummit/docs/cryptography_3_discrete_logarithms_in_cryptography.pdf.

Daileda, Ryan. Primitive roots modulo prime powers. Accessed June 2024.
http://ramanujan.math.trinity.edu/rdaileda/teach/f20/m3341/lectures/lecture16_slides.pdf.

Leon, Jeffrey. “Primitive Roots Mod p.” MCS 425. Accessed June 2024.
<https://homepages.math.uic.edu/leon/mcs425-s08/handouts/PrimitiveElements.pdf>.

Lyons, Chris. Our primitive roots. Accessed June 2024.
https://www.fullerton.edu/math/faculty/clyons/papers/Primitive_roots_talk.pdf.

Bayesian Inference and Tossing Coins

Narnia Poddar

1 What is Bayesian Inference?

We have all spent countless hours, or so it feels, working on a problem regarding a fair coin toss, fair die roll, or a fair draw of cards. But how do we know if the coin being tossed is a fair coin or not? Bayesian inference is a technique that uses Bayes' theorem to help us work towards answering questions of this nature.

2 Bayes' Theorem

Bayes' theorem states that the conditional probability of A given B (the probability of event A given the occurrence of event B) can be calculated from $P(B|A)$, $P(A)$, and $P(B)$ as follows:

$$P(A|B) = \frac{P(B|A)P(A)}{P(B)}$$

2.1 Proof of Bayes' Theorem

To prove Bayes' theorem, we begin with the definition of conditional probability:

$$P(A|B) = \frac{P(A \cap B)}{P(B)}$$

Multiplying both sides by $P(B)$ gives us:

$$P(B)P(A|B) = P(A \cap B)$$

Similarly, we can use the definition of the conditional probability for $P(B|A)$:

$$P(B|A) = \frac{P(B \cap A)}{P(A)}$$

And multiplying both sides by $P(A)$:

$$P(A)P(B|A) = P(B \cap A).$$

However, since $P(A \cap B) = P(B \cap A)$, we can equate the right-hand sides of the equations:

$$P(B)P(A|B) = P(A)P(B|A)$$

Dividing both sides by $P(B)$ gives us Bayes' theorem:

$$P(A|B) = \frac{P(B|A)P(A)}{P(B)}.$$

3 Coin Tossing Example

Suppose we want to determine the fairness of a coin – we want to know the probability of getting heads after tossing the coin. Assume that the coin always lands on either heads or tails, and we have no existing observations about the outcomes of flipping the coin. There are multiple ways to approach this problem: either using the frequentist approach (estimates of probability are gathered after repeating the experiment numerous times) or using Bayesian inference (start with a belief about the probability and keep updating it using Bayes’ Rule).

3.1 Determining the Probability Using the Frequentist Approach

In the frequentist view, the “probability” of an event refers to the proportion of times the event occurs if infinitely many trials are executed:

$$\lim_{n \rightarrow \infty} \frac{n_A}{n},$$

where n_A is the number of occurrences of event A in n trials. In reality, a statistician cannot run infinitely many trials. What a statistician can obtain is an estimate for $P(A)$: $P(A) = \frac{n_A}{n}$, which will get closer to $P(A)$ as n gets larger. In other words, the error (the difference between $P(\hat{A})$ and the true value) associated with our estimate of $P(\hat{A})$ becomes smaller as n becomes large.

The frequentist interpretation of the coin-flipping problem doesn’t make any assumptions about the probability of observing a head or tail before actually flipping the coin – we can’t use our existing knowledge to incorporate a belief that most coins are usually fair or close to fair. To obtain an accurate estimate of the probability of the coin landing on heads, this approach requires tossing the coin a fairly large number of times.

We run into pitfalls with this approach if we can only flip the coin a few times. Although we can normally flip a coin as many times as desired, other statistical problems may have limits on how many trials can be conducted. For example, say we want to determine the probability of a candidate winning an election. The event may only occur a few times, and the observer has no control over how many trials occur. So what can we do when the frequentist approach fails?

3.2 How is Bayesian Inference Different?

In Bayesian inference, the “probability” of an event is considered to be the subjective belief an observer holds about the likelihood that an event would occur. In this point of view, it is fine to have a view about the probability of an event for which no trials have even been conducted. It also allows for multiple observers to hold different beliefs about the probability distribution of the same event.

In this framework, one starts with a subjective belief about the probability of an event, called the prior. As one observes the outcomes of trials, they update their beliefs to incorporate the information obtained from the trials. The updated belief is known as the posterior. The update is done using Bayes Theorem and the process is called Bayesian inference (or Bayesian Learning).

3.3 How is Bayes’ Theorem Used to Update Beliefs?

It is customary in Bayesian Inference to write Bayes’ theorem in the following form:

$$P(\theta|X) = \frac{P(X|\theta)P(\theta)}{P(X)}$$

Here,

- θ represents the object that we are interested in learning about (which in our case would be the probability of the coin landing on heads when it is tossed).
- X represents the observations that we have collected (which in our case would be the sequence of heads and tails observed after the coin was tossed a certain number of times).
- $P(\theta)$, called the prior, is our belief about the probability of the event (landing on heads) before any coin tosses are made.
- $P(X|\theta)$, called the likelihood, is the probability of observing the outcome X given a specific θ – for example, given that the probability of the coin landing on heads is 0.4.
- $P(\theta|X)$, called the posterior, is our updated belief about probability of θ , given the sequence of heads and tails (X) that we have observed so far.

$P(X)$, the unconditional probability of observing X based on prior beliefs, can be computed as a summation of the conditional probabilities $P(X|\theta)$ over all possible distinct values of $\theta \in 1, 2, \dots, n$ (in other words all possibilities based on our prior beliefs),

$$P(X) = \sum_{i=1}^n P(X|\theta_i)P(\theta_i).$$

So we start with a prior belief about the probability distribution of the coin landing on heads when tossed. Then, applying Bayes' Theorem, we update our belief about the probability distribution based on the actual observations collected. Note that when the number of observations becomes sufficiently large, the posterior probability estimate obtained by Bayesian inference would normally converge to a frequentist estimate of the probability.

4 Applying Bayesian Learning in a Discrete Setting

In this section, we explore the application of Bayesian learning to the coin tossing problem where the probability of landing on heads can only take a discrete set of values. For example, say that the probability of landing on heads is either 0.4 or 0.6. So, the coin is biased, but we do not know which has a higher probability: heads or tails. In our prior, we assume both cases to be equally likely:

$$P(\theta = 0.4) = 0.5$$

$$P(\theta = 0.6) = 0.5$$

where θ represents the probability that the coin lands on heads when tossed. Suppose we toss the coin N times and observe h heads. This is our set of observations, X . Then, given a probability θ of observing heads in a single coin toss, and assuming outcome of each toss is independent of other tosses, we can compute the likelihood of h heads in N tosses as:

$$P(X|\theta) = \binom{N}{h} \theta^h (1 - \theta)^{N-h}$$

and

$$P(X) = \sum_{\theta} \binom{N}{h} \theta^h (1 - \theta)^{N-h} \cdot P(\theta)$$

so, our Bayesian posterior belief is given by:

$$P(\theta|X) = \frac{\binom{N}{h} \theta^h (1 - \theta)^{N-h} \cdot P(\theta)}{\sum_{\theta} \binom{N}{h} \theta^h (1 - \theta)^{N-h} \cdot P(\theta)}$$

which lends itself to the intuitive interpretation that the posterior probability of θ is the likelihood of observing h heads in N trials assuming θ is true, multiplied by the prior probability of θ , and divided by the total probability of observing h heads in N tosses given prior beliefs about all possible values of θ .

Specifically let's assume that we toss the coin 5 times and observe 3 heads. We can compute the posterior probabilities $P(\theta = 0.4|h = 3, N = 5)$ and $P(\theta = 0.6|h = 3, N = 5)$ as below:

$$P(h = 3, N = 5|\theta = 0.4) = \binom{5}{3} 0.4^3 (1 - 0.4)^{5-3} = 0.2304$$

$$P(h = 3, N = 5|\theta = 0.6) = \binom{5}{3} 0.6^3 (1 - 0.6)^{5-3} = 0.3456$$

Then, $P(X)$ is:

$$\begin{aligned} P(h = 3, N = 5) &= P(h = 3, N = 5|\theta = 0.4) \cdot P(\theta = 0.4) + P(h = 3, N = 5|\theta = 0.6) \cdot P(\theta = 0.6) \\ &= 0.2304 \times 0.5 + 0.3456 \times 0.5 = 0.2880 \end{aligned}$$

and the posterior beliefs are:

$$P(\theta = 0.4|h = 3, N = 5) = \frac{P(h = 3, N = 5|\theta = 0.4) \cdot P(\theta = 0.4)}{P(h = 3, N = 5)} = \frac{0.2304 \times 0.5}{0.2880} = 0.4$$

$$P(\theta = 0.6|h = 3, N = 5) = \frac{P(h = 3, N = 5|\theta = 0.6) \cdot P(\theta = 0.6)}{P(h = 3, N = 5)} = \frac{0.3456 \times 0.5}{0.2880} = 0.6$$

Given that our observations have a higher number of heads compared to tails, our posterior belief now assigns higher odds to the probability of heads being 0.6 instead of the prior of 0.4 as one would expect.

5 Applying Bayesian Learning in Continuous Settings

In a more general setting, we would like to be able to take on any value within a continuous range. For example, in our coin tossing example, we likely want to allow to take on any value in the range $[0,1]$. In this case will be a continuous variable. Probability distributions of continuous random variables are generally characterized using a probability density function (pdf). The density function at any point describes the probability per unit length around that point. Since any continuous range has infinitely many values, the probability of the variable exactly taking on any single value (a point on the range) is zero. Probability is defined on a range $[a, b]$ as:

$$P(a \leq \theta \leq b) = \int_a^b p(\theta) d\theta$$

where p is the density function and P is the probability. Given that probabilities are fully specified by the density function and can be obtained using the above formula, one can work directly with prior and posterior density functions when applying Bayesian inference on problems with continuous variables.

5.1 Bayes' Theorem for Density Functions

Bayes' theorem for pdfs can be written as:

$$p(\theta|X) = \frac{p(X|\theta) \cdot p(\theta)}{p(X)}$$

where $p(\theta)$ denotes the prior density of θ , $p(X|\theta)$ is the conditional density of X given θ , $p(X)$ is the unconditional density of X , and $p(\theta|X)$ is the posterior density of θ given X . Often the relationship between the prior and posterior is represented as a proportionality:

$$p(\theta|X) \propto p(X|\theta) \cdot p(\theta)$$

The posterior density is proportional to the prior density and the likelihood of the observations under the prior density.

When working with Bayesian inference on a continuous θ , it is often beneficial to select pdfs for priors such that the functional form for the posterior has the same functional form as the prior itself. Such a combination of likelihood and priors are called conjugate priors. It allows us to reuse the Bayesian update algebra over and over as we observe new data and continue to update our beliefs. For example, if we observe a new set of trials Y , after having updated our beliefs to $p(\theta|X)$ from $p(\theta)$ using the observations in X , as long as $p(\theta|X)$ has the same functional form as $p(\theta)$, we can make use of the same algebra or code to obtain $p(\theta|X, Y)$ using the observations in Y and using $p(\theta|X)$ as our current prior.

5.2 Bayesian Inference for Coin Tossing with Continuous θ

In this setting we allow for θ to take any value in the range $[0, 1]$. Note that our likelihood function $p(X|\theta)$ is unchanged from earlier, since for discrete variable X :

$$p(X|\theta) = P(X|\theta) = \binom{N}{h} \theta^h (1 - \theta)^{N-h}.$$

A conjugate prior for this likelihood function is the Beta distribution. The density function of the Beta distribution is given by:

$$p(\theta) = \frac{\theta^{a-1} (1 - \theta)^{b-1}}{B(a, b)}$$

where $B(a, b)$ is the Beta function which ensures that the probability that lies in $[0, 1]$ is 1 and $a, b > 0$ are parameters that determine the shape of the distribution (such as where the peak is, how heavy the tails are, etc). To see why the Beta distribution forms a conjugate prior with our likelihood in this case, we observe:

$$p(\theta|X) \propto p(X|\theta) \cdot p(\theta) \propto \theta^{a-1} (1 - \theta)^{b-1}$$

Simplifying, we obtain:

$$p(\theta|X) \propto \theta^{a+h-1} (1 - \theta)^{b+N-h-1}.$$

So, $p(\theta|X)$ has a density function given by a Beta Distribution with parameters $a' = a + h - 1$, and $b' = b + N - h - 1$.

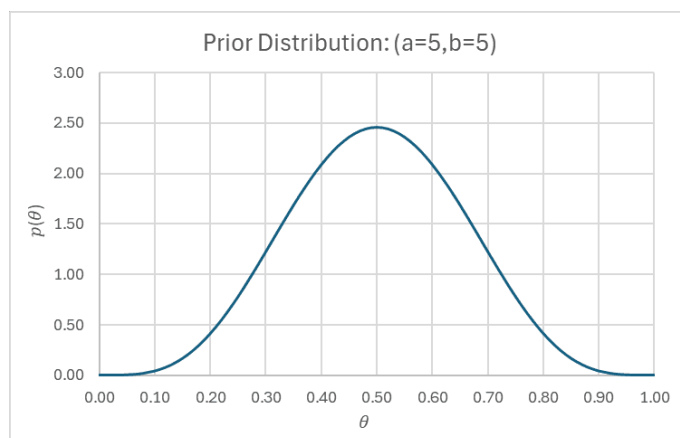
We start with a prior of a Beta distribution with $a = 5, b = 5$.

$$p = \frac{\theta^{5-1}(1-\theta)^{5-1}}{B(5,5)}.$$

This leads to a distribution which peaks at 0.5 but is still quite spread out as shown in the figure below. Under this prior,

$$P(\theta \leq 0.5) = 0.5, P(\theta \geq 0.5) = 0.5.$$

It reflects a belief that the while there is substantial uncertainty about the value of θ , we do not have any view on if the coin is fair or unfair.



Now, we update the belief using the observation $X(N = 5, h = 3)$. The posterior will be a Beta distribution with $a' = a + h - 1 = 7$, and $b' = b + N - h - 1 = 6$. In other words

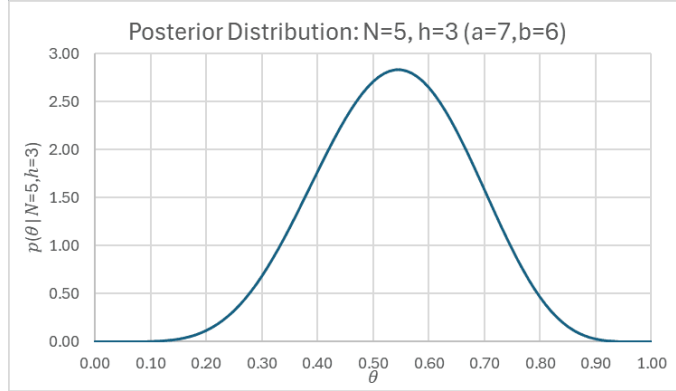
$$p(\theta|N = 5, h = 3) = \frac{\theta^{7-1}(1-\theta)^{6-1}}{B(7,6)}.$$

The plot of the density shows that the peak of the distribution has shifted rightwards from its 0.5 position in the prior and the spread of the distribution has reduced. Under the posterior, the odds of the probability of heads being greater than 0.5 is now higher than odds being lower than 0.5:

$$P(\theta \leq 0.5|N = 5, h = 3) = 0.39$$

$$P(\theta \geq 0.5|N = 5, h = 3) = 0.61$$

The plot below shows the density of the posterior.

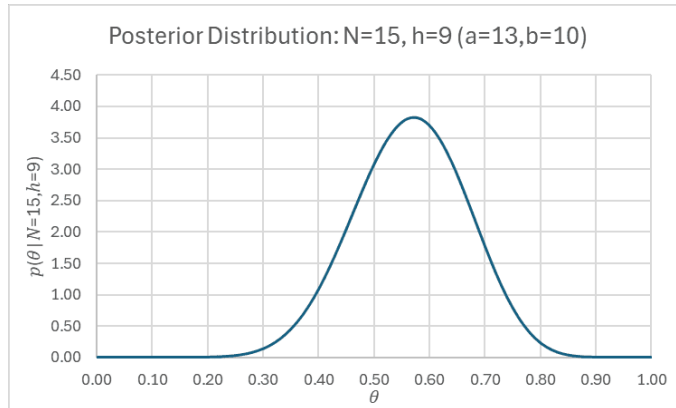


To see how the distribution changes with sample size, we look at another sample with the same proportion of heads but larger sample size: $N = 15, h = 9$. The posterior $p(\theta|N = 15, h = 9)$ is now a Beta distribution with $a' = a + h - 1 = 13$, and $b' = b + N - h - 1 = 10$. The distribution has become narrower and the odds of the probability of heads being greater than 0.5 is now even higher than the $N = 5$ case:

$$P(\theta \leq 0.5|N = 15, h = 9) = 0.26$$

$$P(\theta \geq 0.5|N = 15, h = 9) = 0.74$$

This shows that as the sample size increases, the uncertainty about the value of θ reduces and our belief about starts to center towards the sample estimate.



6 Other Applications

Even though determining the fairness of a coin is something that most people probably don't find themselves needing to do in their daily lives, techniques used for Bayesian inference are useful for a variety of problems. For example, Bayesian inference can be used for updating predictions for weather events based on incoming data or modeling the spread of diseases.

References

Bayesian Inference of a Binomial Proportion - The Analytical Approach. (n.d.). QuantStart.
<https://www.quantstart.com/articles/Bayesian-Inference-of-a-Binomial-Proportion-The>

-Analytical-Approach/.

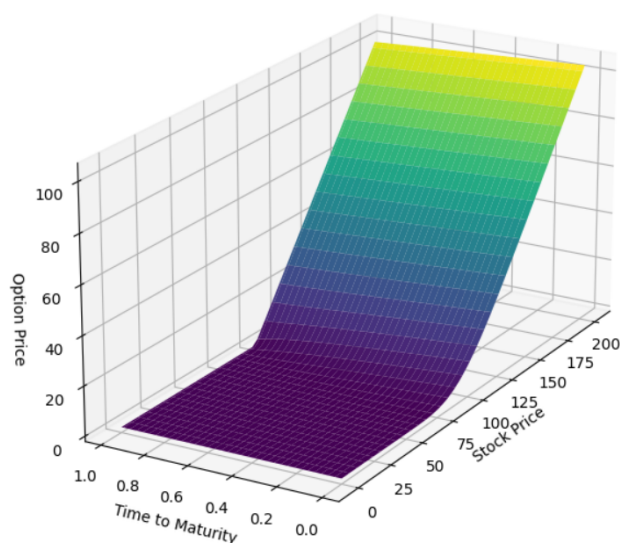
“Bayesian Statistics: A Beginner’s Guide.” QuantStart,
<https://www.quantstart.com/articles/Bayesian-Statistics-A-Beginners-Guide/>.

Lecture 4: Estimating Probabilities from data. (n.d.). CS@Cornell.
<https://www.cs.cornell.edu/courses/cs4780/2018sp/lectures/lecturenote04.html>.

Lancaster, T. (2004). Introduction to Modern Bayesian Econometrics. Wiley.

The Black-Scholes Equation: Transforming Quantitative Finance by Hedging Risk and Catalyzing New Industries

Anastasia Lee



1 Introduction

Around the year 600 BC, ancient Greek mathematician and philosopher Thales of Miletus predicted that the season's olive harvest would be unusually abundant, which would increase the demand for olive presses. Although he could not afford to buy his own olive presses, Thales paid a smaller amount of money to acquire the right to rent the olive presses for a specified price during the harvest season. When the harvest came, there were indeed more olives than expected, so Thales made a profit by renting out the presses at a much higher price than he paid.

Thales was the first reputed buyer of a *call option*, the right but not the obligation to buy something at a later date for a pre-specified price (the strike price). Similarly, a *put option* is the right but not the obligation to sell something at a later date for a pre-specified price (the strike price). So, if one predicts that a stock will increase in value, as Thales did, they would benefit from buying a call option, and if they predict that a stock will decrease in value, they would benefit from buying a put option.

There are several important advantages of options. First, they can often provide higher returns relative to the initial cost. Second, they can offer lower risk (compared to equities) through hedging. Third, options can provide leveraging power through stock replacement. Fourth, options can offer the flexibility needed for strategies such as spreads, straddles, and strangles. Quantitative investors can take advantage of these benefits with a thorough understanding of the mathematics underpinning the use of options in financial economics.

2 Determining the value of an option

The question that all economic markets seek to answer is, “How much should this be worth?” The most common method of determining such a value is summing its future cashflows, discounted to present value. However, the future cashflows of an option are uncertain, because they depend on the future stock price. So, we can attempt to find a trading strategy that will have the same payout as the option, and then set the price of the option equal to the price of the strategy.

Such a trading strategy would need to have the exact same risks as the option. Thus, we can create a portfolio with both the option and a certain number of shares (involved in the trading strategy) in which the risks would be offset, meaning that the portfolio is risk-free.

We denote the price of the option as $V(s, t)$, where s is the stock price and t is time. Moreover, we denote the number of shares as Δ and the share price as S . Then, the portfolio Π is given by $\Pi = V(s, t) - \Delta S$. To analyze the risk factors, we need to look at the evolution of the portfolio, specifically the change in the portfolio value. We have $d\Pi = dV - \Delta dS$, where the d refers to an infinitesimal change in the variable that follows it.

2.1 Change in the stock price

First, let us consider the change in the stock price. Initially, it was believed that since the stock price is dependent on so many variables, it is virtually impossible to predict all of them accurately. So, the best we can do is assume that at any given point in time, the stock price is equally likely to increase or decrease. In other words, the stock price follows a random walk, which can be modeled with **Brownian motion**.

However, this is not accurate, as a stock price often increases over time if the business is doing well or decreases over time if the business is doing poorly. Thus, to account for this drift we choose to model stock prices with **geometric Brownian motion**, which satisfies the equation

$$dS = \mu S dt + \sigma S dW.$$

Here, μ (the percentage drift) and σ (the percentage volatility) are constants, and W represents Brownian motion. Hence, $\mu S dt$ represents the drift, or growth rate, of the stock price, and $\sigma S dW$ represents the volatility of the stock price.

2.2 Change in option price

By Itô's Lemma,

$$dV = \frac{\partial V}{\partial t} dt + \frac{\partial V}{\partial S} dS + \frac{1}{2} \cdot \frac{\partial^2 V}{\partial S^2} dS^2.$$

Substituting our equation for dS into the equation above and noting that in stochastic calculus, $dt \cdot dt = dt \cdot dW = dW \cdot dW = 0$, we obtain

$$dV = \frac{\partial V}{\partial t} dt + \frac{\partial V}{\partial S} dS + \frac{1}{2} \sigma^2 S^2 + \frac{\partial^2 V}{\partial S^2} dt.$$

2.3 The Black-Scholes equation

Substituting the equation above into $d\Pi = dV - \Delta dS$ and factoring out dt and dS , we get

$$d\Pi = \left(\frac{\partial V}{\partial t} + \frac{1}{2} \sigma^2 S^2 \frac{\partial^2 V}{\partial S^2} \right) dt + \left(\frac{\partial V}{\partial S} - \Delta \right) dS.$$

Now, the second term represents the volatility, or risk. Since we established that Π is risk-free, we set $\frac{\partial V}{\partial S} - \Delta = 0$, which yields $\Delta = \frac{\partial V}{\partial S}$.

However, a risk-free portfolio should yield a risk-free rate, so $d\Pi = r \cdot \Pi dt$, where r is the risk-free interest rate. Substituting the expression for Π into this equation, we get

$$d\Pi = \left(rV - rS \frac{\partial V}{\partial S} \right) dt.$$

Finally, we can set our two expressions for $d\Pi$ equal to each other to obtain the famous Black-Scholes equation:

$$\frac{\partial V}{\partial t} + \frac{1}{2} \sigma^2 S^2 \frac{\partial^2 V}{\partial S^2} + rS \frac{\partial V}{\partial S} - rV = 0.$$

2.4 Solving the Black-Scholes equation

In certain cases, such as in the case of a European option, setting V to be equal to a call/put option yields an exact value from the Black-Scholes equation. In other cases, the partial differential equation can be solved using standard methods such as a version of finite differences.

3 Jim Simons: mathematician, professor, code breaker, and “Quant King”

The recent passing of Jim Simons highlights how the Black-Scholes equation and other mathematical models revolutionized trading in the 1980s. After accomplished careers in both academia and government, in 1978, Simons founded an investment firm that would eventually become the hedge fund Renaissance Technologies, which ran one of the most profitable funds of all time, the Medallion Fund. The historic level of returns of this fund effectively disproved the efficient market hypothesis, which states that stock prices reflect all information, and therefore stocks trade at their fair market value. Over a period of 30 years, from 1988 to 2018, the Medallion Fund earned its investors \$100 billion with annual returns of 66 percent (before fees). Simons’ landmark success eclipses that of his contemporaries such as Warren Buffett, George Soros, Peter Lynch, Steve Cohen, and Ray Dalio.

Following in the footsteps of mathematicians from the 19th century through the late 20th century, such as Louis Bachelier, Ed Thorpe, Fischer Black, Myron Sholes, Robert C. Merton, and Leonard Baum, Jim Simons sought to improve and refine trading models by detecting short-term

patterns in option price discovery. He primarily employed mathematicians and scientists (rather than traders) who pioneered the use of algorithmic data analysis to exploit statistical anomalies in the stock market. Quantitative analysis has since spread throughout Wall Street, using machine learning to frequently beat the market.

4 Conclusion

Over the span of the past 50 years, the Black-Scholes equation has contributed to four multi-trillion dollar industries: exchange traded options, over-the-counter derivative securities, credit default swaps, and securitized debt. It fundamentally transformed investors' approach to risk. Some of its limitations have spawned alternative option pricing models, such as the Binomial model and the Monte Carlo simulation model. But the Nobel-prize-winning Black-Scholes Equation's robust mathematical framework remains one of the most impactful applications in the history of finance. By creating entirely new industries, mathematical insight provides unprecedented opportunities to grow investments and improve outcomes and lives all over the world.

References

- "Deriving the Black-Scholes Equation." *QuantStart*, retrieved May 29, 2024.
<https://www.quantstart.com/articles/Deriving-the-Black-Scholes-Equation/>
- Ianieri, Ron. "4 Advantages of Options." *Investopedia*, December 19, 2023.
<https://www.investopedia.com/articles/optioninvestor/06/options4advantages.asp>
- Lalley, Steven P. "Lecture 7: Black-Scholes Theory." University of Chicago Statistics 390/
Mathematical Finance 345, retrieved May 21, 2024.
<https://galton.uchicago.edu/~lalley/Courses/390/Lecture7.pdf>
- Parra, Hair. "A Complete Introduction to the Black-Scholes Model — For Quants: A Mathematical Perspective with Stochastic Calculus and Python." *Medium*, March 7, 2024.
<https://hair-parra.medium.com/a-complete-introduction-to-black-scholes-model-for-quants-65106b33f1ae>
- "The Trillion Dollar Equation." *Veritasium*, February 27, 2024.
<https://youtube.com/@veritasium?si=DffbvfJGkNPlm2lJ>
- Zuckerman, Gregory. "Jim Simons, a Pioneer of Quantitative Trading, Dies at 86:
A mathematician, he helped usher in a revolution in trading, embracing a computer-oriented, quantitative style in the 1980s. *The Wall Street Journal*, May 10, 2024.
<https://www.wsj.com/arts-culture/books/jim-simons-a-pioneer-of-quantitative-trading-dies-6621d66e>

Introduction to Mathematical Modeling

Ethan Sharma

1 Introduction

People make countless decisions every day of their lives. Whether it be from deciding what to have for lunch or who to vote for in the next election, we are constantly faced with choices to make based on limited information. For example, you have no idea who is working at McDonald's or how crowded it is today, but you make the decision to go there for lunch anyway since it's close and you have deals on the app. We create mental models in our minds to make decisions, qualitatively valuing some traits of a choice over another and ignoring others. These mental models work well for simple decisions with few factors to consider, but when systems become more complex and harder to conceptualize, a more advanced system is often needed. In addition, in an attempt to collaborate with others using different mental models with different values, it becomes harder to unify and make one decision that everyone can agree with. The necessity of a uniform, adaptable method of modeling has emerged from a global dependency on important decision making and the attempt to understand increasingly intricate systems around the world.

Mathematical methods of modeling have become popular due to their systematic nature and comprehensibility across different languages. Modeling is a creative process, as one problem can be approached in many different ways using different techniques. However, the principles of modeling ensure that there is little bias or oversights when solving a problem, so that another team can look over a solution and expand on it or take a different approach. This paper will be a brief overview of the process, theory, and techniques behind mathematical modeling.

Much of this will be based off of my own experience from modeling for competitions such as the HiMCM, IMMC, or M3 Challenge. These competitions give teams a multi-step modeling problem and a limited amount of time to write up a solution, following the modeling process, and then those solutions are judged against others across the world. I've learned a lot about different methods and techniques of modeling by doing these competitions.

2 The Process of Mathematical Modeling

A very important element of the modeling process is clarity. Generally, the problem that will be solved is restated so the task of the model is clear.

Each modeling approach requires a team to make simplifying assumptions about how a system operates. All of these assumptions and the justification for them are clearly stated prior to describing the model itself. For example, in trying to model the expected amount of travel time from one place to another by automobile, one could make the assumption that the rate of consumption of fuel is constant. There are many intricacies to cover to exclude extreme cases as well. We'd have to make the assumption that the vehicle is not going to break down, and use a general model of

a car instead of a truck. Every assumption changes what the model is predicting, and serves as an opportunity to extend the model to consider all possibilities. Sometimes assumptions are made because certain events are either negligible or rare enough to be negligible. After assumptions are made, terms and variables used in the model are defined, and then the technique is described and implemented.

Models can be created for multiple purposes. Sometimes, they are made just to understand a system better, which is done in many environmental scenarios. Or, models can be made to explore the effects of certain changes made to a system. This allows for the effects of scenarios to be known without performing them in real life, which can often be costly or inefficient. Another reason may be for management or control of a system, like modeling the performance of workers in an office and determining who isn't doing enough work.

After the model is created and used, it is evaluated using sensitivity analysis. Sensitivity analysis assesses how a model fares outside of situations it was built for, essentially testing how extendable it is against the assumptions that were made in its creation. Evaluation of the model also includes exploring the strengths and weaknesses of the technique used and certain improvements that can be made to it. For example, some methods of modeling may require large amounts of data before being useful, such as machine learning, while others suffer from scale, such as using a computer algorithm with high time complexity. Highlighting where your model works and where it doesn't makes it easier for others to improve upon it.

In competitions, it is often asked for teams to create a letter detailing their model's results and how it can be applied to the real world. This letter often addresses those in power to make certain changes. For example, the 2023 IMMC problem asked teams to determine the best usage for a plot of land and report the findings of their model to the business leaders that have control over that land and the power to implement what the model advised. It's important that the results of the model are clearly communicated, since when solving a real world problem you must provide a clear answer to the problem. Regardless of how many factors you took into consideration for where to go for lunch, the main tangible effect is whether you have to physically go to McDonald's or Whole Foods.

3 The Theory of Mathematical Models

This section will be focused on categorically defining different modeling types/techniques. All models deal with systems, where a system can be defined as a collection of physical objects with known attributes. A modeler can define the boundaries and aspects of a system, how objects within it interact, and how it interacts with anything outside the system (if it does). Systems that interact with their surroundings are called open, while systems that don't are closed. Variables are attributes of a system or its surroundings that have a significant effect on how the system acts. Depending on the purpose of a model, it can have different inputs and outputs. Generally, the inputs are variables that define the traits of the system and the outputs are generated by the initialized system and measured.

Sometimes, models can be based purely on data rather than taking into consideration all of the factors that go into an output. These models are often called black box models or empirical models, and are often used when the science or underlying features of a system are not fully understood.

These types of models are often used on systems in economics, such as moving average models to determine when to buy stocks.

Modeling techniques can either be deterministic or probabilistic. Probabilistic methods implement randomness if it is believed that there is an unpredictability to a system that is not captured in the variables. As a result, running a probabilistic model multiple times will result in different outcomes. Deterministic models are the opposite, there is no randomness and the system is entirely defined by the initial conditions so that the same output will be reached given the same inputs.

Models can also be categorized based on the 'time domain of a system. If time is measured over the real numbers, a system is continuous, while if time is measured over the integers a system is discrete. If the state of a system is dependent on any previous states in a model, that model is dynamic, while a static model does not depend on previous states to determine the next one.

Dynamic systems can either be lumped or distributed. These categories are based off of the function that determines future states based on the current state. This function is a function of time, and it can have other spatial variable inputs that change over time. If this function is only a function of time, the model is lumped, while if it has other inputs then the model is distributed.

Finally, models can either be analytical or numerical. These categories depend on the form of solutions to model the system. In the case of dynamic systems, which are often modelled with differential equations, there is often no closed form for solutions, so solutions are approximated numerically using the differential forms. If closed forms for solutions are available, like in most static systems, the model is analytical, and the solution allows an output to be directly obtained from an input.

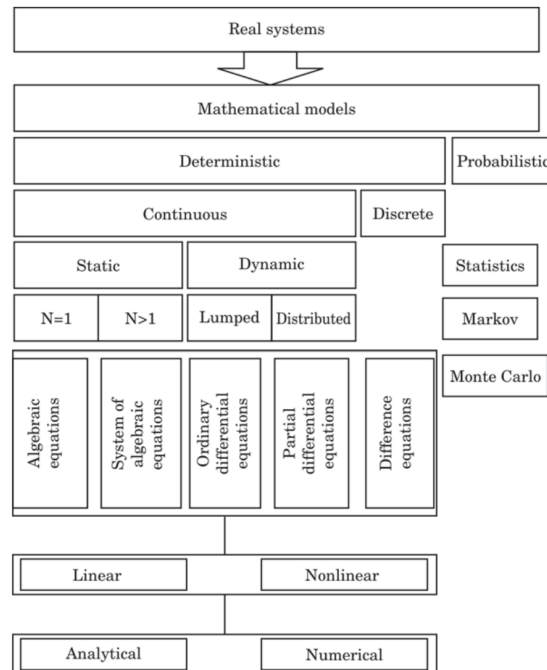


Figure 1: Chart of Types of Mathematical Models (Dr. Seiichiro Yonemur)

3.1 SQM Space Categorization

The SQM Space of Mathematical Models is another way of categorizing models. A mathematical model can be described as a triple (S, Q, M) with a system S , nature of the question Q , and mathematical statements M . This space is used to categorize real world systems in the language of mathematical models. Different systems constitute a different complexity of a model, for example we know much more about the underlying processes about electrical circuits than psychological systems. Questions can also change this, depending on how much control we want to exert over the system. There is much less certainty if we are simply trying to speculate about future states instead of implementing something sure to change future outcomes. Mathematical statements are often matched to the desired system and question. These three traits, as shown in Figure 2, can also describe a model as black box or white box. When we know more about the processes that govern a system, the system is more white box and we can use more deterministic methods for modeling. Understanding SQM space can be useful for determining what mathematical techniques to use when dealing with a certain system and question.

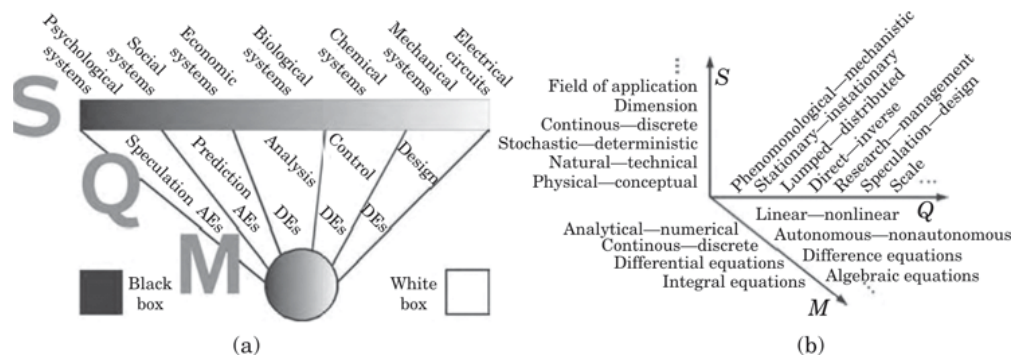


Figure 2: SQM Space, Visualized (Dr. Seiichiro Yonemur)

4 Conclusion

The process of mathematical modeling, even without looking into specific mathematical techniques, is intricate and requires careful work to follow. There is a lot more than math that goes into mathematical modeling. As Stuyvesant mathematics teacher Mr. Honner says,

“Mathematical modeling requires a lot of curiosity and creativity. Modelers must be curious about how the world works, and they must be creative in how they use math to describe it. Modelers have to be great collaborators, as no real-world problem is solved alone. And they must be great communicators, as a proposed solution will only be useful if you can convince decision makers to use it.” - Mr. Honner, 2024

References

- Zeidan, Bakenaz. "Mathematical Modeling of Environment Problems" ISBN Of Series 1-62699-088-3, ISBN OF Volume 1-62699-095-6, Nov. 2015.
https://www.researchgate.net/publication/282206837_Mathematical_Modeling_of_Environment_Problems

Mario Kart Score Distributions

Elijah White

1 Introduction

In a 12-player Mario Kart race, each player is awarded a certain number of points based on their placement in that race. Currently, Mario Kart games use the following point distribution (in placement-descending order):

15
12
10
8
7
6
5
4
3
2
1
0

But is this the best point distribution? It depends. It is impossible to universally define what “best” means here. One could define the best point distribution system to be the one that, when used identically for each race, best predicts the outcomes of the entire n -race competition. One could also define “best” to be the one that most accurately ranks the skill of the racers. Even then, skill is not a quantifiable measure. Similarly, for the formerly-proposed definition, we create a point distribution to determine the outcomes of an n -race competition, so asking which one best “predicts” the outcome brings us straight back to our first question: what is the best point distribution system?

The essential problem rooted within this question is that an identical placement-based point distribution must be used for every race of every n -race competition, meaning that the point system may not use information relating to the players’ finishing times, the players’ positions at times in the race other than at the finish line, etc.

In this article I formulate some alternatives to the games’ point distribution system based on techniques involving Markov chains.

2 Using Markov Chains

If the reader is not familiar with the concept of Markov chains, they will be briefly explained here as we will be using them, though this explanation is far from a comprehensive explanation of Markov chains.

Let P_k denote the player that came in k th place.

We will treat the players as nodes. Each player gets 1 “vote” to distribute between the other players based on the place that they came in. This is represented by a matrix, which we will denote as M , for which the entry in the i th row and j th column (i.e. M_{ij}) is the portion of P_j ’s vote that goes toward P_i . After the first round of “voting,” all of the players get to vote with however many votes they received from the last round. We can raise M to a very high power to determine how many votes each player has after many, many rounds of voting.

Here, we want players to cast more votes to the players that finish ahead of them than to those who finish behind them. There are many ways to accomplish this. Our first Markov chain approach will use the ratios of the placements of each pair of players.

To make it simpler to compare the point distributions, we will formulate each one so that 12th place gets 0 points and so that the sum of all the points distributed is equal to that of the Mario Kart games:

$$15 + 12 + 10 + 8 + 7 + 6 + 5 + 4 + 3 + 2 + 1 + 0 = 73.$$

Additionally, after initially calculating the votes that each player gives every other player, we need to be careful to scale each column so that each player gets exactly 1 vote.

2.1 Ratios of Placements

Here,

$$M_{ij} = \frac{j}{i},$$

e.g. P_9 gives P_4 $\frac{9}{4}$ votes and each player gives themselves 1 vote, which gives us the matrix:

$$\begin{bmatrix} \frac{1}{1} & \frac{2}{1} & \frac{3}{1} & \dots & \frac{12}{1} \\ \frac{1}{2} & \frac{2}{2} & \frac{3}{2} & \dots & \frac{12}{2} \\ \frac{1}{3} & \frac{2}{3} & \frac{3}{3} & \dots & \frac{12}{3} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ \frac{1}{12} & \frac{2}{12} & \frac{3}{12} & \dots & \frac{12}{12} \end{bmatrix}.$$

Now, we must scale each column by 1 over its sum to ensure that each player gets exactly 1 “vote,” which produces the following matrix (each entry is rounded to the nearest thousandth):

[	0.322	0.322	0.322	0.322	0.322	0.322	0.322	0.322	0.322	0.322	0.322	0.322	]
[	0.161	0.161	0.161	0.161	0.161	0.161	0.161	0.161	0.161	0.161	0.161	0.161	]
[	0.107	0.107	0.107	0.107	0.107	0.107	0.107	0.107	0.107	0.107	0.107	0.107	]
[	0.081	0.081	0.081	0.081	0.081	0.081	0.081	0.081	0.081	0.081	0.081	0.081	]
[	0.064	0.064	0.064	0.064	0.064	0.064	0.064	0.064	0.064	0.064	0.064	0.064	]
[	0.054	0.054	0.054	0.054	0.054	0.054	0.054	0.054	0.054	0.054	0.054	0.054	]
[	0.046	0.046	0.046	0.046	0.046	0.046	0.046	0.046	0.046	0.046	0.046	0.046	]
[	0.04	0.04	0.04	0.04	0.04	0.04	0.04	0.04	0.04	0.04	0.04	0.04	]
[	0.036	0.036	0.036	0.036	0.036	0.036	0.036	0.036	0.036	0.036	0.036	0.036	]
[	0.032	0.032	0.032	0.032	0.032	0.032	0.032	0.032	0.032	0.032	0.032	0.032	]
[	0.029	0.029	0.029	0.029	0.029	0.029	0.029	0.029	0.029	0.029	0.029	0.029	]
[	0.027	0.027	0.027	0.027	0.027	0.027	0.027	0.027	0.027	0.027	0.027	0.027	]

Notice that each column is identical (which makes sense: the value of $\frac{j}{\sum_{i=1}^{12} \binom{j}{i}}$ does not depend on j), which means that raising M to a any power would yield itself since M is stochastic. This gives us the point distribution:

$$\begin{bmatrix} 0.322 \\ 0.161 \\ 0.107 \\ 0.081 \\ 0.064 \\ 0.054 \\ 0.046 \\ 0.04 \\ 0.036 \\ 0.032 \\ 0.029 \\ 0.027 \end{bmatrix}.$$

Now, we subtract 0.027 from each entry in this vector and scale it so that the sum of all points distributed is 73. This yields the final distribution of points, with each score rounded to the nearest tenth for ease of readability:

$$\begin{bmatrix} 31.8 \\ 14.5 \\ 8.7 \\ 5.8 \\ 4.0 \\ 2.9 \\ 2.1 \\ 1.4 \\ 1.0 \\ 0.6 \\ 0.3 \\ 0 \end{bmatrix}.$$

Notice that this point distribution is much more heavily skewed towards the higher placements than the one used in-game.

2.2 Ratios of Placement to Sum of Placements

Suppose we want to distribute the initial votes so that each “pair” of votes - that is P_a ’s quantity of votes towards P_b and vice versa - contributes “equally” to the final point distribution. For example, in the last section, the votes between P_1 and P_{12} summed to $12 + \frac{1}{12}$ - much more than the sum of the votes between P_1 and P_2 .

Let us entertain this idea of “equal contribution” by making each pair of votes sum to 1. We can accomplish this by having

$$M_{ij} = \frac{j}{i+j}$$

since it follows from this that $M_{ij} + M_{ji} = 1$ for all i, j where $i \neq j$. For entries on the diagonal (when $i = j$), $M_{ij} = \frac{1}{2}$, i.e. each player gives half a vote to themselves. After scaling each column of M to make M column stochastic, we get:

[	0.229	0.19	0.168	0.154	0.144	0.137	0.131	0.126	0.122	0.119	0.117	0.114	]
[	0.153	0.143	0.135	0.128	0.124	0.12	0.116	0.114	0.111	0.109	0.108	0.106	]
[	0.115	0.114	0.112	0.11	0.108	0.106	0.105	0.103	0.102	0.101	0.1	0.099	]
[	0.092	0.095	0.096	0.096	0.096	0.096	0.095	0.095	0.094	0.094	0.093	0.093	]
[	0.076	0.082	0.084	0.086	0.086	0.087	0.087	0.087	0.087	0.088	0.087	0.087	]
[	0.066	0.071	0.075	0.077	0.079	0.08	0.081	0.081	0.082	0.082	0.082	0.083	]
[	0.057	0.063	0.067	0.07	0.072	0.074	0.075	0.076	0.077	0.077	0.078	0.078	]
[	0.051	0.057	0.061	0.064	0.067	0.068	0.07	0.071	0.072	0.073	0.074	0.074	]
[	0.046	0.052	0.056	0.059	0.062	0.064	0.065	0.067	0.068	0.069	0.07	0.071	]
[	0.042	0.048	0.052	0.055	0.058	0.06	0.062	0.063	0.064	0.066	0.067	0.068	]
[	0.038	0.044	0.048	0.051	0.054	0.056	0.058	0.06	0.061	0.063	0.064	0.065	]
[	0.035	0.041	0.045	0.048	0.051	0.053	0.055	0.057	0.058	0.06	0.061	0.062	]

Once we raise M to a high power and fit the steady state vector so that its 12th entry is 0 and so that the sum of its entries is 73, we get:

$$\begin{bmatrix} 19.5 \\ 14.0 \\ 10.5 \\ 8.1 \\ 6.3 \\ 4.8 \\ 3.7 \\ 2.7 \\ 1.9 \\ 1.2 \\ 0.5 \\ 0 \end{bmatrix}.$$

2.3 Ratios of Placement to Geometric Mean of Placements

The last point distribution could be formulated using the ratios of a racer’s placement to the arithmetic mean of each pair’s placements, since each entry in M would have simply doubled, which would have no bearing on the final result (since we scale each column anyway).

So, now let us try using the geometric mean instead of the arithmetic mean of each pair of racer's placements to quantify how much each "pair" of votes contributes to the final results. This is analogous to using the magnitude of a vector instead of the sum of its components to quantify its size.

Here,

$$M_{ij} = \frac{j}{\sqrt{i^2+j^2}}.$$

This ensures that $(M_{ij})^2 + (M_{ji})^2 = 1$ for all i, j where $i \neq j$. Here $M_{ij} = \frac{1}{\sqrt{2}}$ when $i = j$.

This gives us $M =$

[	0.26	0.196	0.161	0.14	0.127	0.118	0.111	0.106	0.103	0.1	0.097	0.096	]
[	0.164	0.155	0.141	0.129	0.12	0.113	0.108	0.104	0.101	0.098	0.096	0.095	]
[	0.116	0.122	0.12	0.115	0.111	0.107	0.103	0.1	0.098	0.096	0.094	0.093	]
[	0.089	0.098	0.102	0.102	0.101	0.099	0.097	0.096	0.094	0.093	0.092	0.091	]
[	0.072	0.081	0.087	0.09	0.091	0.092	0.091	0.091	0.09	0.09	0.089	0.089	]
[	0.06	0.069	0.076	0.08	0.083	0.084	0.085	0.086	0.086	0.086	0.086	0.086	]
[	0.052	0.06	0.067	0.072	0.075	0.078	0.079	0.081	0.082	0.082	0.083	0.083	]
[	0.046	0.053	0.059	0.065	0.068	0.072	0.074	0.076	0.077	0.078	0.079	0.08	]
[	0.041	0.048	0.054	0.059	0.063	0.066	0.069	0.071	0.073	0.074	0.076	0.077	]
[	0.037	0.043	0.049	0.054	0.058	0.061	0.064	0.067	0.069	0.071	0.072	0.074	]
[	0.033	0.039	0.045	0.049	0.053	0.057	0.06	0.063	0.065	0.067	0.069	0.071	]
[	0.031	0.036	0.041	0.046	0.05	0.053	0.057	0.059	0.062	0.064	0.066	0.068	]

Once we raise M to a high power and fit the steady state vector so that its 12th entry is 0 and so that the sum of its entries is 73, we get:

$$\begin{bmatrix} 18.0 \\ 13.7 \\ 10.6 \\ 8.4 \\ 6.6 \\ 5.1 \\ 3.9 \\ 2.9 \\ 2.0 \\ 1.2 \\ 0.6 \\ 0 \end{bmatrix}.$$

Notice that this distribution is very similar to the last, the difference being that the points here are distributed slightly more evenly.

3 Conclusions and Further Lines of Questioning

Similar lines of reasoning and mathematical techniques could be used more generally for analyzing point distributions in systems of ranking that involve Elo or a similar measure of score that is meant to represent a player's skill, or more generally an item's importance.

This article uses relatively arbitrary methods of formulating a point distribution. Different Markov-chain-related approaches could be applied.

Interestingly the second and third point distributions derived in this article (the ones that use ratio-to-sum and ratio-to-geometric-mean) fit excellently with curves derived through cubic regression, with $R^2 = 0.9976$ and $R^2 = 0.9990$ respectively using the rounded data.

Perhaps a better way to approach this problem would be to use probability distributions of players' times as a function of perceived skill over the course of n races to identify what point distribution most often accurately ranks the players in terms of their skill from the placement data of only one or two races.

The Math Behind Western Standoffs: Bayesian Games

Shaon Anwar

1 Introduction

In the late 1960s, following the explosive development of game theory, Hungarian-American economist John Harsanyi popularized the analysis of situations involving “Bayesian players” distinguished by their lack of complete knowledge of the game. These scenarios, coined “Bayesian games” became incredibly relevant to economic sciences because of their applicability to real-life situations; a model where each player has their own limited set of information avoids many of the unrealistic assumptions taken during a complete-information game.

It is worth noting that in a Bayesian scenario, as with a complete-information game, there exists an optimal solution concept for each player. A common non-Bayesian game solution concept is the Nash equilibrium, where each player cannot benefit from altering their strategy. Likewise, a Bayesian Nash equilibrium solution concept exists for Bayesian games. Since Bayesian players are forced to make assumptions regarding private information, an element of probability is introduced: each player’s strategy is determined by the probabilities of the other player’s potential roles.

2 The Bayesian Game

In game theory, a Bayesian game is a situation defined by the fact that its players hold incomplete information. In other words, players may have access to private information regarding the game, but lack the common knowledge of a complete-information game. To best illustrate this concept of incomplete information, let’s turn to a common example of a Bayesian game: “sheriff’s dilemma.”

In “sheriff’s dilemma,” let there be two players, A and B . Player B assumes the role of the sheriff, while player A assumes the role of either a criminal or an innocent civilian. Importantly, A is aware of their own role as either a criminal or a civilian and the other player’s status as the sheriff. B , on the other hand, is unsure whether or not A is a criminal or a civilian. Thus, B ’s knowledge of their own role is private information, a defining characteristic of Bayesian games.

Assume B decides to confront A , and each player must decide between shooting and not shooting the other. To further establish the players’ intentions, assume that if player A shoots, then player B , the sheriff, prefers to shoot back. Likewise, if A does not shoot, B prefers to not shoot either. On the other hand, if A is a criminal, they prefer to shoot in any case, and if A is a civilian, they prefer to not shoot in any case. Given that p is the probability of player A being a criminal and $1 - p$ is the probability of A being a civilian, we can deduce the scenario that represents Bayesian equilibrium by using expected payoffs and probability.

Each player's expected payoff in each scenario can be modeled by matrices; the upper matrix, C , assumes player A is a civilian, while the lower matrix, D assumes A is a criminal. In each cell, the first number represents player A 's expected payoff, while the second represents that of player B . The headings in the first column correspond to A 's actions, while the headings in the first row correspond to B 's actions. The expected payoffs in each cell are determined by the players' preferences in each scenario.

C	Shoot	Not Shoot
Shoot	-3, -1	-1, -2
Not Shoot	-2, -1	0,0

D	Shoot	Not Shoot
Shoot	0, 0	2, -2
Not Shoot	-2, -1	-1,1

Figure 1: Source: Joonas Vatto, Harvard University

2.1 Matrix C : A is a Civilian

In matrix C , where player A is a civilian, A most prefers to not shoot and not be shot, which would yield a designated expected payoff of 0. A 's next most optimal scenario is to shoot and to not be shot, to which we will assign an expected payoff of -1 . A then prefers to not shoot and be shot, to which we will assign -2 . Finally, A 's worst case scenario is shooting and being shot, yielding an expected payoff of -3 .

As for player B , the optimal scenario is to not shoot in response to not being shot, yielding an expected payoff of 0. B then equally prefers shooting in response to being shot, and shooting in response to not being shot (recall that B does not know if A is a criminal or civilian), with a payoff of -1 . B 's worst-case scenario is being shot without shooting back, with a payoff of -2 .

2.2 Matrix D : A is a Criminal

In matrix D , player A is a criminal, and thus prefers to shoot and not be shot with a payoff of 2. If both parties shoot, B has an expected payoff of 0, which is preferred to neither party shooting (payoff of -1) and being shot at without shooting (payoff of -2).

As the sheriff, B 's preferences generally stay the same, as they do not know if A is a criminal or a civilian. However, if neither party shoots, B receives a payoff of 1, since an optimal scenario results in an arrest after the confrontation.

3 The Optimal Strategies

As demonstrated by the matrices' expected payoffs, A 's strategy simply hinges on their role. If A is a criminal, they should shoot regardless, and if A is a civilian, they should not shoot.

B 's path is naturally reliant on the chances of A being a criminal. Recall that A 's probability of being a criminal is p . We can gauge the expected value by finding the sum of the products of the probability of each case and B 's expected payoffs:

- If the sheriff shoots: $((1 - p) \cdot -1) + (p \cdot 0) = p - 1$
- If the sheriff does not shoot: $((1 - p) \cdot 0) + (p \cdot -2) = -2p$.

Given that the sheriff will shoot if the expected value of shooting outweighs the expected value of not shooting, we can deduce that the sheriff should shoot if $p - 1 > -2p$. After isolating p , we can finally conclude the sheriff shoots when $p > \frac{1}{3}$.

References

- Acemoglu, Daron, and Asu Ozdaglar. "Incomplete Information: Bayesian Nash Equilibria, Auctions ..." MIT Economics, November 23, 2009.
[https://economics.mit.edu/sites/default/files/inline-files/Lectures 19-21 - Bayesian Nash Equilibria, Auctions and Introduction to Social Learning.pdf](https://economics.mit.edu/sites/default/files/inline-files/Lectures%2019-21%20-%20Bayesian%20Nash%20Equilibria,%20Auctions%20and%20Introduction%20to%20Social%20Learning.pdf).
- Hayes, Adam. "Bayes' Theorem: What It Is, the Formula, and Examples." Investopedia, March 30, 2024.
<https://www.investopedia.com/terms/b/bayes-theorem.asp>.
- Obara, Ichiro. "Bayesian Nash Equilibrium." UCLA Economics, February 1, 2012.
<http://www.econ.ucla.edu/iobara/BNE201B.pdf>.
- Vättö, Joonas. Finding Bayes-Nash Equilibria, October 2, 2019.
<https://canvas.harvard.edu/courses/61567/files/8597692/download?verifier=drBHWx30nYWvYVXXzkcahrmfaojZhBiKPQ5y9KTz>.

Lucas Numbers

Elizabeth Burnell

1 Introduction

The Lucas sequence is an intriguing and relatively mysterious series of numbers named after François Édouard Anatole Lucas (1842–91) who studied this sequence in addition to the closely related Fibonacci sequence.

Lucas numbers can be constructed by starting from 2, then 1, and defining the following members of the sequence as the sum of the two prior. This rule is similar to the rule used for Fibonacci sequence which can be thought of starting from 0 and 1 and then any following member is received as the sum of the prior two.

Lucas sequence members are received by taking the prior and following-number Fibonacci sequence member. It can also be thought of as the same-number Fibonacci member plus twice the prior Fibonacci number, see Figure 1.

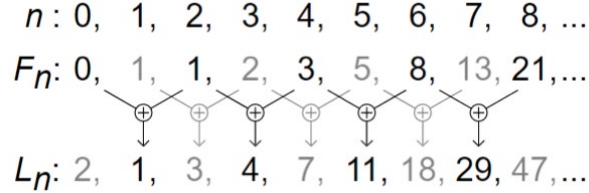


Figure 1: Lucas' relation to the Fibonacci numbers, with $L(n) = F(n-1) + F(n+1)$, or $L(n) = F(n) + 2F(n-1)$. It can be shown that $L(n) = \frac{F(2n)}{F(n)}$

2 Relation to the Golden Ratio

While having complex symmetries, Lucas numbers can be shown as related to the golden ratio and (1-golden ratio ϕ),

$$\phi = \frac{1+\sqrt{5}}{2}, \text{ and}$$

$$(1-\phi) = \frac{2-1-\sqrt{5}}{2} = \frac{1-\sqrt{5}}{2} = \frac{(1-\sqrt{5})(1+\sqrt{5})}{2(1+\sqrt{5})} = -\frac{2}{1+\sqrt{5}} = -\frac{1}{\phi}$$

Denoting $\hat{\phi} = -\frac{1}{\phi}$, we can prove that Lucas sequence members equal to

$$L(n) = \phi^n + (1-\phi)^n = \left(\frac{1+\sqrt{5}}{2}\right)^n + \left(\frac{1-\sqrt{5}}{2}\right)^n = \phi^n + \hat{\phi}^n$$

This proof can be done by mathematical induction.

The basis of induction is $L(0)=1+1=2$ which is true, and $L(1) = \left(\frac{1+\sqrt{5}}{2}\right)^1 + \left(\frac{1-\sqrt{5}}{2}\right)^1 = 1$ which is also true.

As the second part of the proof, induction step, we assume that for any j such that $0 < j \leq k+1$, $L(j) = \phi^j + \hat{\phi}^j$, and we will show that $L(k+2) = \phi^{k+2} + \hat{\phi}^{k+2}$.

We will use an interesting property of the golden ratio, that $\phi^2 = 1 + \phi$, and $\hat{\phi}^2 = 1 + \hat{\phi}$.

In other words, ϕ and $\hat{\phi}$ are roots of the equation $x^2 = 1 + x$.

This is because

$$\phi^2 = \left(\frac{1+\sqrt{5}}{2}\right)^2 = \frac{(1+\sqrt{5})^2}{4} = \frac{6+2\sqrt{5}}{4} = \frac{3+\sqrt{5}}{2} = 1 + \frac{1+\sqrt{5}}{2} = 1 + \phi$$

and

$$\hat{\phi}^2 = \left(\frac{1-\sqrt{5}}{2}\right)^2 = \frac{(1-\sqrt{5})^2}{4} = \frac{6-2\sqrt{5}}{4} = \frac{3-\sqrt{5}}{2} = 1 + \frac{1-\sqrt{5}}{2} = 1 + \hat{\phi}.$$

Then

$$\phi^{k+2} + \hat{\phi}^{k+2} = (1 + \phi)\phi^k + (1 + \hat{\phi})\hat{\phi}^k = (\phi^k + \hat{\phi}^k) + (\phi^{k+1} + \hat{\phi}^{k+1}) = L(k) + L(k+1) = L(k+2)$$

which completes the proof by mathematical induction.

3 Cyclicity

One more interesting property of Lucas numbers is their cyclicity using certain mods (remainders), see Figure 2. For example, for (mod 2) cyclicity, let's apply (mod 2) to the below equality:

$$\begin{aligned} L(n) - L(n-3) &\equiv L(n-1) + L(n-2) - L(n-3) \\ &\equiv 2L(n-2) + L(n-3) - L(n-3) \\ &\equiv 2L(n-2) \end{aligned}$$

We get $L(n) \equiv L(n-3) \pmod{2}$ meaning repetition every 3 numbers.

For (mod 4) cyclicity, let's apply (mod 4) to below equality, which uses above:

$$\begin{aligned} L(n) - L(n-6) &\equiv 2L(n-2) + L(n-3) - L(n-6) \equiv 2L(n-3) + 2L(n-4) + L(n-3) - L(n-6) \\ &\equiv 3L(n-3) + 2L(n-4) - L(n-6) \\ &\equiv 3L(n-3) + L(n-4) + L(n-5) + L(n-6) - L(n-6) \\ &\equiv 3L(n-3) + L(n-3) = 4L(n-3) \end{aligned}$$

$L(n) \equiv L(n-6) \pmod{4}$, meaning repetition every 6 numbers.

For (mod 5) cyclicity, let's apply (mod 5) to below equality, which uses the ideas above plus Fibonacci representation $L(n) = F(n-1) + F(n+1)$:

$$\begin{aligned}
L(n) - L(n-4) &\equiv 2L(n-2) + L(n-3) - L(n-4) \\
&\equiv 3L(n-3) + 2L(n-4) - L(n-4) = 3L(n-3) + L(n-4) \\
&\equiv 3F(n-2) + 3F(n-4) + F(n-3) + F(n-5) \\
&\equiv 3F(n-2) + 2F(n-3) + 2F(n-4) \\
&\equiv 5F(n-2)
\end{aligned}$$

$L(n) \equiv L(n-4) \pmod{5}$, meaning repetition every 4 numbers. Figure 2 also illustrates repetition every 12 numbers by (mod 40) and every 60 numbers by (mod 100).

	L(1)	(Mod 2)	(Mod 4)	(Mod 5)	(Mod 40)	(Mod 100)
1	2	0	2	2	2	2
2	1	1	1	1	1	1
3	3	1	3	3	3	3
4	4	0	0	4	4	4
5	7	1	3	2	7	7
6	11	1	3	1	11	11
7	18	0	2	3	18	18
8	29	1	1	4	29	29
9	47	1	3	2	7	47
10	76	0	0	1	36	76
11	123	1	3	3	3	23
12	199	1	3	4	39	99
13	322	0	2	2	2	22
14	521	1	1	1	1	21
15	843	1	3	3	3	43
16	1,364	0	0	4	4	64
17	2,207	1	3	2	7	7
18	3,571	1	3	1	11	71
19	5,778	0	2	3	18	78
20	9,349	1	1	4	29	49
21	15,127	1	3	2	7	27
22	24,476	0	0	1	36	76
23	39,603	1	3	3	3	3
24	64,079	1	3	4	39	79
25	103,682	0	2	2	2	82
26	167,761	1	1	1	1	61
27	271,443	1	3	3	3	43
28	439,204	0	0	4	4	4
29	710,647	1	3	2	7	47
30	1,149,851	1	3	1	11	51
31	1,860,498	0	2	3	18	98
32	3,010,349	1	1	4	29	49
33	4,870,847	1	3	2	7	47
34	7,881,196	0	0	1	36	96
35	12,752,043	1	3	3	3	43
36	20,633,239	1	3	4	39	39
37	33,385,282	0	2	2	2	82
59	1,322,157,322,203	1	3	3	3	3
60	2,139,295,485,799	1	3	4	39	99
61	3,461,452,808,002	0	2	2	2	2
62	5,600,748,293,801	1	1	1	1	1

Figure 2.

Periodic properties of the Lucas numbers:

- cycles length 3 using (mod 2) colored yellow,

- cycles length 6 using (mod 4) colored green,
- cycles length 4 using (mod 5) colored blue, and
- cycles length 60 using (mod 100) — purple highlights that (mod 100), $L(61) \equiv L(1)$ and $L(62) \equiv L(2)$.

4 Lucas, Fibonacci numbers and Pascal's triangle

Figure 3 illustrates a known representation of Fibonacci numbers using Pascal's triangle discovered by Francois Lucas, algebraically written as

$$F(n) = \sum_{k=0}^{\lfloor (n-1)/2 \rfloor} \binom{n-k-1}{k}$$

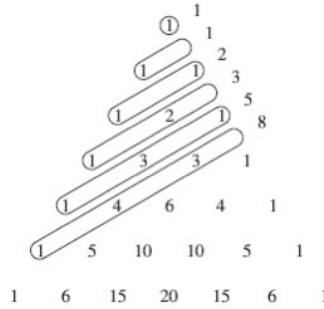


Figure 3: Fibonacci numbers in the Pascal's triangle.

Using the Lucas sequence representation as a sum of prior and the next Fibonacci $L(n) = F(n-1) + F(n+1)$, Pascal's triangle alternate diagonals can be used to receive Lucas numbers, see Figure 4. Algebraically,

$$F(n) = \sum_{k=0}^{\lfloor n/2 \rfloor} \left[\binom{n-k}{k} + \binom{n-k-1}{k-1} \right]$$

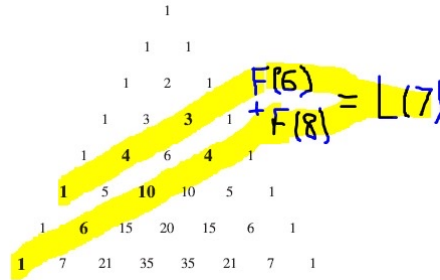


Figure 4: Lucas numbers are sums of alternate diagonals of Pascal's triangle.

This leads to a rewrite of the formula in a combinatorics-like manner.

$$F(n) = \sum_{k=0}^{\lfloor n/2 \rfloor} \left(\frac{n-k}{k} + \frac{n-k-1}{k-1} \right) = \sum_{k=0}^{\lfloor n/2 \rfloor} \frac{n}{n-k} \binom{n-k}{k}$$

5 Notes on Applications

Lucas sequence applications include identifying Mersenne primes (these have form $2^n - 1$), primality tests, solving congruences, integer factorization, solving Diophantine equations and cryptography. As such, Lucas sequences' practical applications are in computational number theory.

The Lucas-Lehmer test for numbers $M_n = 2^n - 1$ was used to identify all known Mersenne primes for $n > 127$.

After applying the theory of his sequences to Mersenne primes, Lucas applied it to primality testing of numbers of the form $Ap^n \pm 1$ where p is prime.

A number of problems concerning Lucas sequences or their possible generalizations remain unsolved or unexplored.

In 1993, a public-key cryptosystem, LUC was invented based on Lucas sequences theory. It was constructed as a viable alternative to RSA encryption. While LUC is somewhat more complex, it has been recommended as a possible authentication system and continues to be the object of active research.

6 Conclusion

It is remarkable that after a lot of activity devoted to a simple pair of sequences, Fibonacci and Lucas, there are still a number of unanswered questions acutely important to today's mathematicians as well as to computer scientists and electrical engineers awaiting solutions. The sequence OEIS number is A000032 and it is listed at <https://oeis.org/A000032>.

References

- Ballot, C.JC., Williams, H.C. (2023). Applications. In: The Lucas Sequences. CMS/CAIMS Books in Mathematics, vol 8. Springer, Cham.
https://doi.org/10.1007/978-3-031-37238-4_3
- “Lucas Numbers: Brilliant Math & Science Wiki.” Brilliant.
<https://brilliant.org/wiki/lucas-numbers/>.
- P. Smith, LUC public-key encryption: a secure alternative to RSA, Dr. Dobb's Journal, 1993, 44–48
- T. Koshy, Fibonacci, Lucas, and Pell Numbers, and Pascal's Triangle. Applied Probability Trust, 2011.
<https://scholar.archive.org/work/yuca6owfj5adtedtstlzx3jd44/access/wayback/http://ms.appliedprobability.org:80/data/files/feature%20articles/43-3-6.pdf>

Cake Cutting

Elizabeth Burnell



1 Piece of cake? How many do you expect?

You must have heard the expression “piece of cake” many times. Have you thought of how many pieces should be expected after making n random cuts? This probability question is easier asked than solved. The answer is $\frac{(n+2)(n+3)}{6}$, which is easy to verify for no cuts, since the cake will be in the original one piece. For one cut, the formula produces 2 pieces: $\frac{(1+2)(1+3)}{6} = 3 \cdot 46 = 2$. The strict probability theory formulation is to produce “the expected number of pieces when a round circular cake is cut with n random cuts”. Let’s count cut-by-cut!

2 Proof

2.1 Cuts do not intersect

First, it is easy to see that if we require for cuts not to intersect, the n th cut will cut one of the existing whole pieces into two new ones, increasing the overall count by 1. Since we started with the whole cake as one piece, after n cuts this will result in $n + 1$ pieces. Therefore, $n + 1$ is a minimum number of pieces from cutting.

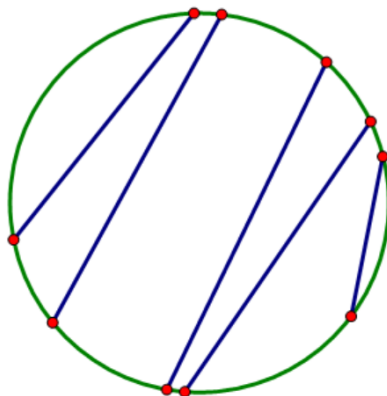


Figure 1: n cuts, no intersections.

2.2 Adding intersections

Second, let's see what is the expected value of pieces added due to an intersection of cuts. Each intersection adds a piece to the total count. If $n = 2$ and the second cut is truly random, what is the expected value of cuts added to the established minimum? To count, please notice that a random cut is equivalent to randomly choosing two points on the circle, which is the circumference of the cake. Then the line through these two points is the random cut.

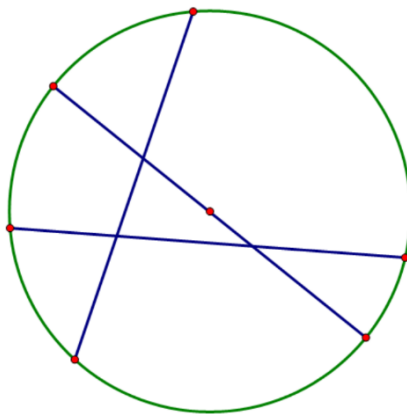


Figure 2: Three cuts, three intersections.

2.3 Case of two cuts intersecting

Two random cuts will be equivalent to picking four random points on the circle. Let's assume that the four points were chosen but the decision about pairing them into straight lines cuts is unknown. Also, without loss of generality, let's pick one of these points, denoting it with A , and other points clockwise as B , C , and D . Then, the probability of $\overline{AB}$ being the cut is $\frac{1}{3}$ (chosen between $\overline{AB}$, $\overline{AC}$, and $\overline{AD}$), and automatically leads to the total of three pieces. This is because the other cut has to be $\overline{CD}$ and they don't intersect.

The choice of $\overline{AC}$ has a $\frac{1}{3}$ probability of being the cut, and results in the total of four pieces, since the other cut has to be $\overline{BD}$ necessarily producing an intersection and one additional, fourth, piece of cake.

The choice of $\overline{AD}$ is symmetric to the choice $\overline{AB}$, has $\frac{1}{3}$ probability of being the cut and results in three pieces total.

The above proves that the expected value of pieces added to the established minimum due to an intersection of two random cuts is $\frac{1}{3} \cdot 1 + \frac{2}{3} \cdot 0 = \frac{1}{3}$.

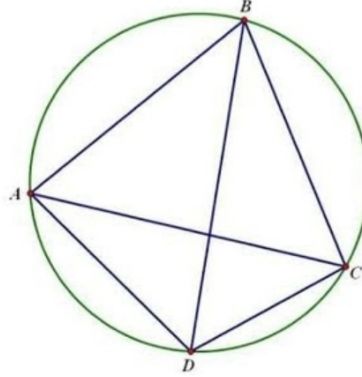


Figure 3: Four points with all possible cuts through them.

3 Generalizations

3.1 N cuts intersecting

With n cuts, let's focus on a certain cut. It could produce an intersection with any of the other cuts, as they are all random, and each pair of cuts is random, similar to the second step above for $n = 2$. There are $(n - 1)$ cut pairs to consider with the cut in focus. Therefore, per the result in 2.3, the expected number of pieces added due to intersections with the cut in focus is *(expected value of pieces added per each pair)(#pairs)* $= \frac{1}{3} \cdot (n - 1) = \frac{n-1}{3}$. However, due to symmetry, this logic applies to each of the n cuts. Applying this logic to every cut of n will double-count each intersection. Therefore, the expected value of additional pieces due to pairwise intersections within n cuts will be

$$E(\#pieces \text{ added due to cuts}) = \frac{n}{2} \cdot \frac{n-1}{3} = \frac{n \cdot (n-1)}{6}.$$

4 Conclusion

Bringing in the first step that without intersections there are $n + 1$ pieces,

$$E(\#pieces \text{ from } n \text{ random cuts}) = n + 1 + E(\#pieces \text{ from intersections}) = n + 1 + \frac{n \cdot (n-1)}{6} = \frac{6 \cdot n + 6 + n^2 - n}{6} = \frac{n^2 + 5 \cdot n + 6}{6} = \frac{(n+2) \cdot (n+3)}{6}.$$

Wasserstein Generative Adversarial Networks (WGANs)

Han Xiao

1 Introduction

Generative Adversarial Networks (GANs) are unsupervised deep learning models that use pre existing data to generate new, realistic data. The generated data has a variety of applications such as acting as training data for networks that lack sufficient data, performing image to image translations, 3D object generation, text to image translations, and more. In this paper, we will continue to talk about the specifics and training algorithm of Wasserstein Generative Adversarial Networks (WGANs), an improved model from basic GANs. For more information about the potential training issues, math, and other intricacies behind GANs, refer to the previous publication.

2 Wasserstein GAN Overview

The foundational structure of a generative adversarial network consists of a generator and a discriminator playing a minmax game with each other. The generator uses noise to produce artificial images that are fed into the discriminator along with images from a real dataset, while the discriminator attempts to assign the correct labels (eg. real or generated) to the received images. The minmax loss function can be simplified as

$$E_{z \sim p(x)}[\log D(x)] + E_{z \sim p(z)}[\log(1 - D(G(z)))]$$

Mathematically, the generator is trying to maximize $D(G(z))$, which ranges between zero and one, thus minimizing the overall function, while the discriminator attempts to do the opposite. Wasserstein networks differ in that they use a Wasserstein loss, or earth mover's distance to minimize function cost.

Furthermore, the discriminator contrasts the generator in that it is a convolutional network, which takes information directly from data, as opposed to a transposed convolutional network, which reconstructs data. As the name suggests, the discriminator reverses the process of the generator.

3 Discriminator: Identifying Data Attributes

A convolutional network is composed of convolutional layers, pooling layers, and a final fully connected layer, as shown below. In image processing, for instance, the convolutional layer consists of an array of weights that act as a small filter placed over a two-dimensional image. It calculates the dot products between the image's pixels, and this is repeated until the entire image has been run through the filter, creating the initial convolutional layer. Additional layers will stem from the first which focuses on different parts of the image. This is useful in situations where the object in the image can be viewed as the sum of its parts. Altogether, the convolutional layers

form a convolutional block. This block is then downsampled by a pooling layer, reducing the dimensionality from five by five to three by three, for example, through a filter of aggregation functions. Aggregation functions take all the features from a region, develop, and apply a singular global feature to the input. The pooling layer can either perform average pooling by calculating the receptive field's average value, or max pooling by applying the maximum value in the receptive field globally. Although this causes some information loss, it creates a more efficient network by reducing input complexity. Finally, the fully connected layer uses features from previous layers and classifies its input using a softmax activation function, which gives a probability between zero and one. This process can be thought of as the discriminator breaking down both generated and real images, and finally assigning a value to the image to classify how likely it is to be a real image.

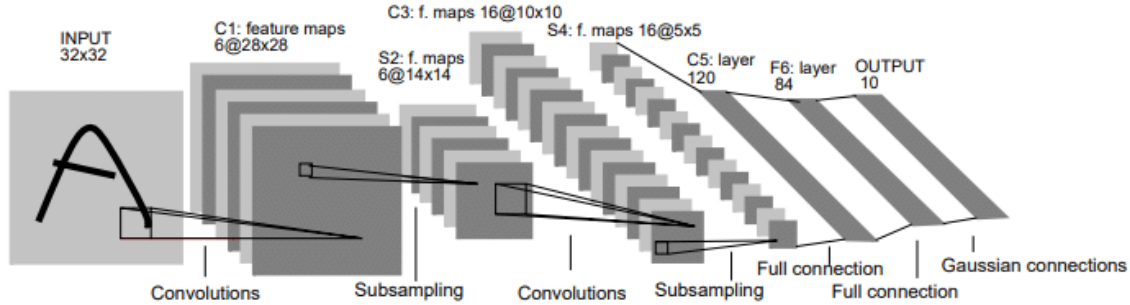


Fig. 2. Architecture of LeNet-5, a Convolutional Neural Network, here for digits recognition. Each plane is a feature map, i.e. a set of units whose weights are constrained to be identical.

4 Generator

Transposed convolution networks, (re)construct images from their input data through upscaling. These networks are capable of producing high resolution images given a low resolution input and performing semantic segregation, which follows the idea previously mentioned: a whole is the sum of its parts. It will classify items in the image and visually label them with color.

After receiving an input with a certain dimension, a filter or kernel, usually a matrix with dimensions 2x2 or 3x3 will slide over the input and perform multiplication. The products of the values are then transferred over to a blank output matrix with the desired dimensions. To further explain using the diagram below, the values of the kernel are multiplied by the value of a pixel of the input and brought over to the output matrix (ex. 0x4, 0x1, 0x2, and 0x3). This is repeated with other pixels within the input before summing up the products that overlap.

$$\begin{array}{c}
 \text{Input} \quad \begin{bmatrix} 0 & 1 \\ 2 & 3 \end{bmatrix} \quad \begin{array}{c} \text{Transposed} \\ \text{Conv} \\ \text{(Stride 1)} \end{array} \quad \text{Kernel} \quad \begin{bmatrix} 4 & 1 \\ 2 & 3 \end{bmatrix} \\
 \\
 = \begin{bmatrix} 0 & 0 & \\ 0 & 0 & \\ & & \end{bmatrix} + \begin{bmatrix} & 4 & 1 \\ & 2 & 3 \\ & & \end{bmatrix} + \begin{bmatrix} & & \\ 8 & 2 & \\ 4 & 6 & \end{bmatrix} + \begin{bmatrix} & & \\ & 12 & 3 \\ & 6 & 9 \end{bmatrix} = \text{Output} \quad \begin{bmatrix} 0 & 4 & 1 \\ 8 & 16 & 6 \\ 4 & 12 & 9 \end{bmatrix}
 \end{array}$$

It is important to note that transposed convolution layers are not the same as deconvolutional layers. Deconvolutional networks utilize the inverse of the multivariate convolutional function

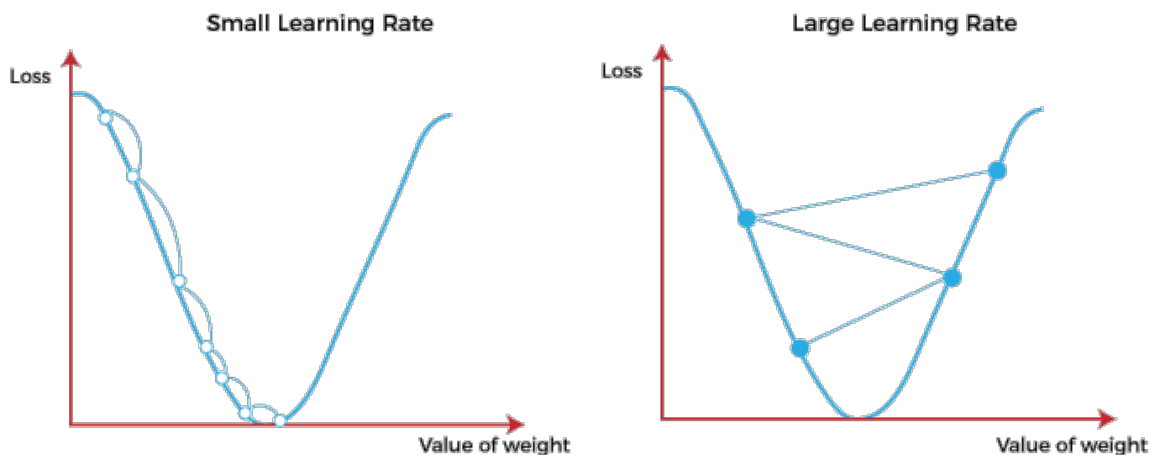
$$h = g \cdot f = \int_x \int_y g(t_1 - x, t_2 - y) f(x, y) dx dy$$

The output of a deconvolutional network is the same as its input. In other words, it will spit out the same image that it was given. On the other hand, transposed convolutional networks produce clearer images that are larger than the one that was input.

5 Network Training

In neural networks, there are hyperparameters such as learning rate, batch size, iterations, epochs, optimizers, and activation functions.

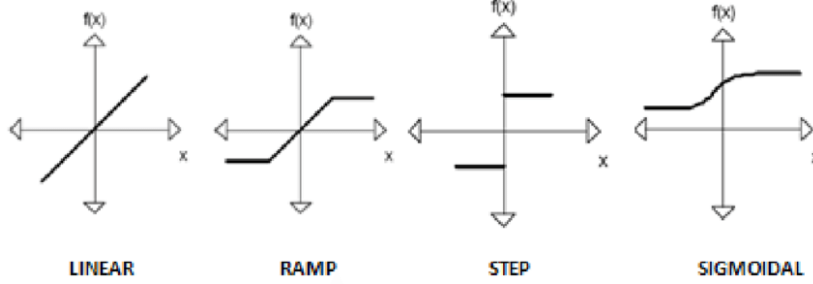
The learning rate is one of the most important hyperparameters for adjusting networks. It sets the pace at which weights are updated in order to decrease network loss. It is important to find an optimal learning rate, as rates that are too small can slow down training and fail to improve the loss function, and rates that are too large can cause divergence, prolonging the training process. To visualize the effects, we can imagine a point descending a curve to reach the global minimum, or convergence. If the learning rate is too small, the point descends too slowly. Contrastly, if the learning rate is too high, the point approaches the global minima quickly but has trouble reaching the minimum as it bounces around continuously. The process of reaching convergence, also known as gradient descent, can become more complicated if there are multiple local minima present in addition to the global minimum.



An epoch is the passing of an entire training set through the network. Usually, an epoch contains too much information to be inputted into the computer at once, so the data is divided into batches. Batch size indicates the number of training samples that should pass through the network before parameters like weights are updated, and iterations involve the number of batches that complete an epoch.

The activation function modifies inputted data to fit within a certain range of values, with the range depending on the type of function used, and establishes a threshold for neuron activation.

Common non-linear activation functions include the binary step function, which labels data as zeros and ones, the softmax function, ranging from zero to one and forcing the sum of all probabilities to equal to one, and the Rectified linear unit (ReLU) function with range zero to infinity.



In most models, a batch normalization layer is present before any activation function is applied. The layer zero-centers and normalizes the input using the batch's mean and standard deviation. Mathematically, the empirical mean of the batch is subtracted from the zero-centered and normalized input, and divided by the square root of the sum of the empirical standard deviation and some minuscule number (usually 10^{-5} , to avoid undefined numbers). Finally, this quotient is multiplied by the scaling parameter for the layer and added to a shifting or offset parameter. This layer further prevents vanishing/exploding gradient problems from arising during training.

6 WGAN Training Algorithm

To elaborate on where the last publication GANs left off, we will be breaking down the original Wasserstein Network algorithm and comparing it to an improved version. Overall, WGANs follow a similar network training process as mentioned above. However, there are a few tweaks to the algorithm other than the unique Wasserstein gradient penalty. The initial proposed algorithm is as follows:

Algorithm 1 WGAN, our proposed algorithm. All experiments in the paper used the default values $\alpha = 0.00005$, $c = 0.01$, $m = 64$, $n_{\text{critic}} = 5$.

Require: : α , the learning rate. c , the clipping parameter. m , the batch size. n_{critic} , the number of iterations of the critic per generator iteration.

Require: : w_0 , initial critic parameters. θ_0 , initial generator's parameters.

```

1: while  $\theta$  has not converged do
2:   for  $t = 0, \dots, n_{\text{critic}}$  do
3:     Sample  $\{x^{(i)}\}_{i=1}^m \sim \mathbb{P}_r$  a batch from the real data.
4:     Sample  $\{z^{(i)}\}_{i=1}^m \sim p(z)$  a batch of prior samples.
5:      $g_w \leftarrow \nabla_w [\frac{1}{m} \sum_{i=1}^m f_w(x^{(i)}) - \frac{1}{m} \sum_{i=1}^m f_w(g_\theta(z^{(i)}))]$ 
6:      $w \leftarrow w + \alpha \cdot \text{RMSProp}(w, g_w)$ 
7:      $w \leftarrow \text{clip}(w, -c, c)$ 
8:   end for
9:   Sample  $\{z^{(i)}\}_{i=1}^m \sim p(z)$  a batch of prior samples.
10:   $g_\theta \leftarrow -\nabla_\theta [\frac{1}{m} \sum_{i=1}^m f_w(g_\theta(z^{(i)}))]$ 
11:   $\theta \leftarrow \theta - \alpha \cdot \text{RMSProp}(\theta, g_\theta)$ 
12: end while

```

The discriminator component uses simple linear activation functions in the form of $y = mx + b$, class labels of one and negative one are assigned for real and fake images, respectively, the Wasserstein loss is applied, weight clipping is applied, and parameters are tweaked with RMSProp gradient descent. It is also notable that the critic is updated more frequently than the generator, shown by the value of n_{critic} , which denotes five discriminator weight updates for every one update to the generator. Other lines of the algorithm show the mathematical processes that utilize the earth mover’s distance.

Some components that distinguish this original algorithm from the rest are weight clipping, a method to keep the Lipschitz constant of the critic under control, and RMSProp (Root Mean Squared Propagation), an optimization algorithm. We see the use of weight clipping in line seven, which constrains all the parameters of the critic to certain set values. Although the proposers of this algorithm recognize the weaknesses of weight clipping, such as prolonging training, it was by far the simplest and best working method at the time of publication. RMSProp optimizes parameter values to make for a time efficient gradient descent and speed up convergence by using exponential decay to only accumulate gradients, or changes in weights, from the most recent iterations. RMSProp was shown to perform better than other optimizers at the time due to its ability to not slow down too fast, which may prevent convergence to the global optimum.

7 Improved WGAN Algorithm and Training

In a slightly newer publication, the authors show the negative effects of weight clipping and suggest using gradient penalty. In addition, Adam, a refined optimization algorithm, is used instead of RMSProp. This is visible in lines seven, nine, and twelve.

Algorithm 1 WGAN with gradient penalty. We use default values of $\lambda = 10$, $n_{critic} = 5$, $\alpha = 0.0001$, $\beta_1 = 0$, $\beta_2 = 0.9$.

Require: The gradient penalty coefficient λ , the number of critic iterations per generator iteration n_{critic} , the batch size m , Adam hyperparameters α, β_1, β_2 .

Require: initial critic parameters w_0 , initial generator parameters θ_0 .

```

1: while  $\theta$  has not converged do
2:   for  $t = 1, \dots, n_{critic}$  do
3:     for  $i = 1, \dots, m$  do
4:       Sample real data  $x \sim \mathbb{P}_r$ , latent variable  $z \sim p(z)$ , a random number  $\epsilon \sim U[0, 1]$ .
5:        $\tilde{x} \leftarrow G_\theta(z)$ 
6:        $\hat{x} \leftarrow \epsilon x + (1 - \epsilon)\tilde{x}$ 
7:        $L^{(i)} \leftarrow D_w(\tilde{x}) - D_w(x) + \lambda(\|\nabla_{\hat{x}} D_w(\hat{x})\|_2 - 1)^2$ 
8:     end for
9:      $w \leftarrow \text{Adam}(\nabla_w \frac{1}{m} \sum_{i=1}^m L^{(i)}, w, \alpha, \beta_1, \beta_2)$ 
10:   end for
11:   Sample a batch of latent variables  $\{z^{(i)}\}_{i=1}^m \sim p(z)$ .
12:    $\theta \leftarrow \text{Adam}(\nabla_\theta \frac{1}{m} \sum_{i=1}^m -D_w(G_\theta(z)), \theta, \alpha, \beta_1, \beta_2)$ 
13: end while
```

The authors of this publication showed that weight clipping, amongst other weight constraints, tend to lead to a failure to fully converge, and predisposes the discriminator towards simpler functions. The alternative way to enforce the Lipschitz constraint is to use gradient penalty, which focuses on constraints along coupled points from the real and generated images. The penalty contains a penalty coefficient that controls the rate of convergence through minimization. Within this change, batch normalization of the critic is removed, since gradient penalty focuses on mapping

individual inputs to individual outputs as opposed to a whole batch of inputs to a whole batch of outputs. Lastly, gradient descent encourages penalty of both the generator and the discriminator, or the norm of the gradient to approach a value of one. Although this has not been further investigated, it seems to improve the results of the network. Adam optimization is similar to RMSProp in that it keeps track of the exponential decay averages of past squared gradients and is an adaptive learning rate algorithm. However, it also tracks the exponential decay average of past gradients. Adam introduces more hyperparameters, β_1 and β_2 .

Other changes include changing class label numbers from one and negative one to zero and one.

8 Conclusion

There are many other ways to improve Wasserstein Generative Adversarial networks, but the improvements may be more difficult to implement or have a high time and cost consumption. In the field of generative networks, it has been proven that if there is sufficient data, networks containing more basic components and computations are able to achieve the same results and even better ones than the improved algorithms we have now.

References

- Arjovsky, Martin, Soumith Chintala, and Léon Bottou. “Wasserstein GAN,” December 6, 2017. <https://arxiv.org/pdf/1701.07875.pdf>.
- Boesch, Gaudenz. “Guide to Generative Adversarial Networks (GANs) in 2023.” viso.ai, February 1, 2023. <https://viso.ai/deep-learning/generative-adversarial-networks-gan/>.
- Brownlee, Jason. “How to Develop a Wasserstein Generative Adversarial Network (WGAN) from Scratch.” Machine Learning Mastery, July 16, 2019. <https://machinelearningmastery.com/how-to-code-a-wasserstein-generative-adversarial-network-wgan-from-scratch/>.
- Datagen. “Convolutional Neural Network: Benefits, Types, and Applications.” Datagen, n.d. <https://datagen.tech/guides/computer-vision/cnn-convolutional-neural-network/>.
- Doshi, Chinesh. “Generalized Way of Interpreting CNNs!!” Medium, August 22, 2020. <https://medium.com/@chinesh4/generalized-way-of-interpreting-cnns-a7d1b0178709>
- Gulrajani, Ishaan, Faruk Ahmed, Martin Arjovsky, Vincent Dumoulin, and Aaron Courville. “Improved Training of Wasserstein GANs,” December 25, 2017. <https://arxiv.org/pdf/1704.00028.pdf>.
- Tsang, Sik-Ho. “Brief Review — WGAN-GP: Improved Training of Wasserstein GANs.” Medium, August 1, 2023. <https://sh-tsang.medium.com/brief-review-wgan-gp-improved-training-of-wasserstein-gans-ae3e2acb25b3>.

Stuyvesant Math and Computer Science Achievements

2023-2024

2023 Congressional App Challenge

John Chandler was a winner of this contest, designed to support STEM and computer coding education for middle and high school students within the 13th Congressional District.

American Regions Mathematics League (ARML) Competition

NYC Math Team's "Tin Man" team finished in 5th place overall. Andrew Li and Mikayla Lin placed 23rd and 41st respectively in the individual round.

Harvard-MIT Mathematics Tournament (HMMT)

NYC Math Team's "Tin Man" team placed 11th overall and 7th in the Team Round. Andrew Li finished in 16th place overall individually, 21st in Algebra/Number Theory, and 4th in Combinatorics. Daniel Potievsky and Aditya Pahuja both tied for 13th in Geometry.

Princeton University Mathematics Competition (PUMaC)

NYC Math Team's "Tin Man" team finished in 2nd place overall and 2nd place in the Power Round. Andrew Li came in 4th in Number Theory.

New York State Mathematics League (NYSML)

NYC Math Team's "Tin Man," "Scarecrow," "Lollipop Guild," and "Lion" won 1st place, 3rd place, 4th place, and 5th place overall. Aditya Pahuja and Ian Buchanan placed 3rd and 4th, respectively, in the Individual Round.

Math Majors of America Tournament for High Schools (MMATHS) @ Yale

NYC Math Team's teams, "Tin Man," "Scarecrow," and "Lion" won 4th place, 7th place and 8th place overall. Aditya Pahuja, Daniel Potievsky, Haokai Ma, and Steven Lou made tie-breakers, and Aditya Pahuja advanced to the Lightning round.

Girls in Math (GiM) @ Yale

Stuyvesant sent 4 teams to the Girls in Math competition at Yale. Mikayla Lin and Lucia Liu both made the tie-breaker round. Mikayla Lin finished in 4th place in the Lightning finals.

Math Prize for Girls (MPfG)

Sophia Jin received honorable mention at Math Prize for Girls held at MIT.

Philadelphia Classic (PClassic)

In the fall contest, team "wellfigureitout" (Andrew Li, Haokai Ma, Aditya Pahuja, and Calvin Zhang) were the 3rd place team in the advanced division. In the spring, Mark Ma was a part of team "WTF Strong!?!?!?" which placed 1st in the advanced division, and team "Mogussy" (Derek Yang, Ethan Liu, Haokai Ma, and Matthew Chen) placed 2nd in the advanced division.

Cornell Coding Competition

Stuyvesant's Competitive Computing Club sent three teams which placed 1st, 2nd and 5th.

Stuyvesant Math Team Contests

The Stuyvesant Math Team held their annual math competition, the Downtown Mathematics Invitational (DMI), with record participation, hosting over 350 students. Members Amanda Louie and Mikayla Lin also co-founded a new competition, Girls Empowered in Mathematics at Stuyvesant (GEMS), aimed at creating a space for girls in NYC to meet and engage with other girls who share the same love for math.

International Mathematical Modeling Challenge (IM²C)

Stuyvesant's Ryan Park, Ethan Sharma, Ethan Sie, and Jackie Zeng were one of two teams representing the USA in the 10th Annual International Mathematical Modeling Contest and received an Honorable Mention, the first Stuyvesant team to do so.

2023 High School Mathematical Contest in Modeling (HiMCM)

Two Stuyvesant teams (Ethan Sharma, Ryan Park, Ethan Sie, and Jackie Zeng; Christopher Louie, Jack Cheng, Andy Huang, and Jaco Xia) earned the "Meritorious" designation and one team received an Honorable Mention (Sascha Gordon-Zolov, Astrid Harrington, Filippas Kogiantis, and Mina Yeh) for their work.

New York National Personal Finance Challenge

Stuyvesant team "Stuybigcompgroup" (Zhihan Yao, Christopher Louie, Otto Buff, and Daniel Gabai) advanced to the state finals held at the NY Federal Reserve Bank.

USA Junior Mathematical Olympiad (USAJMO)/USA Mathematical Olympiad (USAMO)

Enmei Yang was awarded a USAMO Honorable Mention. Sophia Jin was awarded USAJMO Honors and was also selected to attend the Mathematical Olympiad Program (MOP) at Carnegie Mellon University.

USAMO Qualifiers

Brandon Zhang (11th)
Enmei Yang (11th)
Andrew Li (12th)

USAJMO Qualifiers

Sophia Jin (10th)

American Mathematics Competition (AMC) 12B

Enmei Yang was the School Winner with a score of 145.5.

American Mathematics Competition (AMC) 10B

Charles Shen was the School Winner with a score of 123.

American Mathematics Competition (AMC) 12B Honor Roll of Distinction (top 1%)

Ian Buchanan, Enmei Yang

American Mathematics Competition (AMC) 12B Distinction (top 5%)

Andrew Li, Steven Lou, Aditya Pahuja, Noam Pasman, Daniel Potievsky, Stanley Tung, Connor Yau, Jackie Zeng, Brandon Zhang, Qinwen Zheng

American Mathematics Competition (AMC) 10B Distinction (top 5%)

Tanvir Ahmed, Andy Jiang, Sophia Jin, Audrey Jing, Isaac Lin, Narnia Poddar, Henry Santos Hendricks, Charles Shen, Andrew Tsai, Cynthia Wen, Philip Zhang

73 American Invitational Mathematics Examination (AIME) Qualifiers

Tanvir Ahmed, Souvik Basak, Krish Bhandari, Steven Breger, Ian Buchanan, Finn Charest, Matthew Chen, Vincent Chen, Joshua Choi, Richard Delgado, Charlie Ding, Orlin Dyankov, Corwin Eisenbeiss, Caleb Hahn, Astrid Harrington, Hugo Hu, Stanley Huang, Vincent Huang, Aareeb Jamil, Henry Ji, Andy Jiang, David Jiang, Sophia Jin, Audrey Jing, David Jobin, Caden Khuu, Clyde Konty, Anastasia Lee, Hyunseo Lee, Mary Lee, Andrew Li, Aiden Lim, Edwin Lin, Isaac Lin, Mikayla Lin, Steven Lou, Chongtian Ma, Haokai Ma, Thomas Mackey, Aditya Pahuja, Noam Pasman, Ryan Peng (9th), Ryan Peng (12th), Narnia Poddar, Daniel Potievsky, Henry Santos Hendricks, Jai Shah, Ethan Sharma, Charles Shen, Chenkai Shen, Dylan Sohn, Nicholas Trofimov, Andrew Tsai, Stanley Tung, Jerry Wang, Leonna Wang, Yucheng Wang, Cynthia Wen, Leon Wong, Patrick Xu, Enmei Yang, Connor Yau, Eric Ye, Mina Yeh, Jackie Zeng, Brandon Zhang, Calvin Zhang, Philip Zhang, Raymond Zhang, Wen Yuan Zhang, Qinwen Zheng, Ashley Zhu, Ivy Zhu

Mathematics and Computer Science Department Senior Awards

The Dr. Richard Rothenberg Memorial Award for Highest Honors in Mathematics: David Jiang

The Richard B. Geller Memorial Scholarship for Mathematics: Andrew Li, Mikayla Lin

Henrietta O. Midonick Award: Maya Wassercug

David J. Pichler Memorial Scholarship: Shaon Anwar

The Okean Family Memorial Award for Scholarship in Pure or Applied Mathematics: Aditya Pahuja, Noam Pasman

Irene Finkel Memorial Award for Excellence in Mathematics: Calvin Zhang, Daniel Potievsky

Highest Honors in Computer Science: Pranjali Modi, Jason Ng, Randy Sim

High Honors in Computer Science: Zoe Cater, Andrew Che, Anthony Chen, Vincent D'Angelo, Aareeb Jamil, Naowal Rahman, Amanda Tung

Honors in Computer Science: Zidane Karim, Andy Lau, Benjamin Nudelman, Tianlang Qin, Haokun Xu

Golden Ducky for Service to the Computer Science Community: Mary Lee, Jason Lei, Benjamin Nudelman, Jowita Walkup

